Ø¬Ø§Ù…Ø¹Ø© Ø§Ù„Ø£Ù‚Ø³Ø§Ù… Ø§Ù„Ù‡Ù†Ø¯Ø³ÙŠØ©

©Ù‘Ø§Ù„Ø³Ù†Ù‡Ù„Ù‘Ø¬ÙŠØ§

Ù‚Ù„ÙŠÙ„ Ø³Ù†Ù‡Ù„Ù‘Ø¬ÙŠØ§ Ø§Ù„Ù…Ø¹Ù„Ù‘Ù…Ø§Øª Ù‘Ø§Ù„ÙŠÙˆÙ…ÙŠÙ‘Ø© - Ù‚Ø³Ù… Ù‡Ù†Ø¯Ø³Ø© ©Ø§Ù„Ø¨Ø±Ù…Ø¬Ø§Øª Ù‘Ø§Ù„Ø³Ø·Ø·Ù‚Ø§Øª Ø§Ù„Ø°Ù‚ÙŠÙ‚

ðŸ› ï¸ WORK

Ù…Ø´Ø±ÙˆØ¹ Ø§Ù„ØªØ®Ø±Ø¬ Ø§Ù„Ù‡Ù†Ø¯Ø³ÙŠ Ø§Ù„Ù…Ø¹Ù…Ø§Ø±ÙŠ

ðŸ“‹ Ù„Ø¤Ø·Ø¨Ù‚Ø§Øª Ø§Ù„Ø´Ø¨ÙƒØ§Øª Ø§Ù„Ø¹Ø§Ù…Ø© Ù„Ù„Ø¹Ù…Ù„ WORK

Ù…Ù†Ø© Ø³Ø¬Ù„Ø§Øª Ø§Ù„Ø¹Ù…Ù„Ø§Ø¡

Ø§Ù„Ø¹Ù…Ù„Ø©	Ø§Ù„Ø¹Ù…Ù„Ø§Ø¡ Ø§Ù„Ù†Ø´Ø·ÙˆÙ†	Ø§Ù„Ø¹Ù…Ù„Ø§Ø¡ Ø§Ù„Ù…Ø³Ø¬Ù„ÙˆÙ†
Ø§Ù„Ø¹Ù…Ù„Ø© Ø§Ù„Ø£Ø³Ø§Ø³ÙŠØ© Ù„Ù„Ø¹Ù…Ù„	Ø§Ù„Ø¹Ù…Ù„Ø§Ø¡ Ø§Ù„Ø°ÙŠÙ† Ø§Ø´ØªØ±ÙˆØ§ Ø®Ù„Ø§Ù„ Ø§Ù„Ø´Ù‡Ø±	Ø¥Ø¬Ù…Ø§Ù„Ù‰ Ø§Ù„Ø¹Ù…Ù„Ø§Ø¡

Ø§Ù„Ù…ÙŠØ²Ø§Øª Ø§Ù„Ø¥Ø¶Ø§ÙÙŠØ©	**Ø§Ù„ØªÙ‚Ù†ÙŠØ§Øª Ø§Ù„Ù…Ø³ØªØ®Ø¯Ù…Ø©**	**Ø§Ù„Ù…ÙŠØ²Ø§Øª Ø§Ù„Ø£Ø³Ø§Ø³ÙŠØ©**
Smart Price Proximity Algorithm & Base64 Compression Algorithm.	Ø§Ù„Ø®ÙˆØ§Ø±Ø²Ù…ÙŠØ© Ø§Ù„Ø°ÙƒÙŠØ© Ù„Ù„Ø¨ØØ« Ø¹Ù† Ø§Ù„Ø£Ø³Ø¹Ø§Ø± Ø§Ù„Ù‚Ø±ÙŠØ¨Ø©	**Ø§Ù„Ù…Ø®Ø·Ø·Ø§Øª Ø§Ù„Ø¨ÙŠØ§Ù†ÙŠØ©**
Realtime Database NoSQL, Anonymous Authentication, Regional URLs.	Ø§Ù„Ø®Ø¯Ù…Ø§Øª Ø§Ù„Ø³ØØ§Ø¨ÙŠØ© Ø§Ù„Ù…Ø³ØªØ®Ø¯Ù…Ø© Ù…Ù† Google Firebase	**Ø§Ù„Ù…Ø®Ø·Ø·Ø§Øª Ø§Ù„Ø³ØØ§Ø¨ÙŠØ©**
Ù…Ø®Ø·Ø· Ø§Ù„ØªØ³Ù„Ø³Ù„ Ø§Ù„Ø²Ù…Ù†ÙŠ Ù„Ø¹Ù…Ù„ÙŠØ© Ø§Ù„Ø¨ØØ« Ø¹Ù† Ø§Ù„Ø£Ø³Ø¹Ø§Ø± Ø§Ù„Ù‚Ø±ÙŠØ¨Ø©.	Ø§Ù„Ù…Ø®Ø·Ø·Ø§Øª Ø§Ù„Ø³Ù„ÙˆÙƒÙŠØ© Ø§Ù„Ù…Ø³ØªØ®Ø¯Ù…Ø© Ù„ØªÙˆØ¶ÙŠØ Ø¢Ù„ÙŠØ© Ø¹Ù…Ù„ Ø§Ù„Ø®ÙˆØ§Ø±Ø²Ù…ÙŠØ© (UML Class & Sequence Diagrams)	**Ø§Ù„Ù…Ø®Ø·Ø·Ø§Øª Ø§Ù„Ø³Ù„ÙˆÙƒÙŠØ©**
Ø¯Ø¹Ù… Ù„Ù„Ø¹Ù…Ù„ Ø§Ù„Ù€ XML Ø§Ù„Ø®Ø§Øµ Ø§Ù„Ø¨ØØ« Ø§Ù„Ø¬ØºØ±Ø§ÙÙŠ.	Ø§Ù„Ø¹Ù…Ù„ÙŠØ§Øª Ø§Ù„Ø¨Ø±Ù…Ø¬ÙŠØ© Ø§Ù„Ø®Ø§Ø±Ø¬ÙŠØ© Ø§Ù„Ù…Ø³ØªØ®Ø¯Ù…Ø© Ù„Ø¹Ø±Ø¶ Material Design 3	**Ø§Ù„Ù…Ø®Ø·Ø·Ø§Øª Ø§Ù„ØªÙØ§Ø¹Ù„ÙŠØ©**

Ø±Ù‚Ù… Ø§Ù„Ù‚Ø³Ù…	Ø¹Ù†ÙˆØ§Ù† Ø§Ù„Ù‚Ø³Ù… Ø§Ù„Ø±Ø¦ÙŠØ³ÙŠ	Ù†Ø·Ø§Ù‚ Ø§Ù„Ø¹Ù…Ù„ Ø§Ù„Ù…Ø´Ù…ÙˆÙ„
Ø§Ù„Ù‚Ù…Ù‘Ø§Ø³Ø¹Ø©	Ø¯Ù„ÙŠÙ„ Ø§Ù„Ù…Ø³ØªØ®Ø¯Ù… Ø§Ù„Ø´Ø§Ù…Ù„ Ù„Ø¬Ù…ÙŠØ¹ Ø§Ù„Ù…Ù…ÙŠØ²Ø§Øª (User Manual)	Ø¯Ù„ÙŠÙ„ Ø®Ø·ÙˆØ© Ø¨Ø®Ø·ÙˆØ© Ù„Ø´Ø±Ø Ø¬Ù…ÙŠØ¹ Ø´Ø§Ø´Ø§Øª Ø§Ù„ØªØ·Ø¨ÙŠÙ‚ ÙˆÙƒÙŠÙÙŠØ© Ø§Ø³ØªØ®Ø¯Ø§Ù…Ù‡Ø§ Ø¨Ø´ÙƒÙ„ ØµØÙŠØ.
Ø§Ù„Ù‚Ù…Ù‘Ø§Ø±Ø¨Ø¹	Ù‚Ù†ÙˆØ§Øª Ø§Ù„Ø§ØªØµØ§Ù„ Ø¨Ø§Ù„Ø®Ø§Ø¯Ù… Ø¹Ø¨Ø± Ø¨Ø±ÙˆØªÙˆÙƒÙˆÙ„ NoSQL Ø¨Ø§Ù„ØªÙØµÙŠÙ„ Ø§Ù„Ù…Ù…Ù„ Ù„Ù„Ø®Ø¯Ù…Ø©	Ø®Ø¯Ù…Ø§Øª Ø§Ù„Ø¨Ø·Ø§Ù‚Ø§Øª Ø§Ù„Ù…Ø®ØªÙ„ÙØ©: /services, /orders, /chats.
Ø§Ù„Ù‚Ù…Ù‘Ø§Ù„Ø®Ø§Ù…Ø³	Ø§Ù„Ù…Ø±Ø¬Ø¹ Ø§Ù„Ø´Ø§Ù…Ù„ Ù„Ø¬Ù…ÙŠØ¹ Ø§Ù„ÙˆØ§Ø¬Ù‡Ø§Øª Ø§Ù„Ø¨Ø±Ù…Ø¬ÙŠØ© (JavaDocs)	Ø§Ù„ÙˆØ«Ø§Ø¦Ù‚ Ø§Ù„Ø±Ø³Ù…ÙŠØ© Ù„Ù€JavaDoc.
Ø§Ù„Ù‚Ù…Ù‘Ø§Ø¯Ø³	Ø£Ø¯Ù„Ø© Ø§Ù„Ù…Ø·ÙˆØ±ÙŠÙ† Ø§Ù„Ø°ÙŠÙ† ÙŠØ±ÙŠØ¯ÙˆÙ† Ø§Ù„Ø¨Ù†Ø§Ø¡ Ø¹Ù„Ù‰ Ø§Ù„Ù†Ø¸Ø§Ù… Ø£Ùˆ Ø§Ù„ØªÙƒØ§Ù…Ù„ Ù…Ø¹Ù‡	Ø¯Ù„ÙŠÙ„ Ø§Ù„Ù…Ø³Ø§Ù‡Ù…Ø© Ø§Ù„Ø±Ø³Ù…ÙŠØ© ÙˆÙ…Ø¹Ø§ÙŠÙŠØ± Ø§Ù„ÙƒÙˆØ¯ ÙˆØ§Ù„Ø§Ø®ØªØ¨Ø§Ø±Ø§Øª (APA).
Ø§Ù„Ù…Ù„ØÙ‚ (Ø£)	Ø§Ù„Ù…ÙˆØ³Ø¹Ø© Ø§Ù„Ø´Ø§Ù…Ù„Ø© Ù„Ø¬Ù…ÙŠØ¹ Ø§Ù„Ø£Ù‚Ø³Ø§Ù…	Ù†Ø¸Ø±Ø© Ø¹Ø§Ù…Ø© Ø¹Ù„Ù‰ 18 Ù…Ù† Ø§Ù„Ø¯Ø§Ø®Ù„

Ø±Ù‚Ù… Ø§Ù„Ø¯Ø±Ø³	Ø¹Ù†Ù‰Ø§Ù† Ø§Ù„Ø¯Ø±Ø³ Ø§Ù„Ø¹Ù…Ù„Ù‰	Ù†Ø´Ø§Ø· Ø§Ù„Ø·Ø§Ù„Ø¨ Ø§Ù„Ø¹Ù…Ù„Ù‰
	Ù‚Ø§Ù„Ø¨ (Java Source Code)	Ù„Ù‚Ø±Ø§Ø¡Ø© Ù‚Ø§Ù„Ø¨ Ø§Ù„Ø¯Ø±Ø³ ÙˆÙ…Ù†Ø§Ù‚Ø´Ø© Ø§Ù„Ø¹Ù†Ø§ØµØ±
Ø§Ù„Ù…Ù„Ù, (Ø¨)	Ø§Ù„Ø£Ù‚Ø³Ø§Ù… Ø§Ù„Ù…Ø®ØªÙ„ÙØ© Ù„Ù„Ø© Ù„Ù„Ø§Ø³ØªØ®Ø¯Ø§Ù… Ù‚Ø§Ù„Ø¨ XML Layouts	Ø¨Ù†Ø§Ø¡ Ø§Ù„Ø¹Ù†Ø§ØµØ± Ø§Ù„Ø¨ØµØ±ÙŠØ© Ø§Ù„Ù…Ø®ØªÙ„ÙØ© Ø¨Ø§Ø³ØªØ®Ø¯Ø§Ù… Ø§Ù„Ø¨Ø±Ù…Ø¬Ø©

Ø§Ù„Ù…Ø±Ø§ØÙ„ Ø§Ù„Ø£Ù†Ù‰: Ø§Ù„Ù…Ø¬Ù„Ø¯Ø§Øª Ø§Ù„Ø´Ø®ØµÙŠØ© Ø§Ù„Ù…Ø´Ø§Ø±ÙƒØ© Ù„Ù„Ø¹Ø±Ø¶ Ø§Ù„Ø¹Ø§Ù…

Ù…Ù‚Ø¯Ù…Ø© Ù„Ù† ÙŠØ¹Ø±Ù Ø¹Ù…Ù„ Ø¹Ù„Ù‰ 1.1 Ø§Ù„Ù…Ø´Ø±ÙˆØ¹

Ø´Ù ‡ Ø¯ Ø§Ù„Ø¹Ù„Ù… ÙŠÙŠ Ø§Ù„Ø³Ù † Ù‚Ø§Ø¹ Ø§Ù„Ø£Ø®ÙŠØ±Ø© Ø²Ù‚Ù„Ø§Ù‹ Ù‡Ø§Ø¦Ù„Ø§Ù‹ Ù‡Ø°Ø§ Ø§Ù„Ø¬Ø§Ù† Ù‚Ù„Ù„Ø¨ÙˆÙŠØ§Ø§ Ø§Ù„Ø±Ù‚Ù…Ù„Ø© Ù‚Ø§Ù„Ø¨ÙŠØ© Ù‚Ø§Ø¦Ù…Ø© Ø§Ù„Ø¹Ù…Ù„ (Freelancing)Ø Ø¨Ø« Ø£Ø¬Ø¯ Ø£Ø¬Ø§Ø¯ Ø§Ù„Ù…Ù‡ Ù ‡ Ø§Ø±Ø§Ø² Ø§Ù„Ø²Ù† Ù‚Ø§Ù„Ø¨ÙŠØ© Ù‚Ø§Ù„Ø¨ÙŠØ¹ÙŠØ© Ø§Ù„Ø¶Ù„Ù‚ Ø£Ø³Ù„Ù‚Ø¨ Ø§Ù„Ø²Ø®Ø±Ø§Ù Ø§Ù„Ø²Ø®Ø±Ø§Ù Ø§Ù„Ø¹Ø§Ù… Ù‚Ø§Ù„Ø¨ÙŠØ© Ø§Ù„Ø¨Ø±Ø§Ù…Ø¬ Ø§Ù„Ø¹Ø§Ù… Ù‡ Ù ‡ Ø§Ù„Ø¹Ø§Ù… Ø§Ù„Ø¨Ø±Ø§Ù…Ø¬ Ù‚Ø§Ù„Ø¨ÙŠØ© Ø§Ù„Ø¹Ø§Ù… Ø§Ù„Ø¹Ø§Ù„Ù… Ø§Ù„Ø¨Ø±Ø§Ù…Ø¬ Ø§Ù„Ø¹Ø§Ù… Ø§Ù„Ø¹Ø§Ù„Ù…. Ù…Ø¹ Ù‡Ø°Ø§ Ø§Ù„Ø·Ø±Ø Ø£Ø¬Ø¯Ø§ Ø§Ù„Ø¥Ù‡Ø§Ø¦Ù Ø§Ù„Ø°Ù‚Ù„Ø© Ø¯Ù† Ø¸Ù„Ø¨ Ø§Ù„Ø£Ù† Ø¬Ø¯Ø¯ Ù‡ Ù‚Ø§Ù„Ø¨ÙŠØ© Ø§Ù„Ø¨Ø¹Ø¦Ø¨ÙŠØ© Ø§Ù„Ø¹Ø§Ù„Ù… Ù‚Ø³Ø®Ø®Ø© Ù‚Ø§Ù„Ø¨ÙŠØ© Ø§Ù„Ø¨Ø¹Ø¦Ø¨ÙŠØ© Ø§Ù„Ø¹Ø§Ù„Ù… Ù…Ù„Ø§Ù„Ø¨ Ø§Ù„Ø¹Ø§Ù… Ù‡ Ø¯Ø§Ù‡ Ø§Ù„Ø¥Ø¹Ø§Ù„Ø¨ÙŠØ© Ø§Ù„Ø¹Ø§Ù… Ù‚

Ù‡Ù†ÙŠØ©ØŒ Ù‡**Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø© (Provider)** Ø£ØµØ§Ø¯Ø§ Ø§Ù„Ù…Ø±Ø®Øµ Ø§Ù„Ø°ÙŠ ÙŠÙ…ÙƒÙ†Ù‡ Ø¹Ø¨Ø± Ù…Ø§ÙƒÙ†Ù‡ Ù…Ù† Ø®Ù„Ø§Ù„ Ù…Ù†ØµØ©.

Ø®Ù ÙÙŠØ© Ø§Ù Ù…Ø´Ø§Ø¹ Ù‡Ù…Ø§ Ù‡Ù† Ø£ÙŠÙ† 1.2 ©Ø¬Ø§Ø¡Ø§ Ø§Ù ÙÙƒØ±Ø

Ù † Ø´Ø£Ø³ ÙÙƒØ±Ø© Ø³Ø·Ø¨Ù‚ÙŠÙ, **"Ø´Ø°Ù„ - WORK"** Ù…Ù † Ù…Ù„Ø§Ø¡Ù…Ø© Ù…ÙŠØ¯Ø§Ù†ÙŠØ© Ø¯Ù,ÙŠÙ,Ø© Ù„Ù„Ù…Ø´Ø§Ø±ÙƒÙ„ Ø§Ù„Ø°ÙŠ ÙŠÙˆÙØ±Ù‡Ø§ Ø§Ù„Ø´Ø¨ÙƒØ© Ø§Ù„Ø¹Ø§Ù„Ù…ÙŠØ© Ù„Ù„Ø¥Ù†ØªØ±Ù†Øª (Ù…Ø«Ù„ Ù…ÙˆÙ…Ù†ÙŠ Ø§Ù„Ø¬Ø²Ø§Ø¦Ø±ÙŠ Ù…Ø¨Ø§Ø´Ø±Ø© Ø¨Ø§Ù„Ø¹Ø§Ù„Ù… Ø§Ù„Ø®Ø§Ø±Ø¬ÙŠ Ø¹Ø¨Ø± Ø§Ù„Ø¥Ù†ØªØ±Ù†Øª Ø£Ø® Ø°Ù Ø¨Ø¹Ù ‡ Ù ‡ Ø¤Ù„Ø§Ø¡ Ø§Ù„Ø£Ø´Ø®Ø§Øµ Ø¹Ù„Ù‰ Ø§Ù„Ø¥Ø³Ø§Ø¡Ø© Ø£Ùˆ Ø¥Ø³Ø§Ø¡Ø© Ø§Ø³ØªØ®Ø¯Ø§Ù… Ø§Ù„Ø®Ø¯Ù…Ø© Ø¹Ù„Ù‰ Ù†ØÙˆ Ù‡Ø°Ø§ Ø§Ù„Ø´Ø±Ø· Ø£Ùˆ Ø¥Ø°Ø§ Ø¹Ù„Ù…Øª Ø£Ù† Ø¹Ù…Ù„Ø§Ø¡Ù†Ø§ Ø§Ù„Ø°ÙŠÙ† Ø§Ø³ØªØ®Ø¯Ù…Ùˆ Ø§Ù„Ø®Ø¯Ù…Ø© Ø¨Ø·Ø±Ù‚ Ø§Ù„Ø¥Ø³Ø§Ø¡Ø© Ø£Ùˆ Ø¨Ø·Ø±Ù‚ Ù…Ø®Ø§Ù„ÙØ© Ù„Ù„Ø´Ø±ÙˆØ· Ø§Ù„Ù…Ø°ÙƒÙˆØ±Ø© Ø£Ø¹Ù„Ø§Ù‡ Ù‚Ø¯ Ù†Ù‚ÙˆÙ… Ø¨Ø¥Ø¬Ø±Ø§Ø¡ Ø§Ù„Ø¥Ø²Ø§Ù„Ø© Ø§Ù„Ù…Ø¤Ù‚ØªØ© Ø£Ùˆ Ø§Ù„Ø¯Ø§Ø¦Ù…Ø© Ù„Ù„Ù…ØØªÙˆÙ‰ Ø§Ù„Ù…Ø®Ø§Ù„Ù Ø£Ùˆ Ø¥ÙŠÙ‚Ø§Ù ØØ³Ø§Ø¨Ùƒ Ø¬Ø²Ø¦ÙŠÙ‹Ø§ Ø£Ùˆ ÙƒÙ„ÙŠÙ‹Ø§ Ø¯ÙˆÙ† Ø¥Ø´Ø¹Ø§Ø± Ù…Ø³Ø¨Ù‚.

Ù†Ù†Ø§ Ø§Ù„Ù…Ù‚Ø¯Ù…ÙŠÙ† Ù†ØÙ† Ø§Ù„Ø¹Ù…Ù„ Ø§Ù„Ù…Ù…Ø§Ø«Ù„ Ø§Ù„Ù…Ø¹Ø±ÙˆÙ Ø¹Ø§Ù„Ù…ÙŠÙ‹Ø§ Ù†Ø³Ø¹Ù‰ Ø¬Ø§Ù‡Ø¯ÙŠÙ† Ø¥Ù„Ù‰ Ø§Ù„Ø¹Ù…Ù„ Ø¨Ø£Ø¹Ù„Ù‰ Ù…Ø³ØªÙˆÙŠØ§Øª Ø§Ù„Ø¬ÙˆØ¯Ø© Ø§Ù„Ù…Ù‡Ù†ÙŠØ© Ù„ÙƒÙ† Ù†Ø¤Ù…Ù† Ø£Ù† Ø§Ù„Ø®Ø¯Ù…Ø© Ø§Ù„Ù…Ø«Ø§Ù„ÙŠØ© Ù„Ø§ ØªÙˆØ¬Ø¯ØŒ Ù„Ø°Ù„Ùƒ Ù†ØØ±Øµ Ø¹Ù„Ù‰ Ø§Ù„Ø´ÙØ§ÙÙŠØ© Ø§Ù„ÙƒØ§Ù…Ù„Ø© Ø¨Ø´Ø£Ù† Ø·Ø¨ÙŠØ¹Ø© Ø§Ù„Ø®Ø¯Ù…Ø© Ø§Ù„Ù…Ù‚Ø¯Ù…Ø© Ù„Ùƒ. Ù…Ù† Ù‡Ù†Ø§ØŒ Ù†Ø¹Ù„Ù† Ø¹Ù†Ø§ Ø§Ù„Ø¹Ù…Ù„ Ø¨Ø¬Ø¯ Ø§Ù„Ø¹Ø§Ù„Ù… Ø§Ù„Ø°ÙŠ Ù†Ø¹Ø±Ù Ø¨Ø£Ù†Ù†Ø§ Ù„Ø³Ù†Ø§ Ø§Ù„Ø£ÙØ¶Ù„ØŒ Ù„ÙƒÙ†Ù†Ø§ Ù†Ø³Ø¹Ù‰ Ø¬Ø§Ù‡Ø¯ÙŠÙ† Ø¥Ù„Ù‰ Ø£Ù† Ù†ÙƒÙˆÙ† Ø§Ù„Ø®ÙŠØ§Ø± Ø§Ù„Ø£Ù…Ø«Ù„ Ù„Ùƒ Low Friction ÙŠØªÙŠØ-Ù„Ù„Ø¹Ù…Ù„ Ù†Ø§Ù…Ø²Ù‚ Ø§Ù„Ø¬Ø¯ÙŠØ¯ Ø§Ù„Ø°Ù‰ Ø§Ø¨Ø¯Ø£Ù‡ Ø§Ù„Ø¢Ù† Ø´Ø±Ø§ÙƒØ© Ø·ÙˆÙŠÙ„Ø© Ø§Ù„Ø£Ù…Ø¯ Ù…Ø¹ Ø§Ù„Ø¹Ø¯ÙŠØ¯ Ù…Ù† Ø§Ù„Ø¹Ù…Ù„Ø§Ø¡ Ø§Ù„Ø°ÙŠÙ† Ø§Ø³ØªÙ…Ø±ÙˆØ§ Ù…Ø¹Ù†Ø§ Ù„Ø³Ù†ÙˆØ§Øª.

Ø¯Ø±Ø§Ø³Ø© Ø£Ù‚Ù„ Ù‚Ù„ÙŠÙ„Ø© Ù‚Ø§Ù„ Ù‚Ù„ 1.3 Ù…Ø¨Ø§Ø´Ø±Ø© Ø§Ù„Ù…Ø¹Ø§Ù„Ø¬Ø©

Ù‚Ø¨Ù„ Ø§Ù„Ø´Ø±Ù‚ Ù‚ÙŠÙ Ù‚Ø¨Ø§Ø¯Ø© Ø§Ù„Ù‚Ø¨Ø¨ Ø§Ù„Ø¨Ø±Ø¬Ø¹ÙŠØŒ Ø¥Ø¹Ø¯Ø§Ø¯ Ø¥Ø¹Ø¯Ø§Ø¯ Ù…Ø¹Ø§Ù„Ø¬Ø© Ù„Ø¹Ù‚ÙˆØ¯ Ø§Ù„Ø¹Ø¨Ø¯ÙŠØ© Ù„Ø¹Ø¨Ø¯ÙŠØ©. Ù‚Ø¨Ù„ Ø§Ù„Ø¨Ø¯Ø¡ Ø¨Ø§Ù„Ø¹Ù…Ù„ Ø§Ù„Ø¹Ø§Ø¯ÙŠ Ø¨Ø¹Ø¯ Ø§Ù„Ø¹Ø·Ù„ Ø§Ù„Ø±Ø³Ù…ÙŠØ© Ø§Ù„Ø¹Ø§Ù…Ø© Ø§Ù„Ø¹Ø§Ø´Ø±Ø© Ø¸Ù‡Ø±Ø§Ù‹ØŒ Ø§Ù„Ø¬Ù…Ø¹Ø© Ø§Ù„Ø¹Ø§Ø´Ø±Ø© Ø¸Ù‡Ø±Ø§Ù‹ØŒ Ø§Ù„Ø³Ø¨Øª Ø§Ù„Ø¹Ø§Ø´Ø±Ø© Ø¸Ù‡Ø±Ø§Ù‹ØŒ Ø§Ù„Ø£ØØ¯ Ø§Ù„Ø¹Ø§Ø´Ø±Ø© Ø¸Ù‡Ø±Ø§Ù‹ØŒ Ø§Ù„Ø§Ø«Ù†ÙŠÙ† Ø§Ù„Ø¹Ø§Ø´Ø±Ø© Ø¸Ù‡Ø±Ø§Ù‹ØŒ Ø§Ù„Ø«Ù„Ø§Ø«Ø§Ø¡ Ø§Ù„Ø¹Ø§Ø´Ø±Ø© Ø¸Ù‡Ø±Ø§Ù‹ØŒ Ø§Ù„Ø£Ø±Ø¨Ø¹Ø§Ø¡ Ø§Ù„Ø¹Ø§Ø´Ø±Ø© Ø¸Ù‡Ø±Ø§Ù‹ØŒ Ø§Ù„Ø®Ù…ÙŠØ³

Ø§Ù„Ø¨Ø·Ø·Ù‚Ø§Øª Ù…Ø¹ Ø§Ù„Ø¥Ø†ØªØ±Ù†Øª Ø§Ù„Ø¹Ø§Ù„Ù…ÙŠØ©.

- **Ø®Ù‚Ø§Ø²Ù…ÙŠØ© Ø¶Ø¹Ø· Ù‚Ø¨Ù„ Ø§Ù„Ø¥Ø±Ø³Ø§Ù„** **String:** Ø¨Ù‚Ù„ÙŠÙ„ Ø£Ø¨Ø·Ø£ Ø§Ù„Ø¥Ù†ØªØ±Ù†Øª Ù‚Ø¨Ù…Ø§ Ù‚Ø¨ Ø§ Ù„Ù€ JPEG Base64 Ù„Ù…Ø´Ø§Ø±ÙƒØ© Ø¹Ø¨Ø±Ø©.

Ø§Ù„Ø¬Ø²Ø¡ Ø§Ù„Ø±Ø§Ø¨Ø¹: Ø§Ù„Ø¥Ø®Ø±Ø§Ø¬ Ø§Ù„Ù†Ù‡Ø§Ø¦Ù‰ Ù„Ù„Ø¥Ø†ØªØ±Ù†Øª Ø§Ù„Ø¹Ø§Ù„Ù…ÙŠ Ø£Ùˆ Ø§Ù„Ø¥Ù†ØªØ§Ø¬ 1.4

1. **Ø¹Ø¯Ø§Ø¯ Ø§Ù„Ø¨Ù†Ø§Ø¡ Ù„Ù„Ø¥Ø†ØªØ§Ø¬ (Gradle Ù„Ø¥ Ù…Ø´Ù„Ø§Ø©:** Ù… Ø¹Ø§Ù„Ø¬ Ø®Ø·Ø£ com.android.application Ø¨Ø§Ù„Ø¥Ø¶Ø§ÙØ© uild.gradle.kts.
2. **Ø¹Ø¯Ù… Ø·Ø§Ø¨Ù‚ Ø§Ø³Ù… Ø§Ù„ØØ²Ù…Ø© (Package Name Mismatch):** Ù… Ø¹Ø§Ù„Ø¬ Ø®Ø·Ø£ google-services.json Ø¨Ø£Ù†Ø§Ø³Ø¨ Ø§Ø³Ù… Ø§Ù„ØØ²Ù…Ø© com.example.work.
3. **Ø®Ø·Ø£ Ù†Ø·Ø§Ù‚ Ø¹Ù†Ø§ÙˆÙŠÙ† Ø§Ù„Ø´Ø¨ÙƒØ© Ø§Ù„Ø³ØØ§Ø¨ÙŠØ©:** Ø¨Ù† Ø§Ø¡ Ø¬Ù FirebaseHelper Ø§Ù„Ø°ÙƒÙŠ Ù„Ù„Ø±Ø¨Ø· Ø§Ù„Ø¨Ù„Ø¯Ø§Ù† Ø§Ù„Ù…Ø®ØªÙ„ÙØ© (US, EU, Asia).
4. **Ø§Ù„Ø¥Ø®ÙØ§Ù‚ Ø§Ù„Ø®Ø§Ø¯ÙŠ Ù† Ø§Ù„Ø³ØØ§Ø¨ÙŠ Ø¹Ù†Ø¯ Ø¹Ù„Ù‰ Ø§Ù„Ø¬Ù‡Ø§Ø² Ø¨Ø¯Ø§ÙŠØ© Ø§Ù„ØªØ´ØºÙŠÙ„:** Ø¥Ø¹Ø¯Ø§Ø¯ Ø§Ù„Ø¥Ø®ÙØ§Ù‚ Ø§Ù„Ø®Ø§Ø¯ÙŠ Ù† Ø¨Ø¯Ø§ÙŠØ© Ø§Ù„ØªØ´ØºÙŠÙ„ Ù„Ù„Ø¨ØØ« Ø¹Ù† Ø¨Ø¯Ø§Ù„ Ø¯Ø§Ø¦Ù… Ø§Ù„Ø¥Ø®ÙØ§Ù‚ signInAnonymously().

Ø§Ù„Ù‡ÙŒÙ„ Ø§Ù„Ø«Ø§Ù†ÙŠ: Ø§Ù„Ø¯Ø±Ø§Ø³Ø§Øª Ø§Ù„Ø³Ø¨Ù‚Ø© Ùˆ Ø§Ù„Ù…Ø±Ø§Ø¬Ø¹ Ø§Ù„Ù…Ø®ØªÙ„ÙŠÙ†

Ù…Ø±Ø§Ø¬Ø¹Ø© Ø§Ù„Ù…Ø´Ø§Ø±ÙŠØ¹ Ø§Ù„Ø´Ø¨Ù‡Ø© Ùˆ Ø§Ù„Ù…Ù†Ø§ÙØ³Ø© Ø§Ù„Ø¹Ø§Ù„Ù…ÙŠØ© Ùˆ Ø§Ù„Ù…ØÙ„ÙŠØ©

Ø³Ù… Ù‚Ø§Ù„Ù‹ÙŠÙŠÙ… Ø£Ø¨Ø² Ø§Ù„Ø„Ø·Ø·Ø§Ø³ Ù‚Ø§Ù„Ù…Ù† Ø§Ù„Ø¹Ø§Ù„Ù…ÙŠÙ© Ù‚Ø§Ù„Ø¹Ø¨Ù‹Ø© Ø§Ù„Ù‚Ø§Ø¦Ù…Ø© Ù…Ø¬Ø§Ù„ Ø§Ù„Ù‚Ù…Ø§Ù„ Ø§Ù„Ù‚Ø§Ù„Ù…Ø© Ø§Ù„Ù‚Ø§Ù„Ø©:

Ù†Ø©Ø© Ù… Ø³Ø³Ù‚Ù„ / Upwork	Ø³Ø·Ø¨Ù‹ÙŠÙ, Fiverr	Ù…Ù†Ø©Ø© Ø®Ù…Ø³Ø§Øª (Khamsat)	Ù‚Ø¬Ù‡ Ø§Ù„Ù„Ù… ©Ù‚Ø§Ø±Ù†Ø©
Ø¨Ø·Ù‡Ø© (Ø¹Ù‚Ø¯ Ù‚Ù‰Ù„Ø©)	©Ù…Ø³Ø³Ø·Ø·	Ø¨Ø·Ù‡Ø© (Ø³Ø¹Ø¨Ø¹ Ø¹Ø¯Ø¹Ù‰ Ø§Ù„Ø¥Ø³Ø§Ø¦Ù„Ø©)	Ø³Ø±Ø¹Ø© Ø§Ù„Ø„Ø·Ø¨Ø©

Ù…Ø§Ù„Ùˆ Ù… Ø³Ø¹Ù„Ù„ / Upwork	Ø¹Ø·Ø¨Ù„Ù, Fiverr	Ù…Ù†Ø§Ù…Ø© Ø®Ù…Ø³Ø§Øª (Khamsat)	Ù…Ø¹Ø§ Ø§Ù„Ù… Ø´Ø§Ø±Ùƒ
±Ù…Ø¶ Ù…Ù… Ø´Ø§Ø³Ø§Ø³ Ù… Ø´Ø§Ø±Ùƒ	Ù…Ø¦Ø§Ø³ Ø³Ø¹Ø±Ù© Ù…Ø§Ø§Ø©	Ø«Ø§Ø¨Ø§ Ø£Ù… Ù…Ø¦Ø§Ø³ Ø´Ø±Ø§Ø©	Ø§Ù„Ø¨Ø«Ø« Ø§Ù„Ø³Ø¹Ø±Ù©
Ù…Ø§Ø´Ø§Ø¨Ø§ Ø¹Ø§Ø¨Ø©	Ù…Ø²Ø§Ù…Ù†Ø© Ø¬Ø¨Ø§Ø¨Ø© Ø¹Ø¨Ø± Ø±Ø§Ø¨Ø· Ø§Ù„Ù„Ø§Ø¨Ø©	Ù…Ø§Ø¨Ø§Ø¨Ø§ Ø¥Ø¹Ø¨Ø§Ø¨Ø© Ø§Ù„Ø¨Ø§Ø¨Ø©	Ø§Ù„Ù…Ø²Ø§Ù…Ù†Ø© Ø§Ù„Ø¨Ø§Ø¨Ø©
Ø¹Ø§Ù„Ø¬Ø© Ù„Ù„Ù…Ø´Ø§Ø±Ø¹ Ø§Ù„Ø¬Ø¹Ø§Ø¨Ø©	Ù…Ø¹Ø§Ù„Ø¬Ø© Ø§Ù„Ø¹Ø§Ø¨Ø©	Ø¹Ø§Ø¬Ø§Ø¨Ø© Ø¹Ø§Ø¬Ø©	Ø¹Ø§Ù„Ø¬Ø© Ø§Ù„Ø¹Ø§Ø¬Ø©
Ù…Ø¹Ø§Ù„Ø¬Ø© Ø¹Ø§Ø¬Ø© Ø¹Ø§Ø¬Ø© Ø¹Ø§Ø¬Ø©	Ø¹Ø§Ù„Ø¬Ø© Ø¹Ø§Ø¬Ø© Ø¹Ø§Ø¬Ø© Ø¹Ø§Ø¬Ø©	Ø¹Ø§Ù„Ø¬Ø© Ø¹Ø§Ø¬Ø© Ø¹Ø§Ø¬Ø©	Ù…Ø¹Ø§Ù„Ø¬Ø© Ø¹Ø§Ø¬Ø©

Ø¹Ù…ÙŠØ² Ø§Ù„Ø§Ø¨ØªÙƒØ§Ø± Ø§Ù„Ù…Ø·Ù„ÙˆØ¨, ÙÙŠ 2.2 "Ø¹Ø·Ø·ÙŠÙ, "Ø´ÙØºÙ

ÙŠØªÙÙˆÙ, Øªà·Ø¨ÙŠÙŠÙ, **"Ø´ÙØºÙ„"** Ø¨Ùƒ Ù†Ù‡ ÙŠØ±ÙƒØ² Ø¹Ù„Ù ‰ Ø§Ù„Ø³Ø±Ø¹Ø© Ù‚Ø§Ù„Ù„ÙŠØ© Ø§Ù„Ù…Ø¨Ø§Ø´Ø±Ø© ØÙŠØ« ÙŠÙ„Ø¬Ø£ Ø§Ù„Ø¹Ø¨Ø¯ Ø§Ù„Ù‰ Ø§Ù„Ø¥Ø¬Ø§Ø¨Ø© Ø§Ù„Ù…Ø¨Ø§Ø´Ø±Ø© Ù„Ù„Ø¹Ù…Ù„ Ø¥Ù„Ø¯ Ø§Ù„Ø®Ø·Ø£ Ù„Ù…ÙŠØ²Ø§Ø Ø§Ù„Ø®Ø·Ø£ Ø§Ù„Ø¨Ø³ÙŠØ· Ø§Ù„Ø°Ù‰ Ù„Ø§ ÙŠØªØ·Ù„Ø¨ Ø¥Ø¹Ø§Ø¯Ø© Ø¨Ù†Ø§Ø¡ Ø§Ù„Ù…Ø´Ø±ÙˆØ¹ Ø¨Ø£ÙƒÙ…Ù„Ù‡ Ø§Ù„Ø¹Ù…Ù„ Ø¨Ø¹Ø¯ Ø¥Ø¶Ø§ÙØ© Ø§Ù„Ù…ÙŠØ²Ø© Ø§Ù„Ø¬Ø¯ÙŠØ¯Ø© Ø§Ù„Ù…Ø·Ù„ÙˆØ¨Ø© Ù„Ù„Ø¹Ù…Ù„ Ø¨Ø´ÙƒÙ„ Ø³Ù„ÙŠÙ… Ø§Ù„Ø¹Ø¨Ø¯ Ø¨Ø¹Ø¯ Ø§Ù„Ø¥Ø¶Ø§ÙØ© Ø§Ù„Ø¬Ø¯ÙŠØ¯Ø© Ø§Ù„Ù…Ø·Ù„ÙˆØ¨Ø© Ø¨Ø§Ù„Ø¹Ù…Ù„ Ø¨Ø´ÙƒÙ„ Ø³Ù„ÙŠÙ… Ù…Ø¹ Ø¹Ø¯Ù… Ø§Ù„Ø¥Ø®Ù„Ø§Ù„ Ø¨Ø£Ø¯Ø§Ø¡ Ø§Ù„Ù†Ø¸Ø§Ù… (Ø¨Ø·Ø¡ / Ø¨Ø·Ø¡ Ø¨Ø³Ø¨Ø¨ Ø¥Ø¶Ø§ÙØ© Ø¨Ø¹Ø¶ Ø§Ù„Ø¹Ù…Ù„ÙŠØ§Øª Ø§Ù„Ø«Ù‚ÙŠÙ„Ø©).

Ø§Ù„Ù‡ÙØµÙ‡ Ø§Ù„Ø«Ø§Ù„Ø«: Ø¹Ø¨Ù‡ Ø§Ù„Ù…Ø·Ù„Ø¨Ø§Øª Ù‰Ø§Ù„Ø¹Ø§Ù„Ø© Ø§Ù„Ø³Ø¹Ø®Ø¯Ø§Ù… (Use Cases)

Ø§Ù„Ù…Ø·Ù„Ø¨ Ø¨Ø§Ø¹ Ø§Ù„Ù‰Ø§Ø¸ÙŠÙŠÙŠÙ© (Functional 3.1 Requirements)

- **Ø¥Ø¯Ø§Ø±Ø© Ø§Ù„Ø³Ø¨Ø¹Ø¨Ø§Ø¡ Ù‰Ø§Ù„Ø£Ø¨Ù‰Ø±:** Ø¥Ù…Ù‚Ø§Ù † ÙŠØ© Ø¹Ø¨Ø¯Ø¯ Ø¨Ù‰Ø± Ø§Ù„Ù…Ø³Ø¹Ø®Ø¯Ù… (Ø¹Ù…ÙŠÙ„ Ø£Ù‰ Ù…Ù‚Ø¯Ù… Ø®Ø¯Ù…Ø©) Ù‰Ø¹Ø¨Ø¯Ø¯ Ø¨Ù‰Ø± Ø¨ÙŠØ§Ù†Ø§Øª Ø¨ÙŠØ« Ø¨ÙŠØ§Ù†Ø§Øª Ø§Ù„Ø£Ø³Ø¹Ø³ÙŠØ©.
- **Ø¥Ø¯Ø®Ø§Ù„ Ø§Ù„Ø®Ø¯Ù…Ø§Øª:** Ø¥Ù…Ù‚Ø§Ù † ÙŠØ© Ø¥Ø®Ø§Ù„ Ø®Ø¯Ù…Ø© Ø®Ø¨Ø¯Ø© Ø¬Ø¯Ù…Ø© Ø¬Ø¯ÙŠØ¯Ø© Ø§Ù„Ù‰Ø¹Ø§Ù„Ù… Ø§Ù„Ù‰Ø§Ù‚Ø¹ Ù‰Ø§Ù„Ø¹Ø§Ù„Ù… Ø§Ù„Ø®Ø§Ø±Ø¬Ù Ù‰Ø§Ù„Ø¹Ø§Ù„Ù… Ø§Ù„Ø®Ø§Ø±Ø¬Ù Ø¨

Ø§Ù„Ù…ØªØ·Ù„Ø¨Ø§Øª Ø§Ù„Ù‚ÙŠØ± Ø§Ù„ÙˆØ¸ÙŠÙÙŠØ© 3.2 (Non-Functional Requirements)

- **Ø§Ù„Ø£Ø¯Ø§Ø¡ Ùˆ Ø§Ù„Ø³Ø±Ø¹Ø© (Performance):** Ø²Ù…Ù† Ø§Ø³Ø¬Ø§Ø¨Ø© Ø§Ù„Ø´Ø¨ÙƒØ© Ø¹Ù†Ø¯ Ø¬Ù„Ø¨ Ø¹Ø±Ø¶ Ø²Ù…Ù†ÙŠ Ø£Ùˆ Ø¥Ø±Ø³Ø§Ù„ Ø·Ù„Ø¨ Ø¬Ø¯ÙŠØ¯.
- **Ø§Ù„Ù…ÙˆÙ‚Ùˆ Ùˆ Ø§Ù„Ø³Ù„Ø§Ù…Ø© (Reliability):** Ø¹Ù…Ù„ Ø§Ù„Ø¨Ø±Ù†Ø§Ù…Ø¬ Ø¨Ø´ÙƒÙ„ Ù…Ø³ØªÙ…Ø± Ø¯Ø§Ø®Ù„ Ø§Ù„Ø¬Ø§Ù…Ø¹Ø© Ø¯ÙˆÙ† Ø§Ù†Ù‚Ø·Ø§Ø¹.
- **Ø³Ù‡ Ù‘Ù„Ø© Ø§Ù„Ø§Ø³ØªØ®Ø¯Ø§Ù… (Usability):** Ø¹Ù…Ù„ÙŠØ§Øª Ù‘Ø§Ø¨Ù‘Ø© Ø¹Ù„Ù‰ Ù‘Ù† Ø¸ÙŠÙØ© Ø¹Ù„Ù‰ Material Design 3 Ù‘Ø§Ù„Ù„ØºØ© Ø¨Ø§Ù„Ø¹Ø±Ø¨ÙŠØ© Ø¨Ø§Ù„Ø£Ø±Ù‚Ø§Ù….

Ø§Ù„Ù…Ø±Ø§Ø¬Ø¹: Ø§Ù„Ù…Ø±Ø§Ø¬Ø¹ Ù ‡ Ù † Ø¯Ø³Ù„Ø© Ø§Ù„Ù ‡ Ù„ÙŠÙ„Ù Ù‘Ø£Ù † Ù…Ø§Ø·Ø© Ø§Ù„Ø³Ø§Ø¨Ø¹Ø© Ø§Ù„Ø¯Ø±Ø§Ø³ÙŠ

Ù…Ø¹Ù…Ø§Ø±ÙŠØ© Ø§Ù„Ø·Ø¨Ù‚Ø§Øª 4.1 (Clean Architecture) Ù‘Ø§Ù„Ù€ Repository Pattern

Ø³Ù… Ø³Ø·Ø·ÙŠÙ, Ù † Ù…Ø· **Repository Pattern** Ù„Ù„Ø¹Ù„ Ø·Ø¨Ù‚Ø© Ø§Ù„Ø³Ø¨Ø§Ø†Ø§Øª Ø§Ù„Ù † Ø·Ø³Ø¹ Ø¹Ù† Ø·Ø¨Ù‚Ø© Ø§Ù„Ù‘Ø§Ø¬Ù‡ UI. ÙŠØ¶Ù…Ù† Ù ‡ Ø°Ø§ Ø§Ù„Ù † Ù…Ø· Ø¬Ù‘Ø¯Ø© Ø§Ù„Ù‘Ø¨Ø¹ Ø§Ù‘Ù„Ø© Ø§Ù„Ø³Ù‡Ù ‡ Ù‘Ù„Ø© Ø§Ù„Ø¹Ù„Ø§Ù † Ø© Ù‘Ø§Ù„Ø®Ø¨Ø±Ø§Øª ØÙŠØ« ÙŠØ³Ø¹Ø§Ù…Ù„ Ø§Ù„Ù€ Repository Ù‘Ù…Øµ Ù…Ù‘Ø± Ù…Ù„Ø¬Ù‘Ø© Ø§Ù„Ø¹Ù„Ø§Ù † Ø§Ø (Single Source of Truth) ÙŠØ±Ø¬Ø· .Ù‘Ø§Ø¬Ù‡Ø§Øª Ø§Ù„Ø¹Ø¨Ø¬Ù‘ÙŠÙ, Ø¨Ø³Ø§Ø¨Ø© Firebase Realtime Database

Ø£Ù † Ù…Ø§Ø· Ø§Ù„Ø¹Ø§Ù…Ù…ÙŠÙ… Ø§Ù„Ù…Ø³Ø¹Ø¹Ù… 4.2 Ù Ø© (Design Patterns)

- **Singleton Pattern:** Ø³Ù… Ø³Ø·Ø·ÙŠÙ,Ù ‡ ÙÙŠ ØµÙÙ‘Ù `ServiceRepository` Ù‘ `OrderRepository` Ù‘ `FirebaseHelper` Ù„Ø¶Ù…Ø§Ù † Ù‘Ø¬Ø¯ Ù†Ø³Ø®Ø© Ù‘Ù,Ø· ÙŠÙ† Ø¸Ø¨Ø¹ Ø§Ù„Ø§Ø³Ø§Ø¬Ø§Øª Ø¨Ø§Ù‘Ø§Ø¬Ø¹ Ø·Ù‘Ø±Ø© Ø·Ù‘Ø±Ø© Ø·Ù‘Ø±Ø©

- **Observer / Listener Pattern:** Ø§Ø³Ø¹Ø¹Ù…Ù„ Ø¹Ø¨Ø± Ø§Ù„Ù‚Ø§Ø¦Ø§ Ø§Ø Ø§Ù„Ø¨Ø¨Ø±Ù…Ø¬Ù…Ø© OnDataChangeListener Ù‚ OnOrdersChangeListener Ù„Ù„Ø£Ù†ØµØ§Ø§Ø Ù„Ù„Ø¹Ø¹ÙŠÙŠØ±Ø§Ø Ø§Ù„Ø³Ø§Ø§Ø¨ÙŠØ© Ù‚Ø¹Ø§Ø¯ÙŠØ« Ø§Ù„Ù€ UI .›Ù‚Ø±Ø§Ù
- **Adapter Pattern:** Ø¹Ù… Ø¹Ø·Ø¨ÙŠÙ‚Ù ÙŠÙŠ ØµÙ‚Ù OrderAdapter Ù‚ MessageAdapter Ù„Ø¹Ø‚Ù„ Ùƒ Ø§Ø¹ Ø§Ø Ø¥Ù„Ù ‰ Ù…Ùƒ Ø§Ø Ø¹Ø±Ø¶ Ø¨Ø±Ø§Ø¬ RecyclerView.

Ø§Ù„Ù‡ÙµÙ‡ Ø§Ù„Ø®Ø§Ù…Ø³: Ù ‡ ÙŠØ§Ù„Ù‡ Ø§Ù„Ø¨ÙŠÙŠØ§Ù† Ù‰Ø§Ù„Ø®Ù‰Ù‰Ù… ÙŠØ§Ù„Ù…Ø¨ÙŠÙƒ Ù‰Ø§Ù„Ù‡Ø¹Ø-Ù‡ÙŠÙ‡ Ø§Ù„Ù‡Ø¹Ø¹Ù‰Ù‡ÙŠÙ

Ø¹Ø§Ù‡ ÙŠÙ‡ Ù‡ÙŠØ§Ù†Ù‡ Ø§Ù„Ø¨ÙŠÙŠØ§Ù† Ø§Ù„Ù… 5.1 Ø¹Ø§Ù…Ù„Ø© (Data Structures)

Ø¹Ù… Ø§Ø³Ø®Ø¯Ø§Ù… Ø«Ù„Ø§Ø«Ø© Ù‡ÙŠØ§Ù† Ø¨ÙŠÙŠØ§Ù† Ø§Ù„Ø¹Ø±Ø¶ Ø§Ù„Ø¨ÙŠÙŠØ© :¡ÙŠÙŠ Ø§Ù„Ù…Ø´Ø§Ø±ÙƒØ© Ø¨ÙØ³Ø·Ø© Ø§Ù„Ø³Ø±Ø¹Ø© Ù‰Ø§Ù„Ø£Ø¯Ø§Ø

1. **ArrayList:** Ø¹Ø®Ø²ÙŠÙ† Ø§Ù„Ù‚Ø§Ø¦Ù… Ø§Ù„Ù…Ø±Ø¹Ø¨Ø© Ù‰Ù Ø§Ù„Ù…Ø¹Ø¬Ù… Ø¹Ù† Ø§Ø³Ø±Ø¹ Ø§Ù„Ø¹Ø±Ø¶ Ø§Ù„Ø¨ÙŠÙ‰Ø© Ù‰Ù Ø§Ù„Ø¹Ø±Ø¶ RecyclerView.
2. **LinkedHashMap:** Ø±Ø¨Ø· Ø§Ù„Ù…Ø¹Ø±Ù‘Ø§Øª Ø§Ù„Ù‰Ù‰Ø© UUID Ø¨Ø§Ù„Ø®Ø§Ø¯Ù… Ø§Ø³Ø±Ø¹ Ø§Ù‰Ø§Ø¦Ø§Ø¨Ø¹ Ø§Ù„Ù‰Ø³Ø¹ Ø§Ù„Ø¹Ø§Ø¯Ø© Ø§Ù„Ø¹Ø§Ø¯Ø© Ø§Ù„Ø¹Ø§Ø¯Ø© (1)$ Ù…Ø¹ Ø§Ù„Ø¹Ø§Ø¯Ø© Ø§Ù„Ø¹Ø±Ø¶ Ø§Ù„Ø¹Ø±Ø¶ Ø§Ù„Ø®Ø§Ø¯Ù….
3. **HashSet:** Ø¹Ø®Ø²ÙŠÙ† Ù‰Ø§Ø¦Ù…Ø© Ø§Ù„Ø¨Ø±Ø¬Ø¹Ø© Ø§Ù„Ø¹Ø±Ø¶ Ø§Ù„Ø¹Ø±Ø¶ SharedPreferences.

Ø§Ù Ø¹Ø¹Ù ÙŠÙ Ø§Ù Ø¹Ø¹Ù‚Ø¯Ù‘ÙŠ Ø§Ù Ø²Ù…Ù†Ù ÙˆÙ Ù Ø®Ù‘Ø§Ø²Ù…ÙŠÙŠØ§Ø (Big-O Complexity Analysis)

Ø§Ù„Ø´Ø±Ø	Ø§Ù„Ù…Ø³Ø§ØØ© Ø§Ù„Ù…Ø³ØªØ®Ø¯Ù…Ø© (Space Complexity)	Ø§Ù„Ù…Ø¹Ù‚Ø¯Ù‘ Ø§Ù„Ø²Ù…Ù†Ù (Time Complexity)	Ø§Ù„Ø¹Ù…Ù„ÙŠØ© / Ø§Ù„Ø®Ø§Ø²Ù…ÙŠØ©
Ø¥Ø¶Ø§ÙØ© Ù…Ø¨Ø§Ø´Ø±Ø© Ø¥Ù„Ù€ LinkedHashMap Ù‘Ø§Ù„Ø³Ø¬Ø¨Ø© Ø¨Ù¶Ù„ Ø§Ù„Ù…Ø¹Ø±Ø§Ø¶ Ø§Ù„Ø±ÙŠØ¯Ø© .UUID	$(1)	$(1)	Ø¥Ø¶Ø§ÙØ© Ø®Ø¯Ù…Ø© / Ø·Ù„Ø¨ Ø¬Ø¯ÙŠØ¯
Ø§Ù„Ø¹Ø±Ø¶ Ø¹Ù„Ù‰ $ Ù…Ù† Ø§Ù„Ø®Ø¯Ù…Ø§Ø¹ Ø§Ù„Ù…Ø¹Ø§Ø¯Ù„Ø© Ù‘Ø¹Ù…ÙŠÙ $ Ù…Ù† Ø§Ù„Ø®Ø¯Ù…Ø§Ø .©Ø§Ù„Ù…Ø·Ø¨Ù‘Ø¹	$(K)	$(N)	Ø§Ù„Ø¨Ø« Ù‘Ø§Ù„Ù„Ø¹Ø±Ø¶ Ø§Ù„Ø¬ÙŠØ©

Ø§Ù„Ø´Ø±Ø Ø§Ù„Ù…Ø®ØªØµØ±	Ø§Ù„Ù…Ø³Ø§ØØ© Ø§Ù„Ù…Ø³ØªØ®Ø¯Ù…Ø© (Space Complexity)	Ø§Ù„Ø²Ù…Ù† Ø§Ù„Ù…Ø³ØªØºØ±Ù‚ (Time Complexity)	Ø§Ù„Ø§Ø³Ù… / Ø§Ù„Ù…Ø®ØµØ±Ø©
Ø¹Ø±Ø¶ Ø§Ù„Ø¨ÙŠØ§Ù†Ø§Øª Ø§Ù„Ù…Ø±ØªØ¨Ø© Ø¨Ø§Ø³ØªØ®Ø¯Ø§Ù… Ø®ÙˆØ§Ø±Ø²Ù…ÙŠØ© Ø§Ù„ÙØ±Ø² Ø§Ù„Ø³Ø±ÙŠØ¹ Ø§Ù„Ø«Ù†Ø§Ø¦ÙŠ Ø§Ù„Ù…ØÙˆØ±. Dual-Pivot Quicksort	$(K)	$(K \log K)	**Ø§Ù„Ø¹Ø±Ø¶ Ø§Ù„Ø³Ø±ÙŠØ¹ Ø§Ù„Ù…Ø±ØªØ¨**
Ø£Ø¯Ø§Ø© Ù„ØªØÙˆÙŠÙ„ Ø§Ù„ØµÙˆØ± Ø¥Ù„Ù‰ Ù†Øµ H$ Ø¹Ø¨Ø± Ø§Ù„Ø®ÙˆØ§Ø±Ø²Ù…ÙŠØ© Ø§Ù„Ù…ÙˆØØ¯Ø©.	$(W \times H)	$(W \times H)	**Ø¶ØºØ· Ø§Ù„ØµÙˆØ± Base64**

Ø§Ù„Ù…Ù‚Ø§Ø¨Ù„Ø§Øª Ø§Ù„Ø³Ø¯Ø³: Ø§Ù„Ø¹Ù…Ù„ Ù‘Ø§Ù„Ù‘Ø§Ù‚Ø§Ù…Ù‘ Ù…Ø¹ Ø§Ù„Ø®Ù‘Ø§Ø¯Ù… Ø§Ù„Ø³Ø¯Ø¨Ù‘ÙŠÙ‚Ø© Firebase

Ø¹Ù ‡ ÙŠØ¡Ø© Ø§Ù„Ø³Ø¯Ø¨Ø© Ù‘Ø¹Ø³Ø³Ø§Ù‘Ø© Ù‘Ø¹Ø³Ø³Ø§Ù‘Ø© 6.1 Ø§Ù„Ø¯Ø®Ù‘Ù Ø§Ù„Ø¹Ù‘Ù„Ø§Ø¡ Ø§Ù„Ù…Ø¬Ù‡Ù‘Ù„ÙŠÙ†

Ø³Ø³Ù… Ø¹Ù…Ù„ÙŠØ© Ø§Ù„Ø´Ø¹ Ø®Ù„Ù‰ **Firebase** Ø¹Ø¨Ø± Ø§Ù„Ù‘Ø®Ø¯Ù… FirebaseHelper.java. ÙŠØ¶Ù…Ù † Ù ‡Ø°Ø§ Ø§Ù„Ø¬Ù‡Ø§Ø²Ø§Ù„Ø¹Ù…Ù„ Ø§Ù„Ø¹Ø§Ù„Ø¬Ø¡ Ù‘Ø³Ù‘Ø¹Ø¹ Ø¬Ù„Ø³Ø© Ù…Ø¹ Ø§Ù‘Ø« Ù‘Ø§Ù‚Ø¹ Ø¹Ø¨Ø± signInAnonymously() Ù„Ø³Ø¬Ù‘Ø² Ù‚ÙŠÙ‘Ø¯ Ø§Ù„Ù‚Ù‘Ø§Ø¹Ø¯Ø© Ù‘Ø§Ø³Ø³Ø¹Ø¯ Ù‘Ø§Ø³Ø³Ø¹Ø¯Ø© Ø¹Ø±Ø· uth != null Ø¨Ø¯Ù‘Ù† Ø¥Ø²Ø§Ù„Ø© Ø§Ù„Ù…Ø³Ø³Ø®Ø¯Ù… Ø¨Ø´Ø±Ø· Ø§Ù„Ø³Ø³Ø¹Ø¯Ù… Ø§Ù„Ù…Ø³ .Ø¹Ù‚Ø

Ø¥Ø¯Ø§Ø±Ø© Ù‚Ù‘Ø§Ø¹Ø¯ Ø§Ù„Ø¨Ø¹ Ø§Ø§Ø³ 6.2 Ø§Ù„Ø³Ø³Ø§Ø¨Ù‚Ø© Ø¨Ø§Ù„Ø¨Ù‘Ø§Ù‚Ø³ Ù‘Ù‚Ø³ Realtime Sync

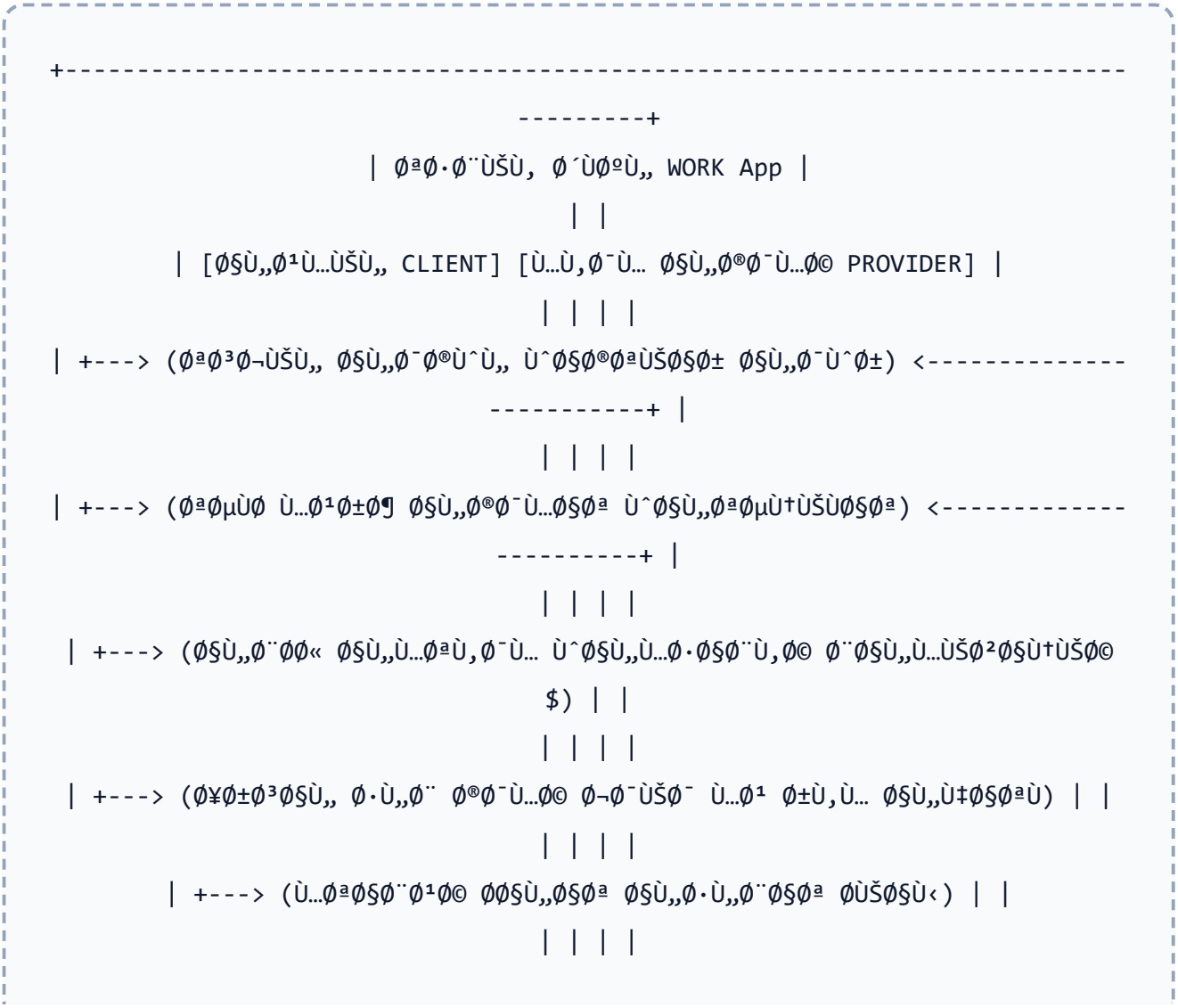
ÙŠÙ‚Ù… Ùƒ Ù…Ù† ServiceRepository Ù‘ OrderRepository Ø¨Ù‘Ø³Ø Ù…Ø³Ø³Ù‚Ø¹ Ø³Ø-Ø§Ø¨Ø¨Ù‘ ØÙŠÙŠ Ø¹Ø¨Ø± ddValueEventListener(). Ø¨Ù…Ø¬Ø¯ Ø¥Ø¬Ø±Ø§Ø¡ Ø£ÙŠ Ùƒ Ø³Ø§Ø¨Ø© Ø¹Ù„Ù‰ Ø§Ù„Ø³Ø³Ø§Ø¨Ø© Ø¹Ø¨Ø± setValue()Œ ÙŠØ³Ù… Ø¥Ø·Ù„Ø§Ù‚ Ø¯Ø¨Ø«

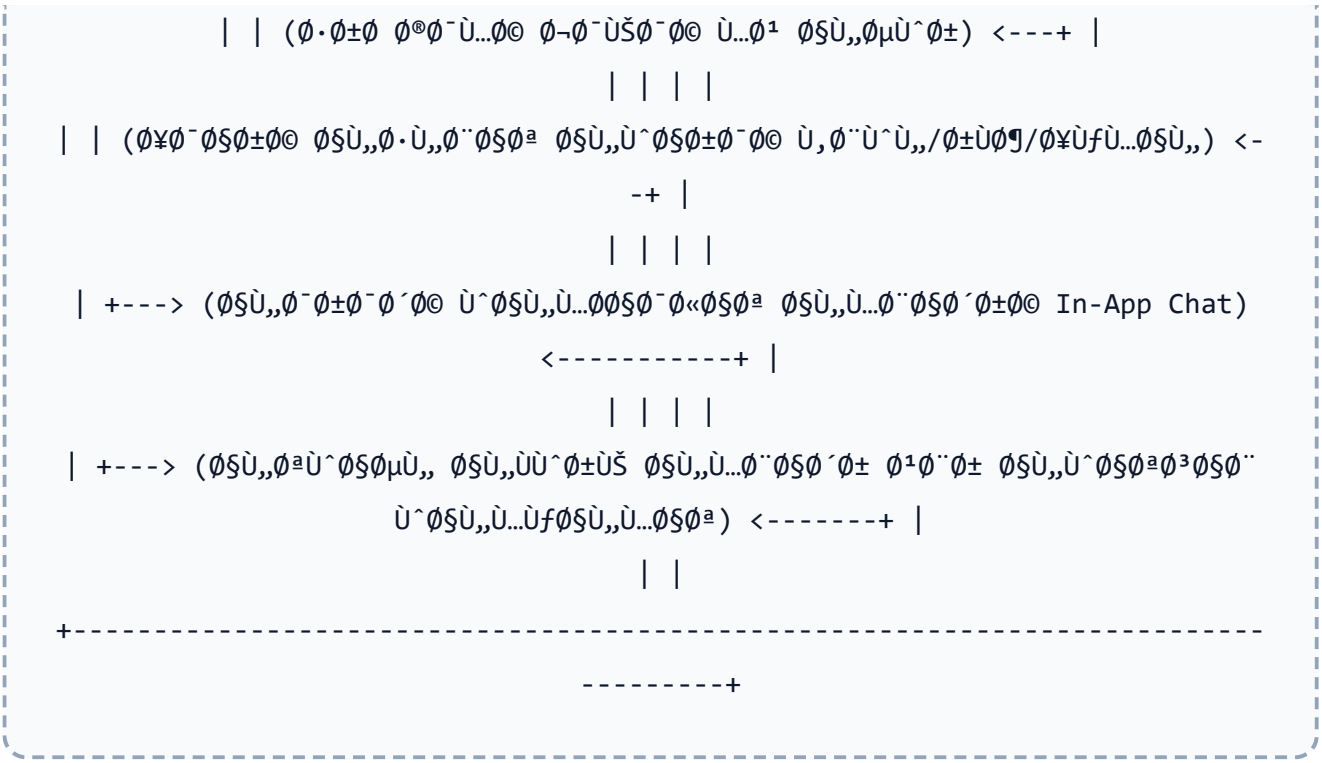
onDataChange() ØªÙ„Ù‚Ø§Ø¦ÙŠØ§Ù‹ Ø¹Ù„Ù‰ Ùƒ Ùƒ Ø§Ù„Ø£Ø¬Ù‡Ø²Ø© Ø§Ù„Ù…ØªØµÙ„Ø©

.Ù„Ø§Ø¨Ù‘ÙŠÙ« Ø§Ù„Ù‚Ø§Ø¬Ù‡Ø§Øª Ù‘Ù‘Ø±Ø§Ø

Ø§Ù„Ø¬Ø²Ø¡ Ø§Ù„Ø³Ø§Ø¨Ø¹: Ø§Ù„Ù…Ø®Ø·Ø·Ø§Øª Ø§Ù„Ø ‡ Ù † Ø¯Ø³Ù„Ø© Ø§Ù„Ø¨Ø±Ù…Ø¬ÙŠØ© (UML Diagrams)

Ù…Ø®Ø·Ø· Ø§Ù„ØØ§Ù„Ø§Øª Ø§Ù„Ø§Ø³ØªØ®Ø¯Ø§Ù… Ø§Ù„Ø 7.1 ´Ø§Ù…Ù (Use Case Diagram)




| | (Ø·Ø±Ø ØªÙ…Ù© Ø¬Ø¯ÙŠØ¯Ø© Ù…Ø¹ Ø§Ù„Ù„ÙˆÙ±) <---+ |
| | | |
| | (Ø¥Ø¯Ø§Ø±Ø© Ø§Ù„Ø·Ù„Ø¨Ø§Øª Ø§Ù„Ù„ÙˆÙ±Ø¯Ø© Ù,Ø¨Ù„Ù,/Ø±ÙØ¶/Ø¥ÙƒÙ…Ø§Ù„) <-
-+ |
| | | |
| +---> (Ø§Ù„Ø¯Ø±Ø¯Ø´Ø© Ù„Ù…ØØ§Ø¯Ø«Ø§Øª Ø§Ù„Ù…Ø¨Ø§Ø´Ø±Ø© In-App Chat)
<-----------+ |
| | | |
| +---> (Ø§Ù„Ø§Ù„ÙˆØ§Ù„ Ø§Ù„Ù„Ù…Ø± Ø§Ù„Ù…Ø¨Ø± Ø¹Ø¨Ø± Ø§Ù„Ø§Ø³Ø³Ø¨
Ù„Ù„Ù…Ø§Ù…Ù„Ø§Ø) <-------+ |
| |
+
---------+

Ø§Ù„Ø¯ÙŒÙ„ Ø§Ù„Ø«Ø§Ù„Ø« : Ø§Ù„Ø¹Ø§Ù…Ù„ Ø§Ù„Ø¨ÙŠØ§Ù†Ù‰ Ù‘Ù † Ø¸Ø§Ù… Ù‘Ø§Ø¬Ù ‡ Ø§Ø² Ø§Ù„Ù…Ø³Ø®Ø¯Ù…

Material Design 3

Ù‘Ù Ø³Ù‘Ø© Ø§Ù„Ø²Ù…Ù…ÙŠÙ… Ù‘Ù † Ø¸Ø§Ù… 8.1 †Ø§Ù Ø£Ù Ù‘Ø§Ù

Ø²Ù… Ø§Ø¹Ø¨Ù‘Ø§Ø± Ø£Ø¯Ø« Ù…Ø¹Ø§ÙŠÙŠØ± **Material Design 3** Ø§Ù„Ø¬Ø§Ø±Ø© Ù„ØªØ·Ø¨ÙŠÙ‚Ø§Øª Google Ù„Ø¨Ù†Ø§Ø¡ Ø²Ø¬Ù…Ù…ÙŠÙ… Ø¹Ø±Ø¶ Ù‘Ù…Ø±Ø¦ÙŠÙ‘:

- **Ø§Ù„Ù„Ù‘Ù † Ø§Ù„Ø¨Ø±Ø²Ù‚Ø§Ù„ÙŠ Ø§Ù„Ø¯Ø¨Ø³Ù‘Ù (#ea580c / #f97316):** ÙŠØ¹Ø¨Ø± Ø¹Ù † Ø§Ù„Ù † Ø´Ø§Ø· Ù‘Ø§Ù„Ø©Ù‘Ù‘Ø© Ù‘Ø§Ù„Ø¹Ù…Ù„ Ø§Ù„Ø±Ø± Ø§Ù„Ø´Ø´Ù‘Ù
- **Ø§Ù„Ù„Ù‘Ù † Ø§Ù„Ø¯Ø§Ù„Ù Ø§Ù„Ø¯Ø§Ù† (#0f172a / #1e293b):** ÙŠØ¹Ù‘Ø³ Ø§Ù„Ø«Ù‚Ø© Ù‘Ø§Ù„Ø£Ø§Ù† Ù‘Ø§Ù„Ø£Ø§Ø³Ù‘ÙŠÙ…Ø© Ø§Ù„Ø±Ù‘Ø© ÙˆÙŠ .©Ø§Ù„Ø¨Ø§Ù†Ø±Ø§Ù…Ø§ Ø§Ù„Ø¹Ù„Ù‘ÙŠØ
- **Ø§Ù„Ø±Ù…Ø§Ø¯Ù‰ Ø§Ù„Ø§Ø¨ Ù„Ù„Ø®Ù„ÙÙŠÙ‘Ø§Øª (#f8fafc / #f1f5f9):** ÙŠØ¶Ù…Ù † ....Ø³Ø¨Ø§ÙŠÙ† Ø§Ù„Ù‘Ø¹Ø§Ø¡Ø© Ù‘Ø¥Ø±Ø§Ø§Ø© Ø¹ÙŠÙ† Ø§Ù„Ù…Ø³Ø®Ø¯Ù
- **Ø§Ø³Ø®Ø¯Ø§Ù… Ø¨Ø·Ø§Ù‚Ø§Øª MaterialCardView:** Ø°Ø§Ø Ø Ù‘Ø§Ù Ù…Ù † Ù† Ø§Ø¹Ù…Ø© Ù‘Ø¥Ø·Ø§Ø© (Corner Radius 12dp to 16dp) Ù…Ø¹ Ø¸Ù„Ø§Ù„ Ù † Ø§Ø¹Ù…Ø© Ù‘Ø¥Ø·Ø§Ø© .©Ø¯Ù‘Ù‘Ù‘Ø

Ø§Ù„Ù…Ø¬Ù„Ø¯ Ø§Ù„Ø«Ø§Ù„Ø«: Ø¯Ù„ÙŠÙ„ Ø§Ù„Ø³Ø§Ø®Ø¯Ø§Ù… Ù„Ù„Ø¹Ù…ÙŠÙ„ Ù„Ø¨ Ø§Ù„Ø´Ø±ÙƒÙŠÙ‡ (User Manual)

Ø¯Ù„ÙŠÙ„ Ø±Ø© Ø§Ù„Ø¹Ù…ÙŠÙ„ (Client 9.1 Walkthrough)

1. Ù‰Ø¹Ø Ø§Ù„Ù„Ø¹Ø·Ø¨Ù‰Ù, Ù‰Ø§Ø®Ø¯Ø§Ù…Ù‡Ø§Ù„Ø§Ø± Ø¯Ù‰Ø± **"Ø¹Ù…Ù„Ù„"** Ø¨Ø¹Ø¯ Ù†Ø¹Ø§Ø§Ø¨Ø© Ø§Ù„Ø¨Ø§Ø¨Ø¹Ø© Ø¨Ù‰Ø¹Ø¯ Ù‰Ø§Ø®Ø¯Ø§Ù… Ø§Ù„Ø¨Ø±Ø§Ù…Ø¬.

2. Ø¹Ø¬Ù‰Ø Ø§Ù„Ø®Ø¯Ù…Ø§Ø¨ Ø¨Ø¨Ø¹Ø¨ Ø§Ù„Ø¹Ø¬Ù†Ù‰Ù (Ø¨Ø±Ù…Ø¬Ø©Œ Ø¹Ø¬Ù…Ù„Ù‰Ø§Ø¹Œ Ø¹Ø±Ø¬Ù‰Ù‰Ù‰Ø¹Œ Ø¹Ø³Ù‰Ù‰Ù‰Ø¹Œ Ø¬Ù‰Ø§Ø§Ù†Ø©) Ø£Ù‰ Ù‰Ø¹Ø Ø´Ø§Ø© Ø§Ù„Ø¨Ø¨Ø« Ø¨Ø§Ø±Ø¬Ø¹Ø¹Ø¹

Ø¯Ù„ÙŠÙ„ Ø±ØÙ„Ø© Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø© 9.2 (Provider Walkthrough)

1. .„Ø§Ø®Ø³ÙŠØ§Ø± Ø¯Ù‘Ù‘Ø± **"Ù…Ù‚Ø¯Ù… Ø®Ø¯Ù…Ø©"** Ù…Ù† Ø´Ø§Ø´Ø© Ø§Ù„Ø¯Ø®Ù‘Ù

2. Ø§Ø³Ø³Ø¹Ø±Ø§Ø¡ Ø³Ø¨Ù‘Ù‘Ø¨ **"Ø¬Ù…Ù‘Ø¹ Ø§Ù„Ø®Ø¯Ù…Ø§Øª ðŸŒ** Ù„Ø±Ø¤Ù‘Ø© .,Ù ƒØ§Ù„Ø®Ø¯Ù…Ø§Øª Ø§Ù„Ù…Ø§Ø§Ø© Ø¨Ø§Ù„Ø³Ù‘Ù

3. Ø§Ù„Ù†Ù‚Ø± Ø¹Ù„Ù‰ **"Ø£Ø¶Ù Ø®Ø¯Ù…Ø© Ù„Ø¹Ø±Ø¶ Ø®Ø¯Ù…Ø© Ø¬Ø¯Ù‘Ø¯Ø© Ù‘Ø¥Ø±Ø§Ù‚ Ø¥Ù‚Ø© Ù…Ù † Ø§Ù„Ù…Ø¹Ø±Ø¶ Ù‘Ø³Ø¬Ø¯Ù‘Ø± .±Ø§Ù„Ø³Ø¹Ø± Ø¨Ø§Ù„Ø¯Ù‚Ø§

4. Ù Ø³Ø¨Ù‘Ù‘Ø± **"Ø§Ù„Ø·Ø¨Ø§Ø¹Ø© Ø§Ù„Ù‘Ø§Ø±Ø© ðŸ“¥"** Ù„Ø±Ø¤Ù‘Ø© Ø·Ù„Ø¨ Ø§Ù„Ø¹Ø§ Ø®Ù‘Ø§Ø¬Ø§Ø¡ **Ù‚Ø¨Ù‘Ù„ Ø§Ù„Ø·Ù„Ø¨ âœ…** Ø£Ù‘ **"Ø±Ù Ø§Ù„Ø·Ù„Ø¨ â€"** Ø£Ù‘ **"Ø¥Ù ƒÙ…Ø§Ù„ Ø§Ù„Ø®Ø¯Ù…Ø© .**"Ù‘Ø³Ø³Ù„Ù‘Ù…Ø§ ðŸ

5. Ø§Ù„Ù † Ù‚Ø± Ø¹Ù„Ù ‰ **"Ù…Ø§Ø³Ù„Ø© Ø¨Ø§Ø·Ù‘Ù, ðŸ'¬"** Ù„Ø¨Ø¯Ø¡ .¨Ø§Ù„Ø±Ø³Ø´Ø§Ø¦Ù© Ø§Ù„Ø¹Ù…Ù„ Ø¹Ø§Ù„Ù…ÙŠ Ø§Ù„Ø·Ù„Ø¨ Ø¨Ø§Ù„Ø·Ù„Ø¨

Ø§Ù„Ù…Ø±Ø§Ø¬Ø¹: Ù‚Ù‘Ø§Ø¹Ø¯Ø© Ø§Ù„Ø¨ÙŠÙŠØ§Ù† Ø§Ø Ù‘Ø§Ù„Ø´Ø¨ÙƒÙŠØ© Ø§Ù„Ø³Ø¬Ø¯ÙŠÙŠØ© NoSQL Ø¨Ø§Ù„Ø³ÙŠÙŠÙŠ

```
      "orderId": "order_01_uuid",
      "serviceItem": { "...": "..." },
      "clientName": "Ø®Ø§Ù„Ø¯ Ø§Ù„Ø¹Ù…ÙŠÙ„",
      "clientEmail": "client@example.com",
      "clientPhone": "0788889999",
      "orderDate": "2026-10-26",
      "status": "ÙŠÙ†ØªØ¸Ø± Ù…Ùˆ Ø§Ù„Ù‚Ø¨Ù… Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø© â³"
    }
  },
  "chats": {
    "chat_order_01_uuid": {
      "messages": {
        "msg_01_uuid": {
          "messageId": "msg_01_uuid",
          "chatId": "chat_order_01_uuid",
          "senderName": "Ø®Ø§Ù„Ø¯ Ø§Ù„Ø¹Ù…ÙŠÙ„",
          "senderRole": "client",
          "text": "Ù…Ø±Ø¨Ø¨Ø§Ù‹ Ø£Ù„ ÙŠÙ…Ù†Ùƒ Ø§Ù†Ù‡Ø§Ø¡ Ø¨Ø§Ù‚Ø© Ø§Ù„Ø¹Ø±Ø¶ Ø¨Ø§Ù„Ø¥Ø¶Ø§ÙØ© Ø¥Ù„Ù‰ Ø§Ù„Ø£Ø³Ø¹Ø§Ø±ØŸ",
          "timestamp": 1792900000000,
          "timeFormatted": "01:15 PM"
        }
      }
    }
  }
}
```

Ø§Ù„Ù…Ù„Ø§ Ø§Ù„Ø¯Ø§Ù‘Ø±: Ø§Ù„Ø³Ù‘Ø«ÙŠÙ‚, Ø§Ù„Ø¨Ø±Ù…Ø¬Ù‘Ø© Ø§Ù„Ù‡Ù„Ù‘Ø§Ø³ (JavaDocs)

Ø³Ù‘Ø«ÙŠÙ‚, Ø§Ù„Ù‡Ù‘Ø¯Ù‘Ø§Ù…Ø³ Ø§Ù„Ù…Ø³Ø¤Ù„ Ø¹Ù† Ø¥Ø¯Ø§Ø±Ø© Ø§Ù„Ø¥Ø®Ø±Ø§Ø¬ 11.1

```
/**
 * OrderRepository singleton managing placement, updates, and realtime synchroni
zation
 * of client orders with Firebase Realtime Database.
 *
 * @author Student Developer
 * @version 1.0
 */
public class OrderRepository {

    /**
     * Listener interface for notifying UI when orders change in Firebase.
     */
    public interface OnOrdersChangeListener {
        /**
         * Fired when Firebase order snapshot updates.
         */
        void onOrdersChanged();
    }

    /**
     * Updates an order's status in Firebase Realtime Database and notifies loca
l listeners.
     *
     * @param orderId   Unique UUID string for the target order.
```

```
     * @param newStatus New status string (e.g. OrderRepository.STATUS_ACCEPTE
D).
     */
    public void updateOrderStatus(String orderId, String newStatus) {
        // Method Implementation
    }
}
```

Ø§Ù„Ù…Ø¨Ø§Ø¯Ø¦ Ø§Ù„Ø«Ø§Ù„Ø«Ø©: Ø£Ø«Ø± Ø§Ù„Ù…Ø´Ø§Ø±Ø© Ø§Ù„Ø¹Ø¨Ø«ÙŠØ© Ù‚Ø§Ù„Ù… Ø§Ù„Ø°Ø§ÙƒØ±Ø©

Ø£Ø«Ø± Ø§Ù„Ù…Ø´Ø§Ø±Ø© Ù‚Ø§Ù„Ù… Ù‡Ø§Ø±Ø¯ Ø©Ø³Ø¹Ø© 12.1 Ø©Ø§Ù„Ø°Ø§ÙƒØ±Ø©

Ø£Ùƒ Ø³Ø¨Ù†ÙŠ Ø³Ø·Ù‰Ø± Ù‡Ø°Ø§ Ø§Ù„Ø¯Ø±Ø³ Ø§Ù„Ø°Ø§ÙƒØ±Ø© Ø§Ù„Ø¹Ø§Ù…Ù„Ø© Ù„Ù„Ø§Ø³ØªØ®Ø¯Ø§Ù… Ø§Ù„Ø¹Ø§Ø¯ÙŠ

Ø£Ø¨Ù„Ø° Ø§Ù„Ø£Ø«Ø± ÙÙŠ Ø¥Ù†Ø¬Ø§Ø² Ù‡Ø°Ø§ Ø§Ù„Ø¹Ù…Ù„ Ø§Ù„Ù…Ø¹ÙØ§Ù…

Ø§Ù„Ù…Ø±Ø§Ø¬Ø¹ Ø§Ù„Ù…Ø³ØªØ®Ø¯Ù…Ø© 12.2 Ø¹Ø¨Ù…Ø¯Ø© (APA Citation Style)

1. Android Developers. (2026). *Build Android apps with Kotlin and Java*. Google Developers. Retrieved from https://developer.android.com/docs

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4. Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley.

Ø§Ù„Ù…Ù…Ù Ø£): Ø§Ù„Ø£Ùƒ Ù‘Ø§Ø¯ Ø§Ù„Ù…Ø¯Ø±Ø±Ø© Ø§Ù„Ù‡Ø§Ù…Ø© Ù Ù Ø¨Ø±Ù…Ø¬ (Java Source Code)

ÙŠØªØ¶Ù…Ù † Ù ‡ Ø°Ø§ Ø§Ù„Ù…Ù„Ù Ø§Ù„Ø´ Ø§Ù„Ù…Ø¹Ø±ÙŠ Ø§Ù„Ùƒ Ø§Ù„Ù…Ù„Ø¡ Ø§Ù„Ù¡Ø§ Ù‘Ø§Ù„Ù‘Ø§Ø¬Ù‡Ø§ Ø§Ù„Ù…Ø¨Ø±Ù…Ø¬ ÙŠÙ Ø´Ø±Ù‘Ø¹ Ù‘Ø§Ù„Ø¨Ø¹Ø¶ Ø§Ù„Ø¨Ø¹Ø¶ :"Ø³Ø·Ø·Ù, "Ø´ÙØºÙ„ - WORK

MainActivity.java .1

```
package com.example.work;

import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private static final int SPLASH_DELAY_MS = 1500;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        new Handler(Looper.getMainLooper()).postDelayed(() -> {
            Intent intent = new Intent(MainActivity.this, RoleSelectionActivity.
class);
```

```
            startActivity(intent);
            finish();
        }, SPLASH_DELAY_MS);
    }
}
```

RoleSelectionActivity.java .2

```
package com.example.work;

import android.content.Intent;
import android.os.Bundle;
import android.util.Patterns;
import android.widget.RadioButton;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import com.google.android.material.button.MaterialButton;
import com.google.android.material.card.MaterialCardView;
import com.google.android.material.textfield.TextInputEditText;
import com.google.android.material.textfield.TextInputLayout;

public class RoleSelectionActivity extends AppCompatActivity {

    private MaterialCardView cardRoleClient;
    private MaterialCardView cardRoleProvider;
    private RadioButton radioClient;
    private RadioButton radioProvider;
    private TextInputLayout tilUserName;
    private TextInputLayout tilUserEmail;
    private TextInputEditText etUserName;
    private TextInputEditText etUserEmail;

    private boolean isClientSelected = true;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_role_selection);
```

```
        initViews();
        setupRoleToggle();
    }

    private void initViews() {
        cardRoleClient = findViewById(R.id.cardRoleClient);
        cardRoleProvider = findViewById(R.id.cardRoleProvider);
        radioClient = findViewById(R.id.radioClient);
        radioProvider = findViewById(R.id.radioProvider);
        tilUserName = findViewById(R.id.tilUserName);
        tilUserEmail = findViewById(R.id.tilUserEmail);
        etUserName = findViewById(R.id.etUserName);
        etUserEmail = findViewById(R.id.etUserEmail);
        MaterialButton btnContinue = findViewById(R.id.btnContinue);

        btnContinue.setOnClickListener(v -> validateAndProceed());
    }

    private void setupRoleToggle() {
        cardRoleClient.setOnClickListener(v -> selectClientRole());
        cardRoleProvider.setOnClickListener(v -> selectProviderRole());
    }

    private void selectClientRole() {
        isClientSelected = true;
        radioClient.setChecked(true);
        radioProvider.setChecked(false);
        cardRoleClient.setStrokeColor(getColor(R.color.orange_primary));
        cardRoleClient.setStrokeWidth(dpToPx(2));
        cardRoleProvider.setStrokeColor(getColor(R.color.card_stroke));
        cardRoleProvider.setStrokeWidth(dpToPx(1));
    }

    private void selectProviderRole() {
        isClientSelected = false;
        radioClient.setChecked(false);
        radioProvider.setChecked(true);
        cardRoleProvider.setStrokeColor(getColor(R.color.orange_primary));
        cardRoleProvider.setStrokeWidth(dpToPx(2));
        cardRoleClient.setStrokeColor(getColor(R.color.card_stroke));
        cardRoleClient.setStrokeWidth(dpToPx(1));
    }
```

```
    private void validateAndProceed() {
        tilUserName.setError(null);
        tilUserEmail.setError(null);

        String name = etUserName.getText() != null ? etUserName.getText().toStri
ng().trim() : "";
        String email = etUserEmail.getText() != null ? etUserEmail.getText().toS
tring().trim() : "";

        boolean nameValid = !name.isEmpty();
        boolean emailValid = !email.isEmpty() && Patterns.EMAIL_ADDRESS.matcher
(email).matches();

        if (!nameValid) {
            tilUserName.setError(getString(R.string.err_empty_name));
        }

        if (!emailValid) {
            tilUserEmail.setError(getString(R.string.err_invalid_email));
        }

        if (!nameValid || !emailValid) {
            Toast.makeText(this, "ÙŠØ±Ø¬Ù‰ Ùƒ Ø§Ø³Ù… Ù‘Ø¨Ø±ÙŠØ¯ Ø¥Ù„Ùƒ
Ø±Ù‘Ùˆ ØµØÙŠØÙ† Ù„Ø§Ø³Ø¬ÙŠÙ„", Toast.LENGTH_SHORT).show();
            return;
        }

        Intent intent;
        if (isClientSelected) {
            intent = new Intent(RoleSelectionActivity.this, ClientMainActivity.c
lass);
        } else {
            intent = new Intent(RoleSelectionActivity.this, ProviderMainActivit
y.class);
        }

        intent.putExtra("USER_NAME", name);
        intent.putExtra("USER_EMAIL", email);
        startActivity(intent);
        finish();
    }

    private int dpToPx(int dp) {
        float density = getResources().getDisplayMetrics().density;
```

```
        return Math.round(dp * density);
    }
}
```

ClientMainActivity.java .3

```
package com.example.work;

import android.content.Intent;
import android.content.res.ColorStateList;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.HorizontalScrollView;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.google.android.material.button.MaterialButton;
import com.google.android.material.card.MaterialCardView;
import com.google.android.material.chip.Chip;
import com.google.android.material.chip.ChipGroup;
import com.google.android.material.floatingactionbutton.FloatingActionButton;

import java.util.List;

public class ClientMainActivity extends AppCompatActivity {

    private ChipGroup chipGroupCategories;
    private HorizontalScrollView hsvCategories;
    private RecyclerView rvServices;
    private RecyclerView rvOrders;
    private TextView tvEmptyState;
    private TextView tvWelcomeHeader;
    private MaterialButton btnTabBrowseServices;
    private MaterialButton btnTabMyOrders;
```

```
    private ServiceAdapter serviceAdapter;
    private OrderAdapter orderAdapter;
    private final ServiceRepository repository = ServiceRepository.getInstance
();
    private final OrderRepository orderRepository = OrderRepository.getInstance
();

    private String selectedCategory = "Ø§Ù„Ùƒ„Ù„";
    private boolean isMyOrdersTabActive = false;

    private final ServiceRepository.OnDataChangeListener dataChangeListener = ()
-> runOnUiThread(this::loadCategoryServices);
    private final OrderRepository.OnOrdersChangeListener ordersChangeListener =
() -> runOnUiThread(this::loadMyOrders);

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_client_main);

        repository.init(this);
        orderRepository.init(this);

        initViews();
        setupUserHeader();
        setupRecyclerViews();
        setupListeners();

        repository.addOnDataChangeListener(dataChangeListener);
        orderRepository.addOnOrdersChangeListener(ordersChangeListener);

        loadCategoryServices();
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        repository.removeOnDataChangeListener(dataChangeListener);
        orderRepository.removeOnOrdersChangeListener(ordersChangeListener);
    }

    private void initViews() {
        tvWelcomeHeader = findViewById(R.id.tvWelcomeHeader);
        chipGroupCategories = findViewById(R.id.chipGroupCategories);
```

```
        hsvCategories = findViewById(R.id.hsvCategories);
        rvServices = findViewById(R.id.rvServices);
        rvOrders = findViewById(R.id.rvOrders);
        tvEmptyState = findViewById(R.id.tvEmptyState);
        btnTabBrowseServices = findViewById(R.id.btnTabBrowseServices);
        btnTabMyOrders = findViewById(R.id.btnTabMyOrders);

        MaterialCardView cardOpenSearchScreen = findViewById(R.id.cardOpenSearch
Screen);
        MaterialButton btnSwitchRole = findViewById(R.id.btnSwitchRole);
        FloatingActionButton fabSearch = findViewById(R.id.fabSearch);

        btnSwitchRole.setOnClickListener(v -> {
            Intent intent = new Intent(ClientMainActivity.this, RoleSelectionAct
ivity.class);
            intent.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TOP | Intent.FLAG_ACTIVIT
Y_NEW_TASK);
            startActivity(intent);
            finish();
        });

        View.OnClickListener openSearchListener = v -> {
            Intent intent = new Intent(ClientMainActivity.this, SearchActivity.c
lass);
            startActivity(intent);
        };

        if (cardOpenSearchScreen != null) {
            cardOpenSearchScreen.setOnClickListener(openSearchListener);
        }

        if (fabSearch != null) {
            fabSearch.setOnClickListener(openSearchListener);
        }
    }

    private void setupUserHeader() {
        String userName = getIntent().getStringExtra("USER_NAME");
        if (userName != null && !userName.trim().isEmpty()) {
            tvWelcomeHeader.setText(String.format("Ø£Ù‡Ù„Ø§Ù‹ Ø¨Ùƒ %s ðŸ‘‹", use
rName.trim()));
        }
    }
```

```
    private void setupRecyclerViews() {
        rvServices.setLayoutManager(new LinearLayoutManager(this));
        serviceAdapter = new ServiceAdapter(service -> {
            Intent intent = new Intent(ClientMainActivity.this, ServiceDetailAct
ivity.class);
            intent.putExtra("SERVICE_ITEM", service);
            startActivity(intent);
        });
        rvServices.setAdapter(serviceAdapter);

        rvOrders.setLayoutManager(new LinearLayoutManager(this));
        orderAdapter = new OrderAdapter(false, new OrderAdapter.OnOrderActionLis
tener() {
            @Override
            public void onOrderContactClick(OrderItem order) {
                if (order.getServiceItem() != null && order.getServiceItem().get
ProviderPhone() != null && !order.getServiceItem().getProviderPhone().trim().isE
mpty()) {
                    String phone = order.getServiceItem().getProviderPhone().tri
m();
                    String digitsOnly = phone.replaceAll("[^0-9]", "");
                    String cleanedPhone = digitsOnly.length() >= 7 ? digitsOnly
: phone;

                    try {
                        String url = "https://api.whatsapp.com/send?phone=" + cl
eanedPhone +
                                "&text=" + Uri.encode("Ù…Ø±Ø¨Ù‰ Ø£Ù Ø§Ù
„Ù…Øª Ø¨Ø¹Ø¯ Ø´Ø£Ù† Ø·Ù„Ø¨ Ù„Ø®Ø¯Ù…Ø©: " + order.getServiceItem().getTitl
e());
                        Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse
(url));
                        startActivity(intent);
                    } catch (Exception e) {
                        try {
                            Intent dialIntent = new Intent(Intent.ACTION_DIAL, U
ri.parse("tel:" + phone));
                            startActivity(dialIntent);
                        } catch (Exception ex) {
                            Toast.makeText(ClientMainActivity.this, "Øª Ø¹Ø°Ø± Ø§
Ù„Øª Ù…Ø¹ Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø©", Toast.LENGTH_SHORT).show();
                        }
                    }
                } else if (order.getServiceItem() != null) {
```

```
                    Intent intent = new Intent(ClientMainActivity.this, ServiceD
etailActivity.class);
                    intent.putExtra("SERVICE_ITEM", order.getServiceItem());
                    startActivity(intent);
                } else {
                    Toast.makeText(ClientMainActivity.this, "Ø¹ÙˆØ§ÙµÙŠÙ„ Ø§Ù„Ø®Ø
¯Ù…Ø© ØºÙŠØ± Ù…Ø¹ÙˆÙ†Ø©", Toast.LENGTH_SHORT).show();
                }
            }

            @Override
            public void onInAppChatClick(OrderItem order) {
                Intent intent = new Intent(ClientMainActivity.this, ChatActivit
y.class);
                intent.putExtra("CHAT_ID", order.getOrderId());
                intent.putExtra("RECEIVER_NAME", order.getServiceItem() != null
? order.getServiceItem().getProviderName() : "Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø©");
                intent.putExtra("SERVICE_TITLE", order.getServiceItem() != null
? order.getServiceItem().getTitle() : "Ø·Ù„Ø¨ Ø®Ø¯Ù…Ø©");
                intent.putExtra("CURRENT_ROLE", "client");
                intent.putExtra("CURRENT_USER_NAME", order.getClientName() != nu
ll ? order.getClientName() : "Ø§Ù„Ø¹Ù…ÙŠÙ„");
                startActivity(intent);
            }

            @Override
            public void onAcceptOrder(OrderItem order) {}

            @Override
            public void onRejectOrder(OrderItem order) {}

            @Override
            public void onCompleteOrder(OrderItem order) {}
        });
        rvOrders.setAdapter(orderAdapter);
    }

    private void setupListeners() {
        btnTabBrowseServices.setOnClickListener(v -> activateBrowseTab());
        btnTabMyOrders.setOnClickListener(v -> activateMyOrdersTab());

        chipGroupCategories.setOnCheckedStateChangeListener((group, checkedIds)
-> {
            if (checkedIds.isEmpty()) {
```

```
                selectedCategory = "Ø§Ù„Ùƒ Ù„";
            } else {
                int checkedId = checkedIds.get(0);
                Chip chip = group.findViewById(checkedId);
                if (chip != null) {
                    selectedCategory = chip.getText().toString();
                } else {
                    selectedCategory = "Ø§Ù„Ùƒ Ù„";
                }
            }
            loadCategoryServices();
        });
    }

    private void activateBrowseTab() {
        isMyOrdersTabActive = false;
        btnTabBrowseServices.setBackgroundTintList(ColorStateList.valueOf(getCol
or(R.color.brand_primary)));
        btnTabBrowseServices.setTextColor(getColor(R.color.white));

        btnTabMyOrders.setBackgroundTintList(ColorStateList.valueOf(getColor(and
roid.R.color.transparent)));
        btnTabMyOrders.setTextColor(getColor(R.color.text_secondary));

        hsvCategories.setVisibility(View.VISIBLE);
        rvServices.setVisibility(View.VISIBLE);
        rvOrders.setVisibility(View.GONE);

        loadCategoryServices();
    }

    private void activateMyOrdersTab() {
        isMyOrdersTabActive = true;
        btnTabMyOrders.setBackgroundTintList(ColorStateList.valueOf(getColor(R.c
olor.brand_primary)));
        btnTabMyOrders.setTextColor(getColor(R.color.white));

        btnTabBrowseServices.setBackgroundTintList(ColorStateList.valueOf(getCol
or(android.R.color.transparent)));
        btnTabBrowseServices.setTextColor(getColor(R.color.text_secondary));

        hsvCategories.setVisibility(View.GONE);
        rvServices.setVisibility(View.GONE);
        rvOrders.setVisibility(View.VISIBLE);
```

```
        loadMyOrders();
    }

    private void loadCategoryServices() {
        if (isMyOrdersTabActive) return;

        List<ServiceItem> results = repository.searchServices("", -1, selectedCa
tegory);

        if (results.isEmpty()) {
            tvEmptyState.setText(R.string.no_services_found);
            tvEmptyState.setVisibility(View.VISIBLE);
            rvServices.setVisibility(View.GONE);
        } else {
            tvEmptyState.setVisibility(View.GONE);
            rvServices.setVisibility(View.VISIBLE);
            serviceAdapter.setServices(results, -1);
        }
    }

    private void loadMyOrders() {
        List<OrderItem> orders = orderRepository.getAllOrders();

        if (orders.isEmpty()) {
            tvEmptyState.setText("Ù„Ù… ØªÙ‚Ù… Ø¨Ø·Ù„Ø¨ Ø£Ù‰ Ø§Ø³ØªØ®Ø¯Ø§Ù… Ø£ÙŠ Ø®Ø¯Ù…Ø© Ø¨Ø¹Ø¯. Ø§Ø¨Ø¯Ø« Ø¹Ù† Ø®Ø¯Ù…Ø© Ùˆ Ø§Ø·Ù„Ø¨Ù‡Ø§ Ù„Ø¹Ø±Ø¶Ù‡Ø§ Ù‡Ù†Ø§!");
            tvEmptyState.setVisibility(View.VISIBLE);
            rvOrders.setVisibility(View.GONE);
        } else {
            tvEmptyState.setVisibility(View.GONE);
            rvOrders.setVisibility(View.VISIBLE);
            orderAdapter.setOrders(orders);
        }
    }

    @Override
    protected void onResume() {
        super.onResume();
        if (isMyOrdersTabActive) {
            loadMyOrders();
        } else {
            loadCategoryServices();
        }
```

```
    }
}
```

ProviderMainActivity.java .4

```
package com.example.work;

import android.content.Intent;
import android.content.res.ColorStateList;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.google.android.material.button.MaterialButton;

import java.util.List;
import java.util.Locale;

public class ProviderMainActivity extends AppCompatActivity {

    private RecyclerView rvProviderServices;
    private RecyclerView rvIncomingOrders;
    private TextView tvEmptyPosted;
    private TextView tvProviderHeader;
    private TextView tvPostedServicesCount;
    private TextView tvSectionTitle;
    private MaterialButton btnTabMyServices;
    private MaterialButton btnTabIncomingOrders;

    private ServiceAdapter serviceAdapter;
    private OrderAdapter orderAdapter;
    private final ServiceRepository repository = ServiceRepository.getInstance
();
    private final OrderRepository orderRepository = OrderRepository.getInstance
```

```
();

    private boolean isOrdersTabActive = false;

    private final ServiceRepository.OnDataChangeListener dataChangeListener = ()
-> runOnUiThread(this::loadData);
    private final OrderRepository.OnOrdersChangeListener ordersChangeListener =
() -> runOnUiThread(this::loadData);

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_provider_main);

        repository.init(this);
        orderRepository.init(this);

        initViews();
        setupUserHeader();
        setupRecyclerViews();

        repository.addOnDataChangeListener(dataChangeListener);
        orderRepository.addOnOrdersChangeListener(ordersChangeListener);

        loadData();
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        repository.removeOnDataChangeListener(dataChangeListener);
        orderRepository.removeOnOrdersChangeListener(ordersChangeListener);
    }

    private void initViews() {
        tvProviderHeader = findViewById(R.id.tvProviderHeader);
        tvPostedServicesCount = findViewById(R.id.tvPostedServicesCount);
        tvSectionTitle = findViewById(R.id.tvSectionTitle);
        rvProviderServices = findViewById(R.id.rvProviderServices);
        rvIncomingOrders = findViewById(R.id.rvIncomingOrders);
        tvEmptyPosted = findViewById(R.id.tvEmptyPosted);

        btnTabMyServices = findViewById(R.id.btnTabMyServices);
        btnTabIncomingOrders = findViewById(R.id.btnTabIncomingOrders);
```

```
        MaterialButton btnSwitchRole = findViewById(R.id.btnSwitchRole);
        MaterialButton btnAddService = findViewById(R.id.btnAddService);

        btnSwitchRole.setOnClickListener(v -> {
            Intent intent = new Intent(ProviderMainActivity.this, RoleSelectionA
ctivity.class);
            intent.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TOP | Intent.FLAG_ACTIVIT
Y_NEW_TASK);
            startActivity(intent);
            finish();
        });

        btnAddService.setOnClickListener(v -> {
            Intent intent = new Intent(ProviderMainActivity.this, AddServiceActi
vity.class);
            startActivity(intent);
        });

        btnTabMyServices.setOnClickListener(v -> activateServicesTab());
        btnTabIncomingOrders.setOnClickListener(v -> activateOrdersTab());
    }

    private void activateServicesTab() {
        isOrdersTabActive = false;
        btnTabMyServices.setBackgroundTintList(ColorStateList.valueOf(getColor
(R.color.brand_primary)));
        btnTabMyServices.setTextColor(getColor(R.color.white));

        btnTabIncomingOrders.setBackgroundTintList(ColorStateList.valueOf(getCol
or(android.R.color.transparent)));
        btnTabIncomingOrders.setTextColor(getColor(R.color.text_secondary));

        if (tvSectionTitle != null) tvSectionTitle.setText("Ùƒ Ø§Ù„Ø© Ø§Ù„Ø®Ø¯Ù…Ø§Øª Ø¨Ùƒ");
        rvProviderServices.setVisibility(View.VISIBLE);
        rvIncomingOrders.setVisibility(View.GONE);

        loadData();
    }

    private void activateOrdersTab() {
        isOrdersTabActive = true;
        btnTabIncomingOrders.setBackgroundTintList(ColorStateList.valueOf(getCol
```

```
or(R.color.brand_primary)));
        btnTabIncomingOrders.setTextColor(getColor(R.color.white));

        btnTabMyServices.setBackgroundTintList(ColorStateList.valueOf(getColor(a
ndroid.R.color.transparent)));
        btnTabMyServices.setTextColor(getColor(R.color.text_secondary));

        if (tvSectionTitle != null) tvSectionTitle.setText("Ø§Ù„Ø·Ù„Ø¨Ø§Øª Ø§Ù„Ù
ˆØ§Ø±Ø¯Ø© Ù„Ø®Ø¯Ù…Ø§ØªÙŠ ðŸ“¥");
        rvProviderServices.setVisibility(View.GONE);
        rvIncomingOrders.setVisibility(View.VISIBLE);

        loadData();
    }

    private void setupUserHeader() {
        String userName = getIntent().getStringExtra("USER_NAME");
        if (userName != null && !userName.trim().isEmpty()) {
            tvProviderHeader.setText(String.format("Ù„Ù‚Ø©© %s ðŸ› ï¸", userNam
e.trim()));
        }
    }

    private void setupRecyclerViews() {
        rvProviderServices.setLayoutManager(new LinearLayoutManager(this));
        serviceAdapter = new ServiceAdapter(service -> {
            Intent intent = new Intent(ProviderMainActivity.this, ServiceDetailA
ctivity.class);
            intent.putExtra("SERVICE_ITEM", service);
            startActivity(intent);
        });
        rvProviderServices.setAdapter(serviceAdapter);

        rvIncomingOrders.setLayoutManager(new LinearLayoutManager(this));
        orderAdapter = new OrderAdapter(true, new OrderAdapter.OnOrderActionList
ener() {
            @Override
            public void onOrderContactClick(OrderItem order) {
                String phone = order.getClientPhone();
                if (phone != null && !phone.trim().isEmpty()) {
                    openWhatsAppOrDial(order, phone.trim());
                } else {
                    final android.widget.EditText input = new android.widget.Edi
tText(ProviderMainActivity.this);
```

```
                    input.setHint("Ø§Ø¯Ø®Ù„ Ø±Ù‚Ù… Ù‡Ø§ØªÙ Ø§Ù„Ø¹Ù…ÙŠÙ„ (Ù…Ø«Ø§Ù„: 0791234567)");
                    input.setInputType(android.text.InputType.TYPE_CLASS_PHONE);
                    input.setPadding(32, 24, 32, 24);

                    new androidx.appcompat.app.AlertDialog.Builder(ProviderMainActivity.this)
                            .setTitle("Ø§Ù„Ø¹Ø§Ù‚Ø§Ø¨ Ù…Ø¹ Ø§Ù„Ø¹Ù…ÙŠÙ„: " + (order.getClientName() != null ? order.getClientName() : "Ø¹Ù…ÙŠÙ„"))
                            .setMessage("Ø£Ø¯Ø®Ù„ Ø±Ù‚Ù… Ù‡Ø§ØªÙ Ø§Ù„Ø¹Ù…ÙŠÙ„ Ù„Ù…Ø±Ø§Ø³Ù„Ø© Ø¹Ù† Ø·Ø±ÙŠÙ‚ Ø§Ù„ØªØ·Ø¨ÙŠÙ‚:")
                            .setView(input)
                            .setPositiveButton("Ù…Ø±Ø§Ø³Ù„Ø© Ø¹Ø¨Ø± Ø§Ù„ØªØ·Ø¨ÙŠÙ‚ ðŸ’¬", (dialog, which) -> {
                                String enteredPhone = input.getText().toString().trim();
                                if (!enteredPhone.isEmpty()) {
                                    order.setClientPhone(enteredPhone);
                                    orderRepository.updateOrderStatus(order.getOrderId(), order.getStatus() != null ? order.getStatus() : OrderRepository.STATUS_PENDING);
                                    openWhatsAppOrDial(order, enteredPhone);
                                } else {
                                    Toast.makeText(ProviderMainActivity.this, "ÙŠØ±Ø¬Ù‰ Ùƒ Ø§Ù„Ø¹Ù…ÙŠÙ„ Ø£ÙˆÙ„Ø§Ù‹", Toast.LENGTH_SHORT).show();
                                }
                            })
                            .setNegativeButton("Ø¥Ù„ØºØ§Ø¡", null)
                            .show();
                }
            }

            @Override
            public void onInAppChatClick(OrderItem order) {
                Intent intent = new Intent(ProviderMainActivity.this, ChatActivity.class);
                intent.putExtra("CHAT_ID", order.getOrderId());
                intent.putExtra("RECEIVER_NAME", order.getClientName() != null ? order.getClientName() : "Ø§Ù„Ø¹Ù…ÙŠÙ„");
                intent.putExtra("SERVICE_TITLE", order.getServiceItem() != null ? order.getServiceItem().getTitle() : "Ø·Ù„Ø¨ Ø®Ø¯Ù…Ø©");
                intent.putExtra("CURRENT_ROLE", "provider");
                intent.putExtra("CURRENT_USER_NAME", order.getServiceItem() != n
```

```
ull ? order.getServiceItem().getProviderName() : "Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø©");
                startActivity(intent);
            }

            @Override
            public void onAcceptOrder(OrderItem order) {
                orderRepository.updateOrderStatus(order.getOrderId(), OrderRepos
itory.STATUS_ACCEPTED);
                Toast.makeText(ProviderMainActivity.this, "Ø³Ù… Ù‚Ø¨Ù„ Ø§Ù„Ø·Ù„Ø¨ Ø¨Ø§Ù„ÙØ¹Ù„! Ø¬Ø§Ø±Ù Ø§Ù„Ø¨Ø¯Ø¡ Ø¨Ø§Ù„Ø³Ø¹Ù„ âš™ï¸", Toast.LENGTH_SHORT).s
how();
            }

            @Override
            public void onRejectOrder(OrderItem order) {
                orderRepository.updateOrderStatus(order.getOrderId(), OrderRepos
itory.STATUS_REJECTED);
                Toast.makeText(ProviderMainActivity.this, "Ø³Ù… Ø±ÙØ¶ Ø§Ù„Ø·Ù„Ø¨", Toast.LENGTH_SHORT).show();
            }

            @Override
            public void onCompleteOrder(OrderItem order) {
                orderRepository.updateOrderStatus(order.getOrderId(), OrderRepos
itory.STATUS_COMPLETED);
                Toast.makeText(ProviderMainActivity.this, "Ø³Ù… Ø¥ÙƒÙ…Ø§Ù„ Ù‚Ø³Ù
³Ù„ÙŠÙ… Ø§Ù„Ø®Ø¯Ù…Ø© Ø¨Ù†Ø¬Ø§Ø! ðŸŽ", Toast.LENGTH_SHORT).show();
            }
        });
        rvIncomingOrders.setAdapter(orderAdapter);
    }

    private void loadData() {
        List<ServiceItem> services = repository.getAllServices();
        List<OrderItem> orders = orderRepository.getAllOrders();

        int completedOrdersCount = 0;
        for (OrderItem o : orders) {
            if (o.getStatus() != null && o.getStatus().contains("Ù…ÙƒØªÙ…Ù„Ø©"))
{
                completedOrdersCount++;
            }
        }
```

```
        if (tvPostedServicesCount != null) {
            if (isOrdersTabActive) {
                tvPostedServicesCount.setText(String.format(Locale.getDefault(),
"%d Ø·Ù„Ø¨Ø§Øª Ù†Ø§Ø±Ø© (%d Ù…Ùƒ Ø§Ù…Ù„Ø©)", orders.size(), completedOrdersCoun
t));
            } else {
                tvPostedServicesCount.setText(String.format(Locale.getDefault(),
"%d Ø®Ø¯Ù…Ø© Ù…Ø¹Ø§Ø§Ø© Ø¨Ø§Ø³Ù†Ø³Ù, | %d Ø·Ù„Ø¨ Ù…Ùƒ Ø§Ù„Ø©", services.size(), co
mpletedOrdersCount));
            }
        }

        if (isOrdersTabActive) {
            if (orders.isEmpty()) {
                tvEmptyPosted.setText("Ù„Ø§ Ø§Ù†Ø¬Ø· Ø·Ù„Ø¨Ø§Øª Ù†Ø§Ø±Ø© Ù…Ù†
Ø§Ù„Ù…Ù„Ø§Ø¡ Ø§Ù„Ø¢Ù† ðŸ“¥");
                tvEmptyPosted.setVisibility(View.VISIBLE);
                rvIncomingOrders.setVisibility(View.GONE);
            } else {
                tvEmptyPosted.setVisibility(View.GONE);
                rvIncomingOrders.setVisibility(View.VISIBLE);
                orderAdapter.setOrders(orders);
            }
        } else {
            if (services.isEmpty()) {
                tvEmptyPosted.setText(R.string.no_posted_services);
                tvEmptyPosted.setVisibility(View.VISIBLE);
                rvProviderServices.setVisibility(View.GONE);
            } else {
                tvEmptyPosted.setVisibility(View.GONE);
                rvProviderServices.setVisibility(View.VISIBLE);
                serviceAdapter.setServices(services, -1);
            }
        }
    }

    @Override
    protected void onResume() {
        super.onResume();
        loadData();
    }

    private void openWhatsAppOrDial(OrderItem order, String phone) {
        if (phone == null || phone.trim().isEmpty()) return;
```

```
        String digitsOnly = phone.replaceAll("[^0-9]", "");
        String cleanedPhone = digitsOnly.length() >= 7 ? digitsOnly : phone.trim
();

        try {
            String url = "https://api.whatsapp.com/send?phone=" + cleanedPhone +
                    "&text=" + Uri.encode("Ù…Ø±Ø¨Ø¨Ø§Ù‹ " + (order.getClientName
() != null ? order.getClientName() : "") + "Ø Ø£Ù†Ø§ " + (order.getServiceItem
() != null ? order.getServiceItem().getProviderName() : "Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø©")
+ " Ø¨Ø´Ø£Ù† Ø·Ù„Ø¨Ùƒ Ù„Ø®Ø¯Ù…Ø©: " + (order.getServiceItem() != null ? order.ge
tServiceItem().getTitle() : ""));
            Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse(url));
            startActivity(intent);
        } catch (Exception e) {
            try {
                Intent dialIntent = new Intent(Intent.ACTION_DIAL, Uri.parse("te
l:" + phone.trim()));
                startActivity(dialIntent);
            } catch (Exception ex) {
                Toast.makeText(ProviderMainActivity.this, "Ø¨Ø¹Ø°Ø± Ùˆ Ø§Ù„Ù‚Ø
§Ø¦Ø§Ø§Ø¨ Ø£Ù‚ Ø§Ù„Ø§Ø§Ø¬Ø„ Ø¨Ø§Ù„Ù…Ù…Ù„„: " + order.getClientName(), Toast.L
ENGTH_SHORT).show();
            }
        }
    }
}
```

SearchActivity.java .5

```
package com.example.work;

import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.view.inputmethod.InputMethodManager;
import android.widget.LinearLayout;
import android.widget.TextView;
```

```
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.google.android.material.button.MaterialButton;
import com.google.android.material.chip.Chip;
import com.google.android.material.chip.ChipGroup;
import com.google.android.material.textfield.TextInputEditText;

import java.util.List;
import java.util.Locale;

public class SearchActivity extends AppCompatActivity {

    private TextInputEditText etSearchQuery;
    private TextInputEditText etTargetPrice;
    private ChipGroup cgSearchHistory;
    private LinearLayout llSearchHistorySection;
    private LinearLayout llSearchResultsHeader;
    private TextView tvResultsHeaderTitle;
    private RecyclerView rvSearchResults;
    private TextView tvSearchEmptyState;

    private ServiceAdapter adapter;
    private final ServiceRepository repository = ServiceRepository.getInstance
();

    private final ServiceRepository.OnDataChangeListener dataChangeListener = ()
-> runOnUiThread(this::executeSearch);

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_search);

        repository.init(this);
        repository.addOnDataChangeListener(dataChangeListener);

        initViews();
        setupRecyclerView();
        loadSearchHistory();
        autoFocusKeyboard();
    }
```

```
    @Override
    protected void onDestroy() {
        super.onDestroy();
        repository.removeOnDataChangeListener(dataChangeListener);
    }

    private void initViews() {
        etSearchQuery = findViewById(R.id.etSearchQuery);
        etTargetPrice = findViewById(R.id.etTargetPrice);
        cgSearchHistory = findViewById(R.id.cgSearchHistory);
        llSearchHistorySection = findViewById(R.id.llSearchHistorySection);
        llSearchResultsHeader = findViewById(R.id.llSearchResultsHeader);
        tvResultsHeaderTitle = findViewById(R.id.tvResultsHeaderTitle);
        rvSearchResults = findViewById(R.id.rvSearchResults);
        tvSearchEmptyState = findViewById(R.id.tvSearchEmptyState);

        findViewById(R.id.btnBack).setOnClickListener(v -> finish());

        TextView tvClearHistory = findViewById(R.id.tvClearHistory);
        MaterialButton btnDoSearch = findViewById(R.id.btnDoSearch);

        btnDoSearch.setOnClickListener(v -> executeSearch());

        tvClearHistory.setOnClickListener(v -> {
            SearchHistoryManager.clearHistory(SearchActivity.this);
            loadSearchHistory();
        });
    }

    private void setupRecyclerView() {
        rvSearchResults.setLayoutManager(new LinearLayoutManager(this));
        adapter = new ServiceAdapter(service -> {
            Intent intent = new Intent(SearchActivity.this, ServiceDetailActivit
y.class);
            intent.putExtra("SERVICE_ITEM", service);
            startActivity(intent);
        });
        rvSearchResults.setAdapter(adapter);
    }

    private void autoFocusKeyboard() {
        etSearchQuery.requestFocus();
        etSearchQuery.postDelayed(() -> {
            InputMethodManager imm = (InputMethodManager) getSystemService(Conte
```

```
xt.INPUT_METHOD_SERVICE);
            if (imm != null) {
                imm.showSoftInput(etSearchQuery, 0);
            }
        }, 200);
    }

    private void loadSearchHistory() {
        cgSearchHistory.removeAllViews();
        List<String> history = SearchHistoryManager.getSearchHistory(this);

        if (history.isEmpty()) {
            llSearchHistorySection.setVisibility(View.GONE);
            return;
        }

        llSearchHistorySection.setVisibility(View.VISIBLE);
        for (String item : history) {
            Chip chip = new Chip(this);
            chip.setText(item);
            chip.setClickable(true);
            chip.setFocusable(true);
            chip.setChipBackgroundColorResource(R.color.orange_chip_bg);
            chip.setTextColor(getColor(R.color.orange_primary));

            chip.setOnClickListener(v -> {
                if (item.contains(" (Ø¨ØØ¯Ùˆ $")) {
                    String[] parts = item.split(" \\(Ø¨ØØ¯Ùˆ \\$");
                    etSearchQuery.setText(parts[0].trim());
                    if (parts.length > 1) {
                        String budgetStr = parts[1].replace(")", "").trim();
                        etTargetPrice.setText(budgetStr);
                    }
                } else if (item.startsWith("Ù…ÙŠØ²Ø§Ù†ÙŠØ© Ø¨Ù€ $")) {
                    etSearchQuery.setText("");
                    etTargetPrice.setText(item.replace("Ù…ÙŠØ²Ø§Ù†ÙŠØ© Ø¨Ù€ $",
"").trim());
                } else {
                    etSearchQuery.setText(item.trim());
                    etTargetPrice.setText("");
                }
                executeSearch();
            });
```

```
            cgSearchHistory.addView(chip);
        }
    }

    private double getTargetPriceValue(String targetPriceStr) {
        if (!targetPriceStr.isEmpty()) {
            try {
                return Double.parseDouble(targetPriceStr);
            } catch (NumberFormatException ignored) {
            }
        }
        return -1;
    }

    private void executeSearch() {
        String query = etSearchQuery.getText() != null ? etSearchQuery.getText
().toString().trim() : "";
        String targetPriceStr = etTargetPrice.getText() != null ? etTargetPrice.
getText().toString().trim() : "";

        double targetPrice = getTargetPriceValue(targetPriceStr);

        if (query.isEmpty() && targetPrice <= 0) {
            tvSearchEmptyState.setText("ÙŠØ±Ø¬Ù‰ Ùƒ Ø§Ø§Ø¨Ø© Ø§Ø³Ù… Ø§Ù„Ø®Ø¯Ù…Ø© Ø£Ù^ Ø§Ù„Ø³Ø¹Ø± Ø§Ù„Ù…Ø·Ù„Ù^Ø¨ Ù„Ù„Ø¨Ø«");
            tvSearchEmptyState.setVisibility(View.VISIBLE);
            rvSearchResults.setVisibility(View.GONE);
            llSearchResultsHeader.setVisibility(View.GONE);
            return;
        }

        // Save to search history
        SearchHistoryManager.addSearchQuery(this, query, targetPrice);
        loadSearchHistory();

        // Perform search
        List<ServiceItem> results = repository.searchServices(query, targetPric
e, "Ø§Ù„Ùƒ Ù„");

        if (results.isEmpty()) {
            tvSearchEmptyState.setText("Ù„Ù… ÙŠØªÙ… Ø§Ù„Ø¹Ø«Ù^Ø± Ø¹Ù„Ù‰ Ø®Ø¯Ù…Ø§Ø§ Ù…Ø·Ø§Ø¨Ù‚Ø© Ù„Ù„Ø¨Ø« Ù^Ø§Ù„Ù…ÙŠØ²Ø§Ù†ÙŠØ© Ø§Ù„Ù…Ø¯Ø¯Ø©. Ø¬Ø±Ø¨ Ø§Ù„Ø¨Ø« Ø¨Ø³Ø±Ø± Ø£Ù^ Ø§Ø³Ù… Ù…Ø®ØªÙ„Ù!");
            tvSearchEmptyState.setVisibility(View.VISIBLE);
```

```
            rvSearchResults.setVisibility(View.GONE);
            llSearchResultsHeader.setVisibility(View.VISIBLE);
            String headerText = String.format(Locale.getDefault(), "Ù†ØªØ§Ø¦Ø¬ Ø
§Ù„Ø¨ØØ« Ù„Ù€: %s",
                    query.isEmpty() ? String.format(Locale.getDefault(), "Ù…ÙŠØ²
Ø§Ù†ÙŠØ© $%.0f", targetPrice) : query);
            tvResultsHeaderTitle.setText(headerText);
        } else {
            tvSearchEmptyState.setVisibility(View.GONE);
            rvSearchResults.setVisibility(View.VISIBLE);
            llSearchResultsHeader.setVisibility(View.VISIBLE);
            String headerText = String.format(Locale.getDefault(), "Ù†ØªØ§Ø¦Ø¬ Ø
§Ù„Ø¨ØØ« (%d Ø®Ø¯Ù…Ø© Ù…Ø·Ø§Ø¨Ù‚Ø© Ù‘Ø³Ø¹Ø± Ù…Ù‚Ø±Ø¨):", results.size());
            tvResultsHeaderTitle.setText(headerText);
            adapter.setServices(results, targetPrice);
        }

        // Hide keyboard after search execution
        InputMethodManager imm = (InputMethodManager) getSystemService(Context.I
NPUT_METHOD_SERVICE);
        if (imm != null) {
            imm.hideSoftInputFromWindow(etSearchQuery.getWindowToken(), 0);
        }
    }
}
```

AddServiceActivity.java .6

```
package com.example.work;

import android.content.Context;
import android.content.Intent;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.graphics.Rect;
import android.net.Uri;
import android.os.Bundle;
import android.util.Base64;
import android.util.Patterns;
import android.view.MotionEvent;
```

```
import android.view.View;
import android.view.inputmethod.InputMethodManager;
import android.widget.ArrayAdapter;
import android.widget.AutoCompleteTextView;
import android.widget.EditText;
import android.widget.ImageView;
import android.widget.Toast;

import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;

import com.google.android.material.button.MaterialButton;
import com.google.android.material.card.MaterialCardView;
import com.google.android.material.textfield.TextInputEditText;

import java.io.ByteArrayOutputStream;
import java.io.InputStream;
import java.util.UUID;

public class AddServiceActivity extends AppCompatActivity {

    private TextInputEditText etTitle;
    private AutoCompleteTextView actvCategory;
    private TextInputEditText etPrice;
    private TextInputEditText etDescription;
    private TextInputEditText etProviderName;
    private TextInputEditText etProviderPhone;
    private ImageView ivSelectedServiceImage;
    private View llImagePlaceholder;

    private Uri selectedImageUri = null;

    private final ActivityResultLauncher<String> imagePickerLauncher =
            registerForActivityResult(new ActivityResultContracts.GetContent(),
uri -> {
                if (uri != null) {
                    selectedImageUri = uri;
                    ivSelectedServiceImage.setImageURI(uri);
                    llImagePlaceholder.setVisibility(View.GONE);
                    try {
                        getContentResolver().takePersistableUriPermission(uri, I
ntent.FLAG_GRANT_READ_URI_PERMISSION);
                    } catch (Exception ignored) {
```

```
                    }
                }
            });

    private final String[] categories = new String[]{"Ø¨Ø±Ù…Ø¬Ø©", "Ø³Ø¨Ø§ÙƒØ©", "Ø³Ø±Ø¬Ø©", "Ø³Ø¤Ø§Ù†ÙˆÙ‚", "Ø¹Ø§Ù…Ø©"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_add_service);

        ServiceRepository.getInstance().init(this);

        initViews();
        setupCategoryDropdown();
        setupTouchToHideKeyboard();
    }

    private void initViews() {
        etTitle = findViewById(R.id.etTitle);
        actvCategory = findViewById(R.id.actvCategory);
        etPrice = findViewById(R.id.etPrice);
        etDescription = findViewById(R.id.etDescription);
        etProviderName = findViewById(R.id.etProviderName);
        etProviderPhone = findViewById(R.id.etProviderPhone);
        ivSelectedServiceImage = findViewById(R.id.ivSelectedServiceImage);
        llImagePlaceholder = findViewById(R.id.llImagePlaceholder);

        MaterialCardView cardSelectServiceImage = findViewById(R.id.cardSelectSe
rviceImage);
        cardSelectServiceImage.setOnClickListener(v -> imagePickerLauncher.launc
h("image/*"));

        findViewById(R.id.btnBack).setOnClickListener(v -> finish());

        MaterialButton btnPublish = findViewById(R.id.btnPublish);
        btnPublish.setOnClickListener(v -> publishService());
    }

    private void setupCategoryDropdown() {
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this, android.R.layou
t.simple_dropdown_item_1line, categories);
        actvCategory.setAdapter(adapter);
```

```
    }

    private void setupTouchToHideKeyboard() {
        View root = findViewById(R.id.rootAddServiceLayout);
        View scrollView = findViewById(R.id.scrollViewContainer);

        View.OnClickListener clickListener = v -> hideSoftKeyboard();

        if (root != null) root.setOnClickListener(clickListener);
        if (scrollView != null) scrollView.setOnClickListener(clickListener);
    }

    @Override
    public boolean dispatchTouchEvent(MotionEvent ev) {
        if (ev.getAction() == MotionEvent.ACTION_DOWN) {
            View v = getCurrentFocus();
            if (v instanceof EditText) {
                Rect outRect = new Rect();
                v.getGlobalVisibleRect(outRect);
                if (!outRect.contains((int) ev.getRawX(), (int) ev.getRawY())) {
                    v.clearFocus();
                    hideSoftKeyboard();
                }
            }
        }
        return super.dispatchTouchEvent(ev);
    }

    private void hideSoftKeyboard() {
        View view = this.getCurrentFocus();
        if (view != null) {
            InputMethodManager imm = (InputMethodManager) getSystemService(Conte
xt.INPUT_METHOD_SERVICE);
            if (imm != null) {
                imm.hideSoftInputFromWindow(view.getWindowToken(), 0);
            }
        }
    }

    private boolean isValidContact(String contact) {
        if (contact == null || contact.trim().isEmpty()) {
            return false;
        }
        String trimmed = contact.trim();
```

```
        boolean isEmail = Patterns.EMAIL_ADDRESS.matcher(trimmed).matches();
        boolean isPhone = trimmed.replaceAll("[^0-9]", "").length() >= 7;
        return isEmail || isPhone;
    }

    private int getCategorySampleImage(String category) {
        if (category.contains("Ø¹ØµÙ…ÙŠÙ…")) {
            return R.drawable.img_work_design;
        } else if (category.contains("Ø¨Ø±Ù…Ø¬Ø©")) {
            return R.drawable.img_work_code;
        } else if (category.contains("Ø¹Ø³Ù^ÙŠÙ,")) {
            return R.drawable.img_work_marketing;
        } else if (category.contains("Ø¹Ø±Ø¬Ù…Ø©")) {
            return R.drawable.img_work_translation;
        } else if (category.contains("ØµÙŠØ§Ù†Ø©")) {
            return R.drawable.img_work_maintenance;
        } else {
            return R.drawable.img_work_code;
        }
    }

    private String uriToBase64(Uri uri) {
        if (uri == null) return null;
        try {
            InputStream inputStream = getContentResolver().openInputStream(uri);
            Bitmap bitmap = BitmapFactory.decodeStream(inputStream);
            if (inputStream != null) inputStream.close();
            if (bitmap == null) return null;

            int maxDim = 500;
            int width = bitmap.getWidth();
            int height = bitmap.getHeight();
            if (width > maxDim || height > maxDim) {
                float ratio = Math.min((float) maxDim / width, (float) maxDim /
height);
                width = Math.round(width * ratio);
                height = Math.round(height * ratio);
                bitmap = Bitmap.createScaledBitmap(bitmap, width, height, true);
            }

            ByteArrayOutputStream baos = new ByteArrayOutputStream();
            bitmap.compress(Bitmap.CompressFormat.JPEG, 70, baos);
            byte[] bytes = baos.toByteArray();
            return Base64.encodeToString(bytes, Base64.DEFAULT);
```

```
        } catch (Exception e) {
            return null;
        }
    }

    private void publishService() {
        hideSoftKeyboard();

        String title = etTitle.getText() != null ? etTitle.getText().toString().
trim() : "";
        String category = actvCategory.getText() != null ? actvCategory.getText
().toString().trim() : "";
        String priceStr = etPrice.getText() != null ? etPrice.getText().toString
().trim() : "";
        String description = etDescription.getText() != null ? etDescription.get
Text().toString().trim() : "";
        String providerName = etProviderName.getText() != null ? etProviderName.
getText().toString().trim() : "";
        String providerPhone = etProviderPhone.getText() != null ? etProviderPho
ne.getText().toString().trim() : "";

        if (title.isEmpty() || category.isEmpty() || priceStr.isEmpty() || descr
iption.isEmpty() || providerName.isEmpty() || providerPhone.isEmpty()) {
            Toast.makeText(this, R.string.err_fill_fields, Toast.LENGTH_SHORT).s
how();
            return;
        }

        if (!isValidContact(providerPhone)) {
            Toast.makeText(this, R.string.err_invalid_contact, Toast.LENGTH_LON
G).show();
            return;
        }

        double price;
        try {
            price = Double.parseDouble(priceStr);
        } catch (NumberFormatException e) {
            Toast.makeText(this, "ÙŠØ±Ø¬Ù‰ Ùƒ Ø§Ø§Ø§Ø¨Ø© Ø±Ù‚Ù… Ø³Ø¹Ø± Ø§Ø§Ø§Ø", Toa
st.LENGTH_SHORT).show();
            return;
        }

        int imageResId = getCategorySampleImage(category);
```

```
        String imageUriStr = selectedImageUri != null ? selectedImageUri.toStrin
g() : null;
        String imageBase64Str = selectedImageUri != null ? uriToBase64(selectedI
mageUri) : null;

        ServiceItem newService = new ServiceItem(
                UUID.randomUUID().toString(),
                title,
                category,
                description,
                price,
                providerName,
                providerPhone,
                5.0f,
                1,
                imageResId,
                imageUriStr,
                imageBase64Str
        );

        ServiceRepository.getInstance().addService(newService);
        Toast.makeText(this, R.string.service_added_success, Toast.LENGTH_SHOR
T).show();
        finish();
    }
}
```

ServiceDetailActivity.java .7

```
package com.example.work;

import android.content.Intent;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.net.Uri;
import android.os.Build;
import android.os.Bundle;
import android.text.InputType;
import android.util.Base64;
import android.widget.EditText;
```

```
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

import com.google.android.material.button.MaterialButton;

import java.util.Locale;

public class ServiceDetailActivity extends AppCompatActivity {

    private ImageView ivDetailWorkPhoto;
    private TextView tvDetailCategory;
    private TextView tvDetailRating;
    private TextView tvDetailTitle;
    private TextView tvDetailPrice;
    private TextView tvDetailDescription;
    private TextView tvDetailProviderName;
    private TextView tvDetailProviderPhone;

    private ServiceItem serviceItem;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_service_detail);

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            serviceItem = getIntent().getSerializableExtra("SERVICE_ITEM", Servi
ceItem.class);
        } else {
            @SuppressWarnings("deprecation")
            ServiceItem item = (ServiceItem) getIntent().getSerializableExtra("S
ERVICE_ITEM");
            serviceItem = item;
        }

        if (serviceItem == null) {
            Toast.makeText(this, "Ø¤Ø¹Ø°Ø± Ø¤ØÙ…ÙŠÙ„ Ø¨ÙŠÙ§Ù†Ø§Øª Ø§Ù„Ø®Ø¯Ù…Ø©",
Toast.LENGTH_SHORT).show();
            finish();
            return;
```

```
        }

        initViews();
        populateData();
    }

    private void initViews() {
        ivDetailWorkPhoto = findViewById(R.id.ivDetailWorkPhoto);
        tvDetailCategory = findViewById(R.id.tvDetailCategory);
        tvDetailRating = findViewById(R.id.tvDetailRating);
        tvDetailTitle = findViewById(R.id.tvDetailTitle);
        tvDetailPrice = findViewById(R.id.tvDetailPrice);
        tvDetailDescription = findViewById(R.id.tvDetailDescription);
        tvDetailProviderName = findViewById(R.id.tvDetailProviderName);
        tvDetailProviderPhone = findViewById(R.id.tvDetailProviderPhone);

        findViewById(R.id.btnBack).setOnClickListener(v -> finish());

        MaterialButton btnContactWhatsApp = findViewById(R.id.btnContactWhatsAp
p);
        MaterialButton btnContactPhone = findViewById(R.id.btnContactPhone);
        MaterialButton btnSendDirectMessage = findViewById(R.id.btnSendDirectMes
sage);
        MaterialButton btnOrderNow = findViewById(R.id.btnOrderNow);

        btnContactWhatsApp.setOnClickListener(v -> openWhatsApp());
        btnContactPhone.setOnClickListener(v -> makePhoneCall());
        btnSendDirectMessage.setOnClickListener(v -> openInAppChat());
        btnOrderNow.setOnClickListener(v -> showOrderConfirmation());
    }

    private void populateData() {
        if (ivDetailWorkPhoto != null) {
            if (serviceItem.getImageBase64() != null && !serviceItem.getImageBas
e64().isEmpty()) {
                try {
                    byte[] decodedBytes = Base64.decode(serviceItem.getImageBase
64(), Base64.DEFAULT);
                    Bitmap bitmap = BitmapFactory.decodeByteArray(decodedBytes,
0, decodedBytes.length);
                    if (bitmap != null) {
                        ivDetailWorkPhoto.setImageBitmap(bitmap);
                    } else {
                        ivDetailWorkPhoto.setImageResource(R.drawable.img_work_c
```

```
ode);
                    }
                } catch (Exception e) {
                    ivDetailWorkPhoto.setImageResource(R.drawable.img_work_cod
e);
                }
            } else if (serviceItem.getImageUri() != null && !serviceItem.getImag
eUri().isEmpty()) {
                try {
                    ivDetailWorkPhoto.setImageURI(Uri.parse(serviceItem.getImage
Uri()));
                } catch (Exception e) {
                    ivDetailWorkPhoto.setImageResource(R.drawable.img_work_cod
e);
                }
            } else if (serviceItem.getImageResId() != 0) {
                ivDetailWorkPhoto.setImageResource(serviceItem.getImageResId());
            } else {
                ivDetailWorkPhoto.setImageResource(R.drawable.img_work_code);
            }
        }

        tvDetailCategory.setText(serviceItem.getCategory());
        tvDetailRating.setText(String.format(Locale.getDefault(), "%.1f", servic
eItem.getRating()));
        tvDetailTitle.setText(serviceItem.getTitle());
        tvDetailPrice.setText(String.format(Locale.getDefault(), "$%.0f", servic
eItem.getPrice()));
        tvDetailDescription.setText(serviceItem.getDescription());
        tvDetailProviderName.setText(String.format("Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø©: %s",
serviceItem.getProviderName()));
        tvDetailProviderPhone.setText(String.format("Ù‡Ø³ÙŠÙ„Ø© Ø§Ù„ØªÙˆØ§ØµÙ„:
%s", serviceItem.getProviderPhone()));
    }

    private void openWhatsApp() {
        String phone = serviceItem.getProviderPhone();
        if (phone == null || phone.trim().isEmpty()) {
            Toast.makeText(this, "Ø±Ù‚Ù… Ø§Ù„ØªÙˆØ§ØµÙ„ ØºÙŠØ± Ù…ØªÙˆÙ±", Toas
t.LENGTH_SHORT).show();
            return;
        }

        String digitsOnly = phone.replaceAll("[^0-9]", "");
```

```
        String cleanedPhone = digitsOnly.length() >= 7 ? digitsOnly : "962791234
567";

        try {
            String url = "https://api.whatsapp.com/send?phone=" + cleanedPhone +
                    "&text=" + Uri.encode("Ù…Ø±ÙØ¨Ù‹Ø Ø£Ù‘Ø¯ Ø§Ù„Ø„Ø³Ø³Ø¹Ù³Ø§Ø
± Ø¹Ù† Ø®Ø¯Ù…Ø©Ùƒ: " + serviceItem.getTitle());
            Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse(url));
            startActivity(intent);
        } catch (Exception e) {
            Toast.makeText(this, "Ø§Ø¹Ø°Ø± Ù‘Ø§Ø Ø§Ø·Ø¨Ù‘Ù‘Ù, Ø§Ù„Ù‘Ø§Ø³Ø§Ø§Ø¨ Ù…Ø¨Ø§Ø´Ø±Ø©", Toast.LENGTH_SHORT).show();
        }
    }

    private void makePhoneCall() {
        String phone = serviceItem.getProviderPhone();
        if (phone != null && !phone.trim().isEmpty()) {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse("tel:" + phone.trim()));
            startActivity(intent);
        } else {
            Toast.makeText(this, "Ø±Ù‚Ù… Ø§Ù„Ø„Ø§Ø§Ø„ Ø¹Ù‘Ø± Ù…Ø§Ù‘Ø±", Toas
t.LENGTH_SHORT).show();
        }
    }

    private void openInAppChat() {
        Intent intent = new Intent(ServiceDetailActivity.this, ChatActivity.clas
s);
        intent.putExtra("CHAT_ID", serviceItem.getId());
        intent.putExtra("RECEIVER_NAME", serviceItem.getProviderName());
        intent.putExtra("SERVICE_TITLE", serviceItem.getTitle());
        intent.putExtra("CURRENT_ROLE", "client");
        intent.putExtra("CURRENT_USER_NAME", "Ø¹Ù…Ù„Ø„ Ù…Ù‡Ø§Ù…");
        startActivity(intent);
    }

    private void showOrderConfirmation() {
        android.widget.LinearLayout layout = new android.widget.LinearLayout(thi
s);
        layout.setOrientation(android.widget.LinearLayout.VERTICAL);
        layout.setPadding(48, 24, 48, 24);
```

```
        final EditText inputName = new EditText(this);
        inputName.setHint("Ø§Ø¯Ø®Ù„ Ø§Ø³Ù…Ùƒ Ø§Ù„ÙƒØ§Ù…Ù„ (Ù…Ø«Ø§Ù„: Ø£ØÙ…Ø¯)");
        inputName.setInputType(InputType.TYPE_CLASS_TEXT | InputType.TYPE_TEXT_F
LAG_CAP_WORDS);
        layout.addView(inputName);

        final EditText inputPhone = new EditText(this);
        inputPhone.setHint("Ø§Ø¯Ø®Ù„ Ø±Ù‚Ù… Ù‡Ø§ØªÙƒ/Ø§Ù„Ù‘Ø§Ø³Ø§Ø¨ (Ù…Ø«Ø§Ù
„: 0791234567)");
        inputPhone.setInputType(InputType.TYPE_CLASS_PHONE);
        inputPhone.setPadding(0, 32, 0, 0);
        layout.addView(inputPhone);

        AlertDialog dialog = new AlertDialog.Builder(this)
                .setTitle("Ø§Ø£Ùƒ Ø·Ù„Ø¨ Ø§Ù„Ø®Ø¯Ù…Ø© ðŸš€")
                .setMessage(String.format(Locale.getDefault(), "Ø®Ø¯Ù…Ø©: %s\nØ§
Ù„Ø³Ø¹Ø±: $%.0f Ø¯Ù‘Ù‘Ù„Ø§Ø±\n\nÙŠØ±Ø¬Ù‰ Ø¥Ø¯Ø®Ø§Ù„ Ø§Ø³Ù…Ùƒ Ù
„Ù„Ø¹Ø¯Ø¡ Ù‘Ø§Ù„Ø¹Ø§Ù‘Ø§Ø¬Ù„ Ù…Ø¹ %s:",
                        serviceItem.getTitle(),
                        serviceItem.getPrice(),
                        serviceItem.getProviderName()))
                .setView(layout)
                .setPositiveButton("Ø¥Ø±Ø³Ø§Ù„ Ø§Ù„Ø·Ù„Ø¨ ðŸš€", null)
                .setNegativeButton("Ø¥Ù„Øºø§Ø¡", null)
                .create();

        dialog.setOnShowListener(dialogInterface -> {
            dialog.getButton(AlertDialog.BUTTON_POSITIVE).setOnClickListener(v -
> {
                String clientName = inputName.getText().toString().trim();
                String clientPhone = inputPhone.getText().toString().trim();

                if (clientName.isEmpty()) {
                    inputName.setError("ÙŠØ±Ø¬Ù‰ Ø¥Ø¯Ø®Ø§Ù„ Ø§Ù„Ø§Ø³Ù… Ø§Ù„Ùƒø§
Ù…Ù„");
                    return;
                }

                if (clientPhone.isEmpty() || clientPhone.length() < 7) {
                    inputPhone.setError("ÙŠØ±Ø¬Ù‰ Ø¥Ø¯Ø®Ø§Ù„ Ø±Ù‚Ù Ø
ŠØ");
                    return;
                }
```

```
                OrderRepository.getInstance().placeOrder(serviceItem, clientNam
e, "", clientPhone);
                Toast.makeText(ServiceDetailActivity.this, "Ø¹Ù… Ø¥Ø±Ø³Ø§Ù„ Ø·Ù„Ø¨Ùƒ Ø¨Ù†Ø§Ø! Ø³ÙŠØµÙ„ Ù„Ù„Ù…Ø´Ø±Ù Ù‚Ø±ÙŠØ¨Ø§Ù‹.", Toast.LENGTH_LONG).show();
                dialog.dismiss();
            });
        });

        dialog.show();
    }
}
```

ChatActivity.java .8

```
package com.example.work;

import android.os.Bundle;
import android.util.Log;
import android.view.View;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.google.android.material.button.MaterialButton;
import com.google.android.material.textfield.TextInputEditText;
import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;

import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Date;
import java.util.List;
```

```
import java.util.Locale;
import java.util.UUID;

public class ChatActivity extends AppCompatActivity {

    private static final String TAG = "ChatActivity";

    private TextView tvChatReceiverName;
    private TextView tvChatServiceTitle;
    private RecyclerView rvChatMessages;
    private TextInputEditText etChatMessageInput;
    private MaterialButton btnSendChatMessage;

    private MessageAdapter messageAdapter;
    private DatabaseReference chatRef;

    private String chatId;
    private String receiverName;
    private String serviceTitle;
    private String currentRole = "client";
    private String currentUserName = "Ø¹Ù…ÙŠÙ„";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_chat);

        chatId = getIntent().getStringExtra("CHAT_ID");
        receiverName = getIntent().getStringExtra("RECEIVER_NAME");
        serviceTitle = getIntent().getStringExtra("SERVICE_TITLE");
        currentRole = getIntent().getStringExtra("CURRENT_ROLE");
        currentUserName = getIntent().getStringExtra("CURRENT_USER_NAME");

        if (chatId == null || chatId.trim().isEmpty()) {
            Toast.makeText(this, "ØªØ¹Ø°Ø± ÙØªØ Ø§Ù„Ù…Ø§Ø¯Ø«Ø© Ù…Ø¹Ù±Ù Ø§Ù„Ù…Ø§Ø¯Ø«Ø© ØºÙŠØ± ØµØ§Ù„ØØ©", Toast.LENGTH_SHORT).show();
            finish();
            return;
        }

        if (currentRole == null) currentRole = "client";
        if (currentUserName == null) currentUserName = currentRole.equals("client") ? "Ø§Ù„Ø¹ÙŠÙ„" : "Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø©";
```

```
        initViews();
        setupRecyclerView();
        connectFirebaseChat();
    }

    private void initViews() {
        tvChatReceiverName = findViewById(R.id.tvChatReceiverName);
        tvChatServiceTitle = findViewById(R.id.tvChatServiceTitle);
        rvChatMessages = findViewById(R.id.rvChatMessages);
        etChatMessageInput = findViewById(R.id.etChatMessageInput);
        btnSendChatMessage = findViewById(R.id.btnSendChatMessage);

        findViewById(R.id.btnBack).setOnClickListener(v -> finish());

        if (tvChatReceiverName != null && receiverName != null) {
            tvChatReceiverName.setText(receiverName);
        }
        if (tvChatServiceTitle != null && serviceTitle != null) {
            tvChatServiceTitle.setText("Ø¨Ø´Ø£Ù†: " + serviceTitle);
        }

        btnSendChatMessage.setOnClickListener(v -> sendMessage());
    }

    private void setupRecyclerView() {
        LinearLayoutManager layoutManager = new LinearLayoutManager(this);
        layoutManager.setStackFromEnd(true);
        rvChatMessages.setLayoutManager(layoutManager);

        messageAdapter = new MessageAdapter(currentRole);
        rvChatMessages.setAdapter(messageAdapter);
    }

    private void connectFirebaseChat() {
        FirebaseHelper.ensureAuthenticated(() -> {
            try {
                FirebaseDatabase db = FirebaseHelper.getDatabase();
                if (db != null) {
                    chatRef = db.getReference("chats").child(chatId).child("mess
ages");
                    listenToMessages();
                }
            } catch (Exception e) {
                Log.e(TAG, "Chat Firebase error: " + e.getMessage());
```

```
            }
        });
    }

    private void listenToMessages() {
        if (chatRef == null) return;

        chatRef.addValueEventListener(new ValueEventListener() {
            @Override
            public void onDataChange(@NonNull DataSnapshot snapshot) {
                List<ChatMessage> list = new ArrayList<>();
                for (DataSnapshot child : snapshot.getChildren()) {
                    ChatMessage msg = child.getValue(ChatMessage.class);
                    if (msg != null) {
                        list.add(msg);
                    }
                }

                list.sort((o1, o2) -> Long.compare(o1.getTimestamp(), o2.getTime
stamp()));

                runOnUiThread(() -> {
                    messageAdapter.setMessages(list);
                    if (!list.isEmpty()) {
                        rvChatMessages.smoothScrollToPosition(list.size() - 1);
                    }
                });
            }

            @Override
            public void onCancelled(@NonNull DatabaseError error) {
                Log.e(TAG, "Chat cancelled: " + error.getMessage());
            }
        });
    }

    private void sendMessage() {
        if (etChatMessageInput == null) return;

        String text = etChatMessageInput.getText() != null ? etChatMessageInput.
getText().toString().trim() : "";
        if (text.isEmpty()) return;

        String msgId = UUID.randomUUID().toString();
```

```
        long now = System.currentTimeMillis();
        String timeFormatted = new SimpleDateFormat("hh:mm a", Locale.getDefault
()).format(new Date(now));

        ChatMessage message = new ChatMessage(
                msgId,
                chatId,
                currentUserName,
                currentRole,
                text,
                now,
                timeFormatted
        );

        etChatMessageInput.setText("");

        FirebaseHelper.ensureAuthenticated(() -> {
            try {
                if (chatRef == null) {
                    FirebaseDatabase db = FirebaseHelper.getDatabase();
                    if (db != null) {
                        chatRef = db.getReference("chats").child(chatId).child
("messages");
                    }
                }
                if (chatRef != null) {
                    chatRef.child(msgId).setValue(message, (error, ref) -> {
                        if (error != null) {
                            Log.e(TAG, "Failed sending message: " + error.getMes
sage());
                            Toast.makeText(ChatActivity.this, "ÙØ´Ù„ Ø¥Ø±Ø³Ø§Ù„
Ø§Ù„Ø±Ø³Ø§Ù„Ø©: " + error.getMessage(), Toast.LENGTH_SHORT).show();
                        } else {
                            Log.d(TAG, "Message sent successfully!");
                        }
                    });
                }
            } catch (Exception e) {
                Log.e(TAG, "Send message exception: " + e.getMessage());
            }
        });
    }
}
```

ServiceRepository.java .9

```
package com.example.work;

import android.content.Context;
import android.os.Handler;
import android.os.Looper;
import android.util.Log;
import android.widget.Toast;

import androidx.annotation.NonNull;

import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;

import java.util.ArrayList;
import java.util.LinkedHashMap;
import java.util.List;
import java.util.Map;

public class ServiceRepository {

    public interface OnDataChangeListener {
        void onDataChanged();
    }

    private static final String TAG = "ServiceRepository";

    private static ServiceRepository instance;
    private final Map<String, ServiceItem> servicesMap = new LinkedHashMap<>();
    private final List<OnDataChangeListener> listeners = new ArrayList<>();
    private DatabaseReference databaseRef;
    private Context appContext;

    private ServiceRepository() {
    }

    public static synchronized ServiceRepository getInstance() {
```

```
        if (instance == null) {
            instance = new ServiceRepository();
        }
        return instance;
    }

    public void init(Context context) {
        if (context != null) {
            this.appContext = context.getApplicationContext();
        }
        FirebaseHelper.ensureAuthenticated(() -> {
            try {
                FirebaseDatabase db = FirebaseHelper.getDatabase();
                if (db != null) {
                    databaseRef = db.getReference("services");
                    attachFirebaseListener();
                }
            } catch (Exception e) {
                Log.e(TAG, "Firebase init error: " + e.getMessage());
            }
        });
    }

    public void addOnDataChangeListener(OnDataChangeListener listener) {
        if (listener != null && !listeners.contains(listener)) {
            listeners.add(listener);
        }
    }

    public void removeOnDataChangeListener(OnDataChangeListener listener) {
        if (listener != null) {
            listeners.remove(listener);
        }
    }

    private void notifyListeners() {
        for (OnDataChangeListener listener : new ArrayList<>(listeners)) {
            if (listener != null) {
                listener.onDataChanged();
            }
        }
    }

    private void attachFirebaseListener() {
```

```
        if (databaseRef == null) return;

        databaseRef.addValueEventListener(new ValueEventListener() {
            @Override
            public void onDataChange(@NonNull DataSnapshot snapshot) {
                Log.d(TAG, "onDataChange: received " + snapshot.getChildrenCount
() + " items from Firebase");
                synchronized (ServiceRepository.this) {
                    for (DataSnapshot child : snapshot.getChildren()) {
                        ServiceItem item = child.getValue(ServiceItem.class);
                        if (item != null && item.getId() != null) {
                            servicesMap.put(item.getId(), item);
                        }
                    }
                }
                notifyListeners();
            }

            @Override
            public void onCancelled(@NonNull DatabaseError error) {
                Log.e(TAG, "Firebase cancelled/error: " + error.getMessage() + "
(code: " + error.getCode() + ")");
                if (appContext != null) {
                    new Handler(Looper.getMainLooper()).post(() ->
                            Toast.makeText(appContext, "Ø®Ø·Ø£ Ø§Ù„Ø§ØªØµØ±Ø¨ÙŠØ
³: " + error.getMessage(), Toast.LENGTH_LONG).show()
                    );
                }
            }
        });
    }

    public synchronized List<ServiceItem> getAllServices() {
        return new ArrayList<>(servicesMap.values());
    }

    public void addService(ServiceItem service) {
        if (service == null || service.getId() == null) return;

        // Immediately store in local memory for instant UI display
        synchronized (this) {
            servicesMap.put(service.getId(), service);
        }
        notifyListeners();
```

```
        // Push to Firebase Realtime Database
        FirebaseHelper.ensureAuthenticated(() -> {
            try {
                if (databaseRef == null) {
                    FirebaseDatabase db = FirebaseHelper.getDatabase();
                    if (db != null) {
                        databaseRef = db.getReference("services");
                    }
                }
                if (databaseRef != null) {
                    databaseRef.child(service.getId()).setValue(service, (error,
ref) -> {
                        if (error != null) {
                            Log.e(TAG, "Firebase setValue failed: " + error.getM
essage() + " (code " + error.getCode() + ")");
                            if (appContext != null) {
                                new Handler(Looper.getMainLooper()).post(() ->
                                        Toast.makeText(appContext, "ØªÙ†Ø¨ÙŠÙ‡ Ø
§Ù„Ø§ØªØµØ¨ÙŠØ³: " + error.getMessage(), Toast.LENGTH_LONG).show()
                                );
                            }
                        } else {
                            Log.d(TAG, "Firebase setValue succeeded for service:
" + service.getTitle());
                            if (appContext != null) {
                                new Handler(Looper.getMainLooper()).post(() ->
                                        Toast.makeText(appContext, "ØªÙ… ØªØ¸ Ø§Ù
„Ø®Ø¯Ù…Ø© Ø¨Ù†Ø¬Ø§Ø ÙŠÙŠ Ø§Ù„Ø³Ø¬Ù„ â˜ï¸", Toast.LENGTH_SHORT).show()
                                );
                            }
                        }
                    });
                }
            } catch (Exception e) {
                Log.e(TAG, "Failed pushing service to Firebase: " + e.getMessage
());
            }
        });
    }

    public synchronized List<ServiceItem> searchServices(String query, double ta
rgetPrice, String category) {
        List<ServiceItem> filtered = new ArrayList<>();
```

```
        String q = query != null ? query.trim().toLowerCase() : "";

        for (ServiceItem item : servicesMap.values()) {
            boolean matchesQuery = q.isEmpty() ||
                    (item.getTitle() != null && item.getTitle().toLowerCase().co
ntains(q)) ||
                    (item.getDescription() != null && item.getDescription().toLo
werCase().contains(q)) ||
                    (item.getCategory() != null && item.getCategory().toLowerCas
e().contains(q));

            boolean matchesCategory = category == null ||
                    category.isEmpty() ||
                    category.equalsIgnoreCase("Ø§Ù„Ùƒ„") ||
                    (item.getCategory() != null && item.getCategory().equalsIgno
reCase(category));

            boolean matchesPriceRange;
            if (targetPrice > 0) {
                double diff = Math.abs(item.getPrice() - targetPrice);
                double maxAllowedDiff = Math.max(25.0, targetPrice * 0.40);
                matchesPriceRange = diff <= maxAllowedDiff;
            } else {
                matchesPriceRange = true;
            }

            if (matchesQuery && matchesCategory && matchesPriceRange) {
                filtered.add(item);
            }
        }

        if (targetPrice > 0) {
            filtered.sort((o1, o2) -> {
                double diff1 = Math.abs(o1.getPrice() - targetPrice);
                double diff2 = Math.abs(o2.getPrice() - targetPrice);
                return Double.compare(diff1, diff2);
            });
        }

        return filtered;
    }
}
```

OrderRepository.java .10

```
package com.example.work;

import android.content.Context;
import android.os.Handler;
import android.os.Looper;
import android.util.Log;
import android.widget.Toast;

import androidx.annotation.NonNull;

import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;

import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Date;
import java.util.LinkedHashMap;
import java.util.List;
import java.util.Locale;
import java.util.Map;
import java.util.UUID;

public class OrderRepository {

    public interface OnOrdersChangeListener {
        void onOrdersChanged();
    }

    public static final String STATUS_PENDING = "ÙŠÙ† Ø§Ù†ØªØ¸Ø§Ø± Ù…Ùˆ Ù‚Ø§Ù„Ù Ø§Ù„Ù…Ø®Ø²Ù…Ù© â³";
    public static final String STATUS_ACCEPTED = "ØªÙ… Ù‚Ø¨Ù„ Ø§Ù„Ø·Ù„Ø¨ - Ù‚Ø¯ Ø§Ù„Ø§Ù†Ø·Ù„Ø§Ù‚ âš™ï¸";
    public static final String STATUS_REJECTED = "ØªÙ… Ø±Ø¶ Ø§Ù„Ø·Ù„Ø¨ âŒ";
    public static final String STATUS_COMPLETED = "Ø®Ø¯Ù…Ø© Ù…ÙƒØªÙ…Ù„Ø© Ø¨Ù†Ø¬Ø§Ø â…";
```

```
    private static final String TAG = "OrderRepository";

    private static OrderRepository instance;
    private final Map<String, OrderItem> ordersMap = new LinkedHashMap<>();
    private final List<OnOrdersChangeListener> listeners = new ArrayList<>();
    private DatabaseReference databaseRef;
    private Context appContext;

    private OrderRepository() {
    }

    public static synchronized OrderRepository getInstance() {
        if (instance == null) {
            instance = new OrderRepository();
        }
        return instance;
    }

    public void init(Context context) {
        if (context != null) {
            this.appContext = context.getApplicationContext();
        }
        FirebaseHelper.ensureAuthenticated(() -> {
            try {
                FirebaseDatabase db = FirebaseHelper.getDatabase();
                if (db != null) {
                    databaseRef = db.getReference("orders");
                    attachFirebaseListener();
                }
            } catch (Exception e) {
                Log.e(TAG, "Firebase init error: " + e.getMessage());
            }
        });
    }

    public void addOnOrdersChangeListener(OnOrdersChangeListener listener) {
        if (listener != null && !listeners.contains(listener)) {
            listeners.add(listener);
        }
    }

    public void removeOnOrdersChangeListener(OnOrdersChangeListener listener) {
        if (listener != null) {
            listeners.remove(listener);
```

```
        }
    }

    private void notifyListeners() {
        for (OnOrdersChangeListener listener : new ArrayList<>(listeners)) {
            if (listener != null) {
                listener.onOrdersChanged();
            }
        }
    }

    private void attachFirebaseListener() {
        if (databaseRef == null) return;

        databaseRef.addValueEventListener(new ValueEventListener() {
            @Override
            public void onDataChange(@NonNull DataSnapshot snapshot) {
                Log.d(TAG, "onDataChange: received " + snapshot.getChildrenCount
() + " orders from Firebase");
                synchronized (OrderRepository.this) {
                    for (DataSnapshot child : snapshot.getChildren()) {
                        OrderItem item = child.getValue(OrderItem.class);
                        if (item != null && item.getOrderId() != null) {
                            ordersMap.put(item.getOrderId(), item);
                        }
                    }
                }
                notifyListeners();
            }

            @Override
            public void onCancelled(@NonNull DatabaseError error) {
                Log.e(TAG, "Firebase cancelled/error: " + error.getMessage());
                if (appContext != null) {
                    new Handler(Looper.getMainLooper()).post(() ->
                            Toast.makeText(appContext, "Ø®Ø·Ø£ Ø§Ù„Ù§ÙŠØ±Ø¨ÙŠØ
³: " + error.getMessage(), Toast.LENGTH_LONG).show()
                    );
                }
            }
        });
    }

    public synchronized List<OrderItem> getAllOrders() {
```

```
        return new ArrayList<>(ordersMap.values());
    }

    public void placeOrder(ServiceItem service, String clientName, String client
Email) {
        placeOrder(service, clientName, clientEmail, null);
    }

    public void placeOrder(ServiceItem service, String clientName, String client
Email, String clientPhone) {
        String name = clientName != null && !clientName.trim().isEmpty() ? clien
tName.trim() : "Ø¹Ù…ÙŠÙ„ Ù…Ø³Ø¬Ù„";
        String email = clientEmail != null && !clientEmail.trim().isEmpty() ? cl
ientEmail.trim() : "client@example.com";
        String phone = clientPhone != null && !clientPhone.trim().isEmpty() ? cl
ientPhone.trim() : "";

        OrderItem order = new OrderItem(
                UUID.randomUUID().toString(),
                service,
                name,
                email,
                phone,
                getCurrentFormattedDate(),
                STATUS_PENDING
        );

        synchronized (this) {
            ordersMap.put(order.getOrderId(), order);
        }
        notifyListeners();

        FirebaseHelper.ensureAuthenticated(() -> {
            try {
                if (databaseRef == null) {
                    FirebaseDatabase db = FirebaseHelper.getDatabase();
                    if (db != null) {
                        databaseRef = db.getReference("orders");
                    }
                }
                if (databaseRef != null) {
                    databaseRef.child(order.getOrderId()).setValue(order, (erro
r, ref) -> {
                        if (error != null) {
```

```
                            Log.e(TAG, "Firebase setValue order failed: " + erro
r.getMessage());
                            if (appContext != null) {
                                new Handler(Looper.getMainLooper()).post(() ->
                                        Toast.makeText(appContext, "Ø³Ù‡Ø¨ÙŠÙŠ Ø
§Ù„Ø§Ù†ØªØ±Ù†Øª: " + error.getMessage(), Toast.LENGTH_LONG).show()
                                );
                            }
                        } else {
                            Log.d(TAG, "Firebase setValue order succeeded!");
                        }
                    });
                }
            } catch (Exception e) {
                Log.e(TAG, "Failed pushing order to Firebase: " + e.getMessage
());
            }
        });
    }

    public void updateOrderStatus(String orderId, String newStatus) {
        if (orderId == null || newStatus == null) return;

        synchronized (this) {
            OrderItem order = ordersMap.get(orderId);
            if (order != null) {
                order.setStatus(newStatus);
            }
        }
        notifyListeners();

        FirebaseHelper.ensureAuthenticated(() -> {
            try {
                if (databaseRef == null) {
                    FirebaseDatabase db = FirebaseHelper.getDatabase();
                    if (db != null) {
                        databaseRef = db.getReference("orders");
                    }
                }
                if (databaseRef != null) {
                    databaseRef.child(orderId).child("status").setValue(newStatu
s, (error, ref) -> {
                        if (error != null) {
                            Log.e(TAG, "Firebase updateOrderStatus failed: " + e
```

```
rror.getMessage());
                            if (appContext != null) {
                                new Handler(Looper.getMainLooper()).post(() ->
                                        Toast.makeText(appContext, "Ø¤Ù†Ø¨ÙŠÙ‡ Ø
§Ù„Ù„Ø§ÙŠØ±Ø¨ÙŠØ³: " + error.getMessage(), Toast.LENGTH_LONG).show()
                                );
                            }
                        } else {
                            Log.d(TAG, "Firebase updateOrderStatus succeeded to:
" + newStatus);
                        }
                    });
                }
            } catch (Exception e) {
                Log.e(TAG, "Failed updating order status in Firebase: " + e.getM
essage());
            }
        });
    }

    private String getCurrentFormattedDate() {
        SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd", Locale.getDefa
ult());
        return sdf.format(new Date());
    }
}
```

FirebaseHelper.java .11

```
package com.example.work;

import android.util.Log;

import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.database.FirebaseDatabase;

public class FirebaseHelper {
    private static final String TAG = "FirebaseHelper";
    private static final String DB_URL_US = "https://work-4451a-default-rtdb.fir
ebaseio.com";
```

```
    private static final String DB_URL_EU = "https://work-4451a-default-rtdb.eur
ope-west1.firebasedatabase.app";
    private static final String DB_URL_ASIA = "https://work-4451a-default-rtdb.a
sia-southeast1.firebasedatabase.app";
    private static final String DB_URL_ALT = "https://work-4451a.firebaseio.co
m";

    private static FirebaseDatabase databaseInstance;

    public interface OnAuthCompleteListener {
        void onAuthSuccess();
    }

    public static synchronized FirebaseDatabase getDatabase() {
        if (databaseInstance == null) {
            String[] urls = new String[]{DB_URL_US, DB_URL_EU, DB_URL_ASIA, DB_U
RL_ALT};
            for (String url : urls) {
                try {
                    FirebaseDatabase db = FirebaseDatabase.getInstance(url);
                    if (db != null) {
                        databaseInstance = db;
                        break;
                    }
                } catch (Exception ignored) {
                }
            }

            if (databaseInstance == null) {
                try {
                    databaseInstance = FirebaseDatabase.getInstance();
                } catch (Exception e) {
                    Log.e(TAG, "Error getting default FirebaseDatabase instance:
" + e.getMessage());
                }
            }

            if (databaseInstance != null) {
                try {
                    databaseInstance.setPersistenceEnabled(true);
                } catch (Exception ignored) {
                }
            }
        }
```

```
        return databaseInstance;
    }

    public static void ensureAuthenticated(OnAuthCompleteListener listener) {
        try {
            FirebaseAuth auth = FirebaseAuth.getInstance();
            if (auth.getCurrentUser() != null) {
                if (listener != null) listener.onAuthSuccess();
                return;
            }

            auth.signInAnonymously().addOnCompleteListener(task -> {
                if (task.isSuccessful()) {
                    Log.d(TAG, "Signed in anonymously to Firebase Auth successfu
lly!");
                } else {
                    Log.e(TAG, "Anonymous sign in failed: " + (task.getException
() != null ? task.getException().getMessage() : "Unknown"));
                }
                if (listener != null) listener.onAuthSuccess();
            });
        } catch (Exception e) {
            Log.e(TAG, "Auth error: " + e.getMessage());
            if (listener != null) listener.onAuthSuccess();
        }
    }
}
```

SearchHistoryManager.java .12

```
package com.example.work;

import android.content.Context;
import android.content.SharedPreferences;

import java.util.ArrayList;
import java.util.HashSet;
import java.util.List;
import java.util.Set;
```

```
public class SearchHistoryManager {

    private static final String PREF_NAME = "work_search_history";
    private static final String KEY_HISTORY = "history_items";

    public static void addSearchQuery(Context context, String query, double budg
et) {
        String entry;
        String trimmed = query != null ? query.trim() : "";
        if (trimmed.isEmpty()) {
            if (budget <= 0) return;
            entry = "Ù…ÙŠØ²Ø§Ù†ÙŠØ© Ø¨Ù€ $" + ((int) budget);
        } else if (budget > 0) {
            entry = trimmed + " (Ø¨Ø¯Ù‘Ø¯ $" + ((int) budget) + ")";
        } else {
            entry = trimmed;
        }

        SharedPreferences prefs = context.getSharedPreferences(PREF_NAME, Contex
t.MODE_PRIVATE);
        Set<String> set = prefs.getStringSet(KEY_HISTORY, new HashSet<>());
        Set<String> newSet = new HashSet<>(set);
        newSet.add(entry);

        prefs.edit().putStringSet(KEY_HISTORY, newSet).apply();
    }

    public static List<String> getSearchHistory(Context context) {
        SharedPreferences prefs = context.getSharedPreferences(PREF_NAME, Contex
t.MODE_PRIVATE);
        Set<String> set = prefs.getStringSet(KEY_HISTORY, new HashSet<>());
        return new ArrayList<>(set);
    }

    public static void clearHistory(Context context) {
        SharedPreferences prefs = context.getSharedPreferences(PREF_NAME, Contex
t.MODE_PRIVATE);
        prefs.edit().remove(KEY_HISTORY).apply();
    }
}
```

ServiceItem.java .13

```
package com.example.work;

import org.json.JSONObject;

import java.io.Serializable;

public class ServiceItem implements Serializable {
    private String id;
    private String title;
    private String category;
    private String description;
    private double price;
    private String providerName;
    private String providerPhone;
    private float rating;
    private int reviewsCount;
    private int imageResId;
    private String imageUri;
    private String imageBase64;

    // Default constructor required for Firebase Realtime Database
    public ServiceItem() {
    }

    public ServiceItem(String id, String title, String category, String descript
ion, double price, String providerName, String providerPhone, float rating, int
reviewsCount, int imageResId, String imageUri, String imageBase64) {
        this.id = id;
        this.title = title;
        this.category = category;
        this.description = description;
        this.price = price;
        this.providerName = providerName;
        this.providerPhone = providerPhone;
        this.rating = rating;
        this.reviewsCount = reviewsCount;
        this.imageResId = imageResId;
        this.imageUri = imageUri;
        this.imageBase64 = imageBase64;
```

```
    }

    public ServiceItem(String id, String title, String category, String descript
ion, double price, String providerName, String providerPhone, float rating, int
reviewsCount, int imageResId, String imageUri) {
        this(id, title, category, description, price, providerName, providerPhon
e, rating, reviewsCount, imageResId, imageUri, null);
    }

    public ServiceItem(String id, String title, String category, String descript
ion, double price, String providerName, String providerPhone, float rating, int
reviewsCount, int imageResId) {
        this(id, title, category, description, price, providerName, providerPhon
e, rating, reviewsCount, imageResId, null, null);
    }

    public ServiceItem(String id, String title, String category, String descript
ion, double price, String providerName, String providerPhone, float rating, int
reviewsCount) {
        this(id, title, category, description, price, providerName, providerPhon
e, rating, reviewsCount, R.drawable.img_work_code, null, null);
    }

    public JSONObject toJsonObject() {
        JSONObject json = new JSONObject();
        try {
            json.put("id", id);
            json.put("title", title);
            json.put("category", category);
            json.put("description", description);
            json.put("price", price);
            json.put("providerName", providerName);
            json.put("providerPhone", providerPhone);
            json.put("rating", rating);
            json.put("reviewsCount", reviewsCount);
            json.put("imageResId", imageResId);
            if (imageUri != null) json.put("imageUri", imageUri);
            if (imageBase64 != null) json.put("imageBase64", imageBase64);
        } catch (Exception e) {
            e.printStackTrace();
        }
        return json;
    }
```

```
    public static ServiceItem fromJsonObject(JSONObject json) {
        if (json == null) return null;
        return new ServiceItem(
                json.optString("id", ""),
                json.optString("title", ""),
                json.optString("category", ""),
                json.optString("description", ""),
                json.optDouble("price", 0.0),
                json.optString("providerName", ""),
                json.optString("providerPhone", ""),
                (float) json.optDouble("rating", 5.0),
                json.optInt("reviewsCount", 1),
                json.optInt("imageResId", R.drawable.img_work_code),
                json.has("imageUri") ? json.optString("imageUri", null) : null,
                json.has("imageBase64") ? json.optString("imageBase64", null) :
null
        );
    }

    public String getId() {
        return id;
    }

    public void setId(String id) {
        this.id = id;
    }

    public String getTitle() {
        return title;
    }

    public void setTitle(String title) {
        this.title = title;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public String getDescription() {
```

```
        return description;
    }

    public void setDescription(String description) {
        this.description = description;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }

    public String getProviderName() {
        return providerName;
    }

    public void setProviderName(String providerName) {
        this.providerName = providerName;
    }

    public String getProviderPhone() {
        return providerPhone;
    }

    public void setProviderPhone(String providerPhone) {
        this.providerPhone = providerPhone;
    }

    public float getRating() {
        return rating;
    }

    public void setRating(float rating) {
        this.rating = rating;
    }

    public int getReviewsCount() {
        return reviewsCount;
    }

    public void setReviewsCount(int reviewsCount) {
```

```
        this.reviewsCount = reviewsCount;
    }

    public int getImageResId() {
        return imageResId;
    }

    public void setImageResId(int imageResId) {
        this.imageResId = imageResId;
    }

    public String getImageUri() {
        return imageUri;
    }

    public void setImageUri(String imageUri) {
        this.imageUri = imageUri;
    }

    public String getImageBase64() {
        return imageBase64;
    }

    public void setImageBase64(String imageBase64) {
        this.imageBase64 = imageBase64;
    }
}
```

OrderItem.java .14

```
package com.example.work;

import org.json.JSONObject;

import java.io.Serializable;

public class OrderItem implements Serializable {
    private String orderId;
    private ServiceItem serviceItem;
    private String clientName;
```

```
    private String clientEmail;
    private String clientPhone;
    private String orderDate;
    private String status;

    public OrderItem() {
    }

    public OrderItem(String orderId, ServiceItem serviceItem, String clientName,
String clientEmail, String clientPhone, String orderDate, String status) {
        this.orderId = orderId;
        this.serviceItem = serviceItem;
        this.clientName = clientName;
        this.clientEmail = clientEmail;
        this.clientPhone = clientPhone;
        this.orderDate = orderDate;
        this.status = status;
    }

    public OrderItem(String orderId, ServiceItem serviceItem, String clientName,
String clientEmail, String orderDate, String status) {
        this(orderId, serviceItem, clientName, clientEmail, null, orderDate, sta
tus);
    }

    public JSONObject toJsonObject() {
        JSONObject json = new JSONObject();
        try {
            json.put("orderId", orderId);
            if (serviceItem != null) {
                json.put("serviceItem", serviceItem.toJsonObject());
            }
            json.put("clientName", clientName);
            json.put("clientEmail", clientEmail);
            json.put("clientPhone", clientPhone);
            json.put("orderDate", orderDate);
            json.put("status", status);
        } catch (Exception e) {
            e.printStackTrace();
        }
        return json;
    }

    public static OrderItem fromJsonObject(JSONObject json) {
```

```
        if (json == null) return null;
        ServiceItem service = null;
        if (json.has("serviceItem")) {
            service = ServiceItem.fromJsonObject(json.optJSONObject("serviceIte
m"));
        }
        return new OrderItem(
                json.optString("orderId", ""),
                service,
                json.optString("clientName", ""),
                json.optString("clientEmail", ""),
                json.optString("clientPhone", ""),
                json.optString("orderDate", ""),
                json.optString("status", "")
        );
    }

    public String getOrderId() {
        return orderId;
    }

    public void setOrderId(String orderId) {
        this.orderId = orderId;
    }

    public ServiceItem getServiceItem() {
        return serviceItem;
    }

    public void setServiceItem(ServiceItem serviceItem) {
        this.serviceItem = serviceItem;
    }

    public String getClientName() {
        return clientName;
    }

    public void setClientName(String clientName) {
        this.clientName = clientName;
    }

    public String getClientEmail() {
        return clientEmail;
    }
```

```
    public void setClientEmail(String clientEmail) {
        this.clientEmail = clientEmail;
    }

    public String getClientPhone() {
        return clientPhone;
    }

    public void setClientPhone(String clientPhone) {
        this.clientPhone = clientPhone;
    }

    public String getOrderDate() {
        return orderDate;
    }

    public void setOrderDate(String orderDate) {
        this.orderDate = orderDate;
    }

    public String getStatus() {
        return status;
    }

    public void setStatus(String status) {
        this.status = status;
    }
}
```

ChatMessage.java .15

```
package com.example.work;

import java.io.Serializable;

public class ChatMessage implements Serializable {
    private String messageId;
    private String chatId;
    private String senderName;
```

```
    private String senderRole; // "client" or "provider"
    private String text;
    private long timestamp;
    private String timeFormatted;

    public ChatMessage() {
    }

    public ChatMessage(String messageId, String chatId, String senderName, Strin
g senderRole, String text, long timestamp, String timeFormatted) {
        this.messageId = messageId;
        this.chatId = chatId;
        this.senderName = senderName;
        this.senderRole = senderRole;
        this.text = text;
        this.timestamp = timestamp;
        this.timeFormatted = timeFormatted;
    }

    public String getMessageId() {
        return messageId;
    }

    public void setMessageId(String messageId) {
        this.messageId = messageId;
    }

    public String getChatId() {
        return chatId;
    }

    public void setChatId(String chatId) {
        this.chatId = chatId;
    }

    public String getSenderName() {
        return senderName;
    }

    public void setSenderName(String senderName) {
        this.senderName = senderName;
    }

    public String getSenderRole() {
```

```
        return senderRole;
    }

    public void setSenderRole(String senderRole) {
        this.senderRole = senderRole;
    }

    public String getText() {
        return text;
    }

    public void setText(String text) {
        this.text = text;
    }

    public long getTimestamp() {
        return timestamp;
    }

    public void setTimestamp(long timestamp) {
        this.timestamp = timestamp;
    }

    public String getTimeFormatted() {
        return timeFormatted;
    }

    public void setTimeFormatted(String timeFormatted) {
        this.timeFormatted = timeFormatted;
    }
}
```

ServiceAdapter.java .16

```
package com.example.work;

import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.net.Uri;
import android.util.Base64;
```

```
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.ArrayList;
import java.util.List;
import java.util.Locale;

public class ServiceAdapter extends RecyclerView.Adapter<ServiceAdapter.ServiceV
iewHolder> {

    public interface OnServiceClickListener {
        void onServiceClick(ServiceItem service);
    }

    private List<ServiceItem> services = new ArrayList<>();
    private final OnServiceClickListener listener;
    private double currentTargetPrice = -1;

    public ServiceAdapter(OnServiceClickListener listener) {
        this.listener = listener;
    }

    public void setServices(List<ServiceItem> newServices, double targetPrice) {
        this.services = newServices != null ? newServices : new ArrayList<>();
        this.currentTargetPrice = targetPrice;
        notifyDataSetChanged();
    }

    @NonNull
    @Override
    public ServiceViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int v
iewType) {
        View view = LayoutInflater.from(parent.getContext()).inflate(R.layout.it
em_service, parent, false);
        return new ServiceViewHolder(view);
    }

    @Override
```

```
    public void onBindViewHolder(@NonNull ServiceViewHolder holder, int positio
n) {
        ServiceItem item = services.get(position);
        holder.bind(item, currentTargetPrice, listener);
    }

    @Override
    public int getItemCount() {
        return services.size();
    }

    static class ServiceViewHolder extends RecyclerView.ViewHolder {
        private final ImageView ivServiceThumbnail;
        private final TextView tvCategoryBadge;
        private final TextView tvRating;
        private final TextView tvReviewsCount;
        private final TextView tvServiceTitle;
        private final TextView tvProviderName;
        private final TextView tvPriceValue;
        private final TextView tvPriceSimilarityBadge;

        public ServiceViewHolder(@NonNull View itemView) {
            super(itemView);
            ivServiceThumbnail = itemView.findViewById(R.id.ivServiceThumbnail);
            tvCategoryBadge = itemView.findViewById(R.id.tvCategoryBadge);
            tvRating = itemView.findViewById(R.id.tvRating);
            tvReviewsCount = itemView.findViewById(R.id.tvReviewsCount);
            tvServiceTitle = itemView.findViewById(R.id.tvServiceTitle);
            tvProviderName = itemView.findViewById(R.id.tvProviderName);
            tvPriceValue = itemView.findViewById(R.id.tvPriceValue);
            tvPriceSimilarityBadge = itemView.findViewById(R.id.tvPriceSimilarit
yBadge);
        }

        public void bind(ServiceItem item, double targetPrice, OnServiceClickLis
tener listener) {
            if (ivServiceThumbnail != null) {
                if (item.getImageBase64() != null && !item.getImageBase64().isEm
pty()) {
                    try {
                        byte[] decodedBytes = Base64.decode(item.getImageBase64
(), Base64.DEFAULT);
                        Bitmap bitmap = BitmapFactory.decodeByteArray(decodedByt
es, 0, decodedBytes.length);
```

```
                        if (bitmap != null) {
                            ivServiceThumbnail.setImageBitmap(bitmap);
                        } else {
                            ivServiceThumbnail.setImageResource(R.drawable.img_w
ork_code);
                        }
                    } catch (Exception e) {
                        ivServiceThumbnail.setImageResource(R.drawable.img_work_
code);
                    }
                } else if (item.getImageUri() != null && !item.getImageUri().isE
mpty()) {
                    try {
                        ivServiceThumbnail.setImageURI(Uri.parse(item.getImageUr
i()));
                    } catch (Exception e) {
                        ivServiceThumbnail.setImageResource(R.drawable.img_work_
code);
                    }
                } else if (item.getImageResId() != 0) {
                    ivServiceThumbnail.setImageResource(item.getImageResId());
                } else {
                    ivServiceThumbnail.setImageResource(R.drawable.img_work_cod
e);
                }
            }

            tvCategoryBadge.setText(item.getCategory());
            tvRating.setText(String.format(Locale.getDefault(), "%.1f", item.get
Rating()));
            tvReviewsCount.setText(String.format(Locale.getDefault(), "(%d)", it
em.getReviewsCount()));
            tvServiceTitle.setText(item.getTitle());
            tvProviderName.setText(String.format("Ø¨Ùˆ Ø§Ø³Ø·Ø©: %s", item.getPro
viderName()));
            tvPriceValue.setText(String.format(Locale.getDefault(), "%.0f", ite
m.getPrice()));

            // Display price similarity indicator if user specified a budget
            if (targetPrice > 0) {
                tvPriceSimilarityBadge.setVisibility(View.VISIBLE);
                double diff = item.getPrice() - targetPrice;
                if (Math.abs(diff) < 0.01) {
                    tvPriceSimilarityBadge.setText("Ø³Ø¹Ø± Ù…Ø·Ø§Ø¨Ù‚, Ù„Ù…ÙŠØ²Ø§Ù†
```

```
Ù†ÙŠØªÙƒ Ø¨Ø§Ù„Ø¶Ø¨Ø·! ðŸŽ¯");
                } else if (diff < 0) {
                    tvPriceSimilarityBadge.setText(String.format(Locale.getDefau
lt(), "Ø£Ù„Ù„ Ù…Ù† Ù…ÙŠØ²Ø§Ù†ÙŠØªÙƒ Ø¨Ù€ $%.0f ðŸ’¡", Math.abs(diff)));
                } else {
                    tvPriceSimilarityBadge.setText(String.format(Locale.getDefau
lt(), "Ø£Ø¹Ù„Ù‰ Ù…Ù† Ù…ÙŠØ²Ø§Ù†ÙŠØªÙƒ Ø¨Ù€ $%.0f", diff));
                }
            } else {
                tvPriceSimilarityBadge.setVisibility(View.GONE);
            }

            itemView.setOnClickListener(v -> {
                if (listener != null) {
                    listener.onServiceClick(item);
                }
            });
        }
    }
}
```

OrderAdapter.java .17

```
package com.example.work;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import com.google.android.material.button.MaterialButton;

import java.util.ArrayList;
import java.util.List;
import java.util.Locale;

public class OrderAdapter extends RecyclerView.Adapter<OrderAdapter.OrderViewHol
```

```
der> {

    public interface OnOrderActionListener {
        void onOrderContactClick(OrderItem order);
        void onInAppChatClick(OrderItem order);
        void onAcceptOrder(OrderItem order);
        void onRejectOrder(OrderItem order);
        void onCompleteOrder(OrderItem order);
    }

    private List<OrderItem> orders = new ArrayList<>();
    private final OnOrderActionListener listener;
    private boolean isProviderMode = false;

    public OrderAdapter(OnOrderActionListener listener) {
        this.listener = listener;
    }

    public OrderAdapter(boolean isProviderMode, OnOrderActionListener listener)
{
        this.isProviderMode = isProviderMode;
        this.listener = listener;
    }

    public void setOrders(List<OrderItem> newOrders) {
        this.orders = newOrders != null ? newOrders : new ArrayList<>();
        notifyDataSetChanged();
    }

    @NonNull
    @Override
    public OrderViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int vie
wType) {
        View view = LayoutInflater.from(parent.getContext()).inflate(R.layout.it
em_order, parent, false);
        return new OrderViewHolder(view);
    }

    @Override
    public void onBindViewHolder(@NonNull OrderViewHolder holder, int position)
{
        OrderItem item = orders.get(position);
        holder.bind(item, isProviderMode, listener);
    }
```

```
    @Override
    public int getItemCount() {
        return orders.size();
    }

    static class OrderViewHolder extends RecyclerView.ViewHolder {
        private final TextView tvOrderStatus;
        private final TextView tvOrderDate;
        private final TextView tvOrderServiceTitle;
        private final TextView tvOrderProviderName;
        private final TextView tvOrderClientName;
        private final TextView tvOrderPrice;
        private final View llProviderActions;
        private final MaterialButton btnAcceptOrder;
        private final MaterialButton btnRejectOrder;
        private final MaterialButton btnCompleteOrder;
        private final MaterialButton btnInAppChat;
        private final MaterialButton btnOrderContactProvider;

        public OrderViewHolder(@NonNull View itemView) {
            super(itemView);
            tvOrderStatus = itemView.findViewById(R.id.tvOrderStatus);
            tvOrderDate = itemView.findViewById(R.id.tvOrderDate);
            tvOrderServiceTitle = itemView.findViewById(R.id.tvOrderServiceTitl
e);
            tvOrderProviderName = itemView.findViewById(R.id.tvOrderProviderNam
e);
            tvOrderClientName = itemView.findViewById(R.id.tvOrderClientName);
            tvOrderPrice = itemView.findViewById(R.id.tvOrderPrice);
            llProviderActions = itemView.findViewById(R.id.llProviderActions);
            btnAcceptOrder = itemView.findViewById(R.id.btnAcceptOrder);
            btnRejectOrder = itemView.findViewById(R.id.btnRejectOrder);
            btnCompleteOrder = itemView.findViewById(R.id.btnCompleteOrder);
            btnInAppChat = itemView.findViewById(R.id.btnInAppChat);
            btnOrderContactProvider = itemView.findViewById(R.id.btnOrderContact
Provider);
        }

        public void bind(OrderItem order, boolean isProviderMode, OnOrderActionL
istener listener) {
            ServiceItem service = order.getServiceItem();

            tvOrderStatus.setText(order.getStatus() != null ? order.getStatus()
```

```
: OrderRepository.STATUS_PENDING);
            tvOrderDate.setText(String.format("Ø²Ø§Ø±ÙŠØ® Ø§Ù„Ø·Ù„Ø¨: %s", orde
r.getOrderDate()));

            if (service != null) {
                tvOrderServiceTitle.setText(service.getTitle());
                tvOrderProviderName.setText(String.format("Ø§Ù„Ù…Ø²Ùˆ Ø¯: %s", se
rvice.getProviderName()));
                tvOrderPrice.setText(String.format(Locale.getDefault(), "$%.0f",
service.getPrice()));
            }

            if (isProviderMode) {
                if (tvOrderClientName != null) {
                    tvOrderClientName.setVisibility(View.VISIBLE);
                    String clientInfo = String.format("Ø§Ù„Ø¹Ù…ÙŠÙ„ Ø§Ù„Ø·Ø§Ù„Ø¨
¨: %s (%s)",
                            order.getClientName() != null ? order.getClientName
() : "Ø¹Ù…ÙŠÙ„",
                            order.getClientEmail() != null ? order.getClientEmai
l() : "");
                    tvOrderClientName.setText(clientInfo);
                }

                String status = order.getStatus();
                if (status == null || status.contains("ÙŠÙ† Ø§Ù†Ø¸Ø§Ø±")) {
                    llProviderActions.setVisibility(View.VISIBLE);
                    btnCompleteOrder.setVisibility(View.GONE);
                } else if (status.contains("Ø¬Ø… Ù‚Ø¨Ù„") || status.contains
("Ù‚ÙŠØ¯ Ø§Ù„Ø¬Ø§Ø±ÙŠÙŠØ°")) {
                    llProviderActions.setVisibility(View.GONE);
                    btnCompleteOrder.setVisibility(View.VISIBLE);
                } else {
                    llProviderActions.setVisibility(View.GONE);
                    btnCompleteOrder.setVisibility(View.GONE);
                }
            } else {
                if (tvOrderClientName != null) {
                    tvOrderClientName.setVisibility(View.GONE);
                }
                llProviderActions.setVisibility(View.GONE);
                btnCompleteOrder.setVisibility(View.GONE);
            }
```

```
            btnAcceptOrder.setOnClickListener(v -> {
                if (listener != null) listener.onAcceptOrder(order);
            });

            btnRejectOrder.setOnClickListener(v -> {
                if (listener != null) listener.onRejectOrder(order);
            });

            btnCompleteOrder.setOnClickListener(v -> {
                if (listener != null) listener.onCompleteOrder(order);
            });

            btnOrderContactProvider.setOnClickListener(v -> {
                if (listener != null) listener.onOrderContactClick(order);
            });

            btnInAppChat.setOnClickListener(v -> {
                if (listener != null) listener.onInAppChatClick(order);
            });
        }
    }
}
```

MessageAdapter.java .18

```
package com.example.work;

import android.content.res.ColorStateList;
import android.graphics.Color;
import android.view.Gravity;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.LinearLayout;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.core.content.ContextCompat;
import androidx.recyclerview.widget.RecyclerView;
```

```
import com.google.android.material.card.MaterialCardView;

import java.util.ArrayList;
import java.util.List;

public class MessageAdapter extends RecyclerView.Adapter<MessageAdapter.MessageV
iewHolder> {

    private List<ChatMessage> messages = new ArrayList<>();
    private String currentUserRole = "client";

    public MessageAdapter(String currentUserRole) {
        this.currentUserRole = currentUserRole != null ? currentUserRole : "clie
nt";
    }

    public void setMessages(List<ChatMessage> newMessages) {
        this.messages = newMessages != null ? newMessages : new ArrayList<>();
        notifyDataSetChanged();
    }

    @NonNull
    @Override
    public MessageViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int v
iewType) {
        View view = LayoutInflater.from(parent.getContext()).inflate(R.layout.it
em_chat_message, parent, false);
        return new MessageViewHolder(view);
    }

    @Override
    public void onBindViewHolder(@NonNull MessageViewHolder holder, int positio
n) {
        ChatMessage message = messages.get(position);
        holder.bind(message, currentUserRole);
    }

    @Override
    public int getItemCount() {
        return messages.size();
    }

    static class MessageViewHolder extends RecyclerView.ViewHolder {
        private final LinearLayout llMessageContainer;
```

```
        private final MaterialCardView cardMessageBubble;
        private final TextView tvMessageSender;
        private final TextView tvMessageText;
        private final TextView tvMessageTime;

        public MessageViewHolder(@NonNull View itemView) {
            super(itemView);
            llMessageContainer = itemView.findViewById(R.id.llMessageContainer);
            cardMessageBubble = itemView.findViewById(R.id.cardMessageBubble);
            tvMessageSender = itemView.findViewById(R.id.tvMessageSender);
            tvMessageText = itemView.findViewById(R.id.tvMessageText);
            tvMessageTime = itemView.findViewById(R.id.tvMessageTime);
        }

        public void bind(ChatMessage message, String currentUserRole) {
            boolean isMyMessage = currentUserRole.equalsIgnoreCase(message.getSe
nderRole());

            tvMessageSender.setText(message.getSenderName() != null ? message.ge
tSenderName() : "");
            tvMessageText.setText(message.getText() != null ? message.getText()
: "");
            tvMessageTime.setText(message.getTimeFormatted() != null ? message.g
etTimeFormatted() : "");

            if (isMyMessage) {
                llMessageContainer.setGravity(Gravity.END);
                cardMessageBubble.setCardBackgroundColor(ColorStateList.valueOf
(ContextCompat.getColor(itemView.getContext(), R.color.orange_primary)));
                tvMessageSender.setTextColor(Color.WHITE);
                tvMessageText.setTextColor(Color.WHITE);
                tvMessageTime.setTextColor(Color.parseColor("#FEF08A"));
            } else {
                llMessageContainer.setGravity(Gravity.START);
                cardMessageBubble.setCardBackgroundColor(ColorStateList.valueOf
(ContextCompat.getColor(itemView.getContext(), R.color.surface_white)));
                tvMessageSender.setTextColor(ContextCompat.getColor(itemView.get
Context(), R.color.orange_primary));
                tvMessageText.setTextColor(ContextCompat.getColor(itemView.getCo
ntext(), R.color.text_primary));
                tvMessageTime.setTextColor(ContextCompat.getColor(itemView.getCo
ntext(), R.color.text_secondary));
            }
        }
```

```
        }
    }
```

Ø§Ù„Ù…Ù…Ù‰Ø, (Ø¨): Ø§Ù„Ø£Ù†Ù‰Ø§Ø¹ Ø§Ù„Ù…Ø¯Ø±Ø±Ù© Ø§Ù„Ù†Ø§Ø© Ù‰Ø³Ø¹Ø©Ù… Ø§Ù„Ù‰Ø§ Ø§Ø§Ø¹ (XML Layouts)

ÙŠØªØ¶Ù…Ù † Ù ‡ Ø°Ø§ Ø§Ù„Ù…Ø„Ù‰ Ø§Ù„Ù† Øµ Ø§Ù„Ù†Ø§Ø© Ø§Ù„Ø¹Ø§Ù† Ù„Ù…Ø§Ø§Ø© Ø§Ù„Ù…Ø±Ø¦ÙŠ XML Ù†Ø§Ø§Ù‘Ø© :Ø§Ù„Ù‰Ø§Ø¬Ù‡Ø§Ø Ù‰Ø§Ù„Ù…Ù†Ù†Ø§Ø

activity_main.xml .1

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/bg_splash"
    android:gravity="center"
    android:padding="24dp">

    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:gravity="center"
        android:orientation="vertical">

        <!-- Animated / Highlighted Logo Container -->
        <FrameLayout
```

```
            android:layout_width="120dp"
            android:layout_height="120dp"
            android:background="@drawable/bg_button_navy"
            android:elevation="8dp"
            android:padding="20dp">

            <ImageView
                android:layout_width="match_parent"
                android:layout_height="match_parent"
                android:contentDescription="@string/app_name"
                android:src="@drawable/ic_work_logo" />
        </FrameLayout>

        <!-- App Name -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="20dp"
            android:letterSpacing="0.1"
            android:text="@string/app_name"
            android:textColor="@color/white"
            android:textSize="36sp"
            android:textStyle="bold" />

        <!-- Tagline -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="8dp"
            android:text="@string/app_tagline"
            android:textColor="#CBD5E1"
            android:textSize="15sp" />

        <!-- Loading Spinner -->
        <ProgressBar
            style="?android:attr/progressBarStyleSmall"
            android:layout_width="32dp"
            android:layout_height="32dp"
            android:layout_marginTop="40dp"
            android:indeterminateTint="@color/orange_primary" />
    </LinearLayout>

    <TextView
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:layout_alignParentBottom="true"
        android:layout_centerHorizontal="true"
        android:layout_marginBottom="24dp"
        android:text="Ø§Ù„Ø¥Ø¸Ø¯Ø§Ø± 1.0 â€¢ WORK App"
        android:textColor="#64748B"
        android:textSize="12sp" />

</RelativeLayout>
```

activity_role_selection.xml .2

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/bg_app"
    android:orientation="vertical">

    <!-- Top Branding Header -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@drawable/bg_header_navy"
        android:gravity="center"
        android:orientation="vertical"
        android:paddingTop="36dp"
        android:paddingBottom="28dp">

        <ImageView
            android:layout_width="72dp"
            android:layout_height="72dp"
            android:contentDescription="@string/app_name"
            android:src="@drawable/ic_work_logo" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="8dp"
```

```
            android:text="@string/app_name"
            android:textColor="@color/white"
            android:textSize="28sp"
            android:textStyle="bold" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="2dp"
            android:text="@string/app_tagline"
            android:textColor="#94A3B8"
            android:textSize="13sp" />
    </LinearLayout>

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:fillViewport="true">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:padding="20dp">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="@string/welcome_title"
                android:textColor="@color/text_primary"
                android:textSize="20sp"
                android:textStyle="bold" />

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginTop="4dp"
                android:layout_marginBottom="16dp"
                android:text="@string/welcome_subtitle"
                android:textColor="@color/text_secondary"
                android:textSize="13sp" />

            <!-- Login Registration Inputs -->
```

```
        <!-- Full Name Field -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginBottom="4dp"
            android:text="@string/enter_name_label"
            android:textColor="@color/text_primary"
            android:textSize="13sp"
            android:textStyle="bold" />

        <com.google.android.material.textfield.TextInputLayout
            android:id="@+id/tilUserName"
            style="@style/Widget.Material3.TextInputLayout.OutlinedBox"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginBottom="12dp"
            app:boxCornerRadiusBottomEnd="10dp"
            app:boxCornerRadiusBottomStart="10dp"
            app:boxCornerRadiusTopEnd="10dp"
            app:boxCornerRadiusTopStart="10dp"
            app:errorEnabled="true">

            <com.google.android.material.textfield.TextInputEditText
                android:id="@+id/etUserName"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="@string/enter_name_hint"
                android:inputType="textPersonName"
                android:textColor="#000000"
                android:textColorHint="@color/text_hint"
                android:textSize="14sp" />
        </com.google.android.material.textfield.TextInputLayout>

        <!-- Verified Email Field -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginBottom="4dp"
            android:text="@string/enter_email_label"
            android:textColor="@color/text_primary"
            android:textSize="13sp"
            android:textStyle="bold" />

        <com.google.android.material.textfield.TextInputLayout
```

```
        android:id="@+id/tilUserEmail"
        style="@style/Widget.Material3.TextInputLayout.OutlinedBox"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"
        app:boxCornerRadiusBottomEnd="10dp"
        app:boxCornerRadiusBottomStart="10dp"
        app:boxCornerRadiusTopEnd="10dp"
        app:boxCornerRadiusTopStart="10dp"
        app:errorEnabled="true">

        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/etUserEmail"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="@string/enter_email_hint"
            android:inputType="textEmailAddress"
            android:textColor="#000000"
            android:textColorHint="@color/text_hint"
            android:textSize="14sp" />
    </com.google.android.material.textfield.TextInputLayout>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="8dp"
        android:text="Ø§Ø®ØªØ± Ùˆ Ø¸ÙŠÙØªÙƒ:"
        android:textColor="@color/text_primary"
        android:textSize="14sp"
        android:textStyle="bold" />

    <!-- Role Card 1: Client -->
    <com.google.android.material.card.MaterialCardView
        android:id="@+id/cardRoleClient"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="12dp"
        android:clickable="true"
        android:focusable="true"
        app:cardBackgroundColor="@color/surface_white"
        app:cardCornerRadius="14dp"
        app:cardElevation="2dp"
        app:strokeColor="@color/orange_primary"
        app:strokeWidth="2dp">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center_vertical"
    android:orientation="horizontal"
    android:padding="16dp">

    <ImageView
        android:layout_width="48dp"
        android:layout_height="48dp"
        android:contentDescription="@string/role_client_title"
        android:src="@drawable/ic_client" />

    <LinearLayout
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginStart="12dp"
        android:layout_marginEnd="8dp"
        android:layout_weight="1"
        android:orientation="vertical">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/role_client_title"
            android:textColor="@color/text_primary"
            android:textSize="16sp"
            android:textStyle="bold" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="2dp"
            android:text="@string/role_client_desc"
            android:textColor="@color/text_secondary"
            android:textSize="12sp" />
    </LinearLayout>

    <RadioButton
        android:id="@+id/radioClient"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:checked="true"
```

```
                android:clickable="false"
                android:focusable="false" />
        </LinearLayout>
    </com.google.android.material.card.MaterialCardView>

    <!-- Role Card 2: Service Provider -->
    <com.google.android.material.card.MaterialCardView
        android:id="@+id/cardRoleProvider"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:clickable="true"
        android:focusable="true"
        app:cardBackgroundColor="@color/surface_white"
        app:cardCornerRadius="14dp"
        app:cardElevation="2dp"
        app:strokeColor="@color/card_stroke"
        app:strokeWidth="1dp">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center_vertical"
            android:orientation="horizontal"
            android:padding="16dp">

            <ImageView
                android:layout_width="48dp"
                android:layout_height="48dp"
                android:contentDescription="@string/role_provider_title"
                android:src="@drawable/ic_provider" />

            <LinearLayout
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_marginStart="12dp"
                android:layout_marginEnd="8dp"
                android:layout_weight="1"
                android:orientation="vertical">

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="@string/role_provider_title"
```

```
                            android:textColor="@color/text_primary"
                            android:textSize="16sp"
                            android:textStyle="bold" />

                        <TextView
                            android:layout_width="wrap_content"
                            android:layout_height="wrap_content"
                            android:layout_marginTop="2dp"
                            android:text="@string/role_provider_desc"
                            android:textColor="@color/text_secondary"
                            android:textSize="12sp" />
                    </LinearLayout>

                    <RadioButton
                        android:id="@+id/radioProvider"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:checked="false"
                        android:clickable="false"
                        android:focusable="false" />
                </LinearLayout>
            </com.google.android.material.card.MaterialCardView>

            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnContinue"
                android:layout_width="match_parent"
                android:layout_height="54dp"
                android:backgroundTint="@color/orange_primary"
                android:text="@string/btn_continue"
                android:textColor="@color/white"
                android:textSize="15sp"
                android:textStyle="bold"
                app:cornerRadius="12dp" />
        </LinearLayout>
    </ScrollView>
</LinearLayout>
```

activity_client_main.xml .3

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:android="http://schem
as.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/bg_app">

    <!-- Collapsible AppBar Layout (Hides automatically when scrolling down) -->
    <com.google.android.material.appbar.AppBarLayout
        android:id="@+id/appBarLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@android:color/transparent"
        app:elevation="0dp">

        <!-- Top Navy Header Banner & Search Card (Collapses away on Scroll Dow
n) -->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:background="@drawable/bg_header_navy"
            android:orientation="vertical"
            android:padding="20dp"
            app:layout_scrollFlags="scroll|enterAlways|snap">

            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:gravity="center_vertical"
                android:orientation="horizontal">

                <ImageView
                    android:layout_width="32dp"
                    android:layout_height="32dp"
                    android:contentDescription="@string/app_name"
                    android:src="@drawable/ic_work_logo" />

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:layout_marginStart="8dp"
                    android:text="@string/app_name"
```

```
                android:textColor="@color/white"
                android:textSize="20sp"
                android:textStyle="bold" />

            <View
                android:layout_width="0dp"
                android:layout_height="0dp"
                android:layout_weight="1" />

            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnSwitchRole"
                style="@style/Widget.Material3.Button.OutlinedButton"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="@string/switch_role"
                android:textColor="@color/white"
                android:textSize="12sp"
                app:strokeColor="#475569" />
        </LinearLayout>

        <TextView
            android:id="@+id/tvWelcomeHeader"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="12dp"
            android:text="@string/client_header_title"
            android:textColor="@color/white"
            android:textSize="18sp"
            android:textStyle="bold" />

        <!-- Search Bar Banner Card -->
        <com.google.android.material.card.MaterialCardView
            android:id="@+id/cardOpenSearchScreen"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="16dp"
            android:clickable="true"
            android:focusable="true"
            app:cardBackgroundColor="@color/surface_white"
            app:cardCornerRadius="14dp"
            app:cardElevation="4dp"
            app:strokeWidth="0dp">

            <LinearLayout
```

```
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:gravity="center_vertical"
                    android:orientation="horizontal"
                    android:padding="14dp">

                    <ImageView
                        android:layout_width="24dp"
                        android:layout_height="24dp"
                        android:contentDescription="Ø¨ØØ«"
                        android:src="@drawable/ic_search" />

                    <TextView
                        android:layout_width="0dp"
                        android:layout_height="wrap_content"
                        android:layout_marginStart="12dp"
                        android:layout_weight="1"
                        android:text="Ø¨ØØ« Ø¹Ù† Ø®Ø¯Ù…Ø© Ø¨Ø§Ù„Ø§Ø³Ù… ÙˆØ§Ù„Ø³Ø… ðŸ”"
                        android:textColor="@color/navy_secondary"
                        android:textSize="14sp"
                        android:textStyle="bold" />

                    <com.google.android.material.button.MaterialButton
                        android:id="@+id/btnOpenSearchScreen"
                        android:layout_width="wrap_content"
                        android:layout_height="38dp"
                        android:backgroundTint="@color/orange_primary"
                        android:clickable="false"
                        android:focusable="false"
                        android:text="Ø¨ØØ« ðŸ”"
                        android:textColor="@color/white"
                        android:textSize="12sp"
                        android:textStyle="bold"
                        app:cornerRadius="8dp" />
                </LinearLayout>
            </com.google.android.material.card.MaterialCardView>
        </LinearLayout>

        <!-- Main View Mode Selector Tabs (Browse Services vs My Orders Used) --
>
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```
            android:background="@color/bg_app"
            android:orientation="vertical"
            android:paddingHorizontal="16dp"
            android:paddingTop="12dp"
            android:paddingBottom="4dp">

            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="48dp"
                android:background="@drawable/bg_category_badge"
                android:orientation="horizontal"
                android:padding="4dp">

                <com.google.android.material.button.MaterialButton
                    android:id="@+id/btnTabBrowseServices"
                    style="@style/Widget.Material3.Button.UnelevatedButton"
                    android:layout_width="0dp"
                    android:layout_height="match_parent"
                    android:layout_weight="1"
                    android:backgroundTint="@color/brand_primary"
                    android:text="Ø²Ø µÙØ Ø§Ù„Ø®Ø¯Ù…Ø§Øª ðŸ”"
                    android:textColor="@color/white"
                    android:textSize="13sp"
                    android:textStyle="bold"
                    app:cornerRadius="8dp" />

                <com.google.android.material.button.MaterialButton
                    android:id="@+id/btnTabMyOrders"
                    style="@style/Widget.Material3.Button.TextButton"
                    android:layout_width="0dp"
                    android:layout_height="match_parent"
                    android:layout_weight="1"
                    android:text="Ø®Ø¯Ù…Ø§ØªÙŠ Ø§Ù„ØØ§Ù„ÙŠØ© Ùˆ Ø§Ù„Ø³Ø§Ø¨Ù‚Ø© ðŸ“¦"
                    android:textColor="@color/text_secondary"
                    android:textSize="12sp"
                    android:textStyle="bold"
                    app:cornerRadius="8dp" />
            </LinearLayout>

            <!-- Categories Filter Horizontal Scroll (Shown in Browse Tab) -->
            <HorizontalScrollView
                android:id="@+id/hsvCategories"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
```

```
                android:layout_marginTop="8dp"
                android:scrollbars="none">

                <com.google.android.material.chip.ChipGroup
                    android:id="@+id/chipGroupCategories"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    app:singleLine="true"
                    app:singleSelection="true">

                    <com.google.android.material.chip.Chip
                        android:id="@+id/chipAll"
                        style="@style/Widget.Material3.Chip.Filter"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:checked="true"
                        android:text="@string/all_categories" />

                    <com.google.android.material.chip.Chip
                        android:id="@+id/chipProgramming"
                        style="@style/Widget.Material3.Chip.Filter"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:text="@string/cat_programming" />

                    <com.google.android.material.chip.Chip
                        android:id="@+id/chipDesign"
                        style="@style/Widget.Material3.Chip.Filter"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:text="@string/cat_design" />

                    <com.google.android.material.chip.Chip
                        android:id="@+id/chipTranslation"
                        style="@style/Widget.Material3.Chip.Filter"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:text="@string/cat_translation" />

                    <com.google.android.material.chip.Chip
                        android:id="@+id/chipMarketing"
                        style="@style/Widget.Material3.Chip.Filter"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
```

```
                        android:text="@string/cat_marketing" />

                    <com.google.android.material.chip.Chip
                        android:id="@+id/chipMaintenance"
                        style="@style/Widget.Material3.Chip.Filter"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:text="@string/cat_maintenance" />
                </com.google.android.material.chip.ChipGroup>
            </HorizontalScrollView>

        </LinearLayout>
    </com.google.android.material.appbar.AppBarLayout>

    <!-- Main Content Area: RecyclerView for Services OR Orders -->
    <FrameLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:layout_behavior="@string/appbar_scrolling_view_behavior">

        <!-- Services Browse List -->
        <androidx.recyclerview.widget.RecyclerView
            android:id="@+id/rvServices"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:clipToPadding="false"
            android:paddingTop="8dp"
            android:paddingBottom="24dp" />

        <!-- Orders Used List -->
        <androidx.recyclerview.widget.RecyclerView
            android:id="@+id/rvOrders"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:clipToPadding="false"
            android:paddingTop="8dp"
            android:paddingBottom="24dp"
            android:visibility="gone" />

        <TextView
            android:id="@+id/tvEmptyState"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center"
```

```
            android:padding="24dp"
            android:text="@string/no_services_found"
            android:textAlignment="center"
            android:textColor="@color/text_secondary"
            android:textSize="15sp"
            android:visibility="gone" />
    </FrameLayout>

    <!-- Floating Action Button for Search (Auto-Hides on Scroll Down) -->
    <com.google.android.material.floatingactionbutton.FloatingActionButton
        android:id="@+id/fabSearch"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="bottom|end"
        android:layout_margin="20dp"
        android:contentDescription="Ø¨ØØ«"
        android:src="@drawable/ic_search"
        app:backgroundTint="@color/orange_primary"
        app:layout_behavior="com.google.android.material.behavior.HideBottomView
OnScrollBehavior"
        app:tint="@color/white" />

</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

activity_provider_main.xml .4

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/bg_app"
    android:orientation="vertical">

    <!-- Top Navy Header Banner -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@drawable/bg_header_navy"
        android:orientation="vertical"
```

```
        android:padding="20dp">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center_vertical"
            android:orientation="horizontal">

            <ImageView
                android:layout_width="32dp"
                android:layout_height="32dp"
                android:contentDescription="@string/app_name"
                android:src="@drawable/ic_work_logo" />

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginStart="8dp"
                android:text="@string/app_name"
                android:textColor="@color/white"
                android:textSize="20sp"
                android:textStyle="bold" />

            <View
                android:layout_width="0dp"
                android:layout_height="0dp"
                android:layout_weight="1" />

            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnSwitchRole"
                style="@style/Widget.Material3.Button.OutlinedButton"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="@string/switch_role"
                android:textColor="@color/white"
                android:textSize="12sp"
                app:strokeColor="#475569" />
        </LinearLayout>

        <TextView
            android:id="@+id/tvProviderHeader"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="16dp"
```

```
            android:text="@string/provider_header_title"
            android:textColor="@color/white"
            android:textSize="20sp"
            android:textStyle="bold" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="4dp"
            android:text="@string/provider_header_subtitle"
            android:textColor="#94A3B8"
            android:textSize="13sp" />

        <!-- Add Service Call-to-Action Card -->
        <com.google.android.material.card.MaterialCardView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="16dp"
            app:cardBackgroundColor="@color/surface_white"
            app:cardCornerRadius="16dp"
            app:cardElevation="4dp"
            app:strokeWidth="0dp">

            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:gravity="center"
                android:orientation="vertical"
                android:padding="20dp">

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Ù‡Ù„ Ù„Ø¯ÙŠÙƒ Ù…Ù‡Ø§Ø±Ø© Ø£Ùˆ Ø®Ø¯Ù…Ø© Ø¥Ø±ÙŠØ¯ Ø¨ÙŠØ¹Ù‡Ø§Ø؟"
                    android:textColor="@color/text_primary"
                    android:textSize="16sp"
                    android:textStyle="bold" />

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:layout_marginTop="4dp"
                    android:layout_marginBottom="16dp"
```

```
                    android:text="Ø£Ø¶Ù Ø®Ø¯Ù…Ø©Ù Ù„Ù„Ø¹Ù…Ù„Ø§Ø¡ Ù‘Ø£Ø¯Ø¯ Ø§Ù„Ø³Ø¹Ø± Ø¨Ø§Ù„Ø¯Ù‘Ù‘Ù„Ø§Ø± ($)"
                    android:textColor="@color/text_secondary"
                    android:textSize="13sp" />

                <com.google.android.material.button.MaterialButton
                    android:id="@+id/btnAddService"
                    android:layout_width="match_parent"
                    android:layout_height="52dp"
                    android:backgroundTint="@color/brand_primary"
                    android:text="@string/btn_add_new_service"
                    android:textColor="@color/white"
                    android:textSize="16sp"
                    android:textStyle="bold"
                    app:cornerRadius="12dp"
                    app:icon="@drawable/ic_add"
                    app:iconGravity="textStart"
                    app:iconPadding="8dp" />
            </LinearLayout>
        </com.google.android.material.card.MaterialCardView>
    </LinearLayout>

    <!-- Navigation Tabs (My Services vs Incoming Orders) -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginHorizontal="20dp"
        android:layout_marginTop="16dp"
        android:layout_marginBottom="8dp"
        android:background="@drawable/bg_card_rounded"
        android:orientation="horizontal"
        android:padding="4dp">

        <com.google.android.material.button.MaterialButton
            android:id="@+id/btnTabMyServices"
            style="@style/Widget.Material3.Button.UnelevatedButton"
            android:layout_width="0dp"
            android:layout_height="44dp"
            android:layout_weight="1"
            android:backgroundTint="@color/brand_primary"
            android:text="Ø¬Ù…ÙŠØ¹ Ø§Ù„Ø®Ø¯Ù…Ø§Øª ðŸŒ"
            android:textColor="@color/white"
            android:textSize="13sp"
            android:textStyle="bold"
```

```
            app:cornerRadius="10dp" />

        <com.google.android.material.button.MaterialButton
            android:id="@+id/btnTabIncomingOrders"
            style="@style/Widget.Material3.Button.UnelevatedButton"
            android:layout_width="0dp"
            android:layout_height="44dp"
            android:layout_weight="1"
            android:backgroundTint="@android:color/transparent"
            android:text="Ø§Ù„Ø·Ù„Ø¨Ø§Øª Ø§Ù„Ù‚Ø§Ø¯Ù…Ø© ðŸ“¥"
            android:textColor="@color/text_secondary"
            android:textSize="13sp"
            android:textStyle="bold"
            app:cornerRadius="10dp" />
    </LinearLayout>

    <!-- Provider Header Info Bar -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center_vertical"
        android:orientation="horizontal"
        android:paddingHorizontal="20dp"
        android:paddingBottom="8dp">

        <TextView
            android:id="@+id/tvSectionTitle"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="@string/my_posted_services"
            android:textColor="@color/text_primary"
            android:textSize="16sp"
            android:textStyle="bold" />

        <TextView
            android:id="@+id/tvPostedServicesCount"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:background="@drawable/bg_category_badge"
            android:paddingHorizontal="10dp"
            android:paddingVertical="4dp"
            android:text="0 Ø®Ø¯Ù…Ø§Øª"
            android:textColor="@color/brand_primary"
```

```
            android:textSize="12sp"
            android:textStyle="bold" />
    </LinearLayout>

    <FrameLayout
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1">

        <androidx.recyclerview.widget.RecyclerView
            android:id="@+id/rvProviderServices"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:clipToPadding="false"
            android:paddingBottom="16dp" />

        <androidx.recyclerview.widget.RecyclerView
            android:id="@+id/rvIncomingOrders"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:clipToPadding="false"
            android:paddingBottom="16dp"
            android:visibility="gone" />

        <TextView
            android:id="@+id/tvEmptyPosted"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center"
            android:padding="24dp"
            android:text="@string/no_posted_services"
            android:textAlignment="center"
            android:textColor="@color/text_secondary"
            android:textSize="14sp"
            android:visibility="gone" />
    </FrameLayout>
</LinearLayout>
```

activity_search.xml .5

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/bg_app"
    android:orientation="vertical">

    <!-- Top Navigation Bar -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/surface_white"
        android:elevation="2dp"
        android:gravity="center_vertical"
        android:orientation="horizontal"
        android:padding="16dp">

        <ImageButton
            android:id="@+id/btnBack"
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:contentDescription="Ø±Ø¬Ùˆ Ø¹"
            android:src="@drawable/ic_arrow_back" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="12dp"
            android:text="Ø§Ù„Ø¨Øø« Ø¹Ù† Ø®Ø¯Ù…Ø© Ø¨Ø§Ù„Ù‚Ø±Ø¨ Ù†Ø§Ù„Ù…Ø²Ø§Ø¬ ðŸ”"
            android:textColor="@color/text_primary"
            android:textSize="18sp"
            android:textStyle="bold" />
    </LinearLayout>

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:fillViewport="true">

        <LinearLayout
```

```
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical">

            <!-- Search Form Card -->
            <com.google.android.material.card.MaterialCardView
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_margin="16dp"
                app:cardBackgroundColor="@color/surface_white"
                app:cardCornerRadius="16dp"
                app:cardElevation="3dp"
                app:strokeColor="@color/card_stroke"
                app:strokeWidth="1dp">

                <LinearLayout
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:orientation="vertical"
                    android:padding="20dp">

                    <TextView
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:text="Ø§Ùƒ Ø¨Ø§Ø§Ù„Ø¯ÙŠ Ø§Ù„Ø®Ø¯Ù…Ø© Ù…Ù†Ù‡Ø§ Ø§Ù„Ø¥Ø·Ù„Ø§Ù‚:"
                        android:textColor="@color/text_primary"
                        android:textSize="14sp"
                        android:textStyle="bold" />

                    <!-- Service Name Input -->
                    <com.google.android.material.textfield.TextInputLayout
                        style="@style/Widget.Material3.TextInputLayout.OutlinedBox"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:layout_marginTop="12dp"
                        app:boxCornerRadiusBottomEnd="10dp"
                        app:boxCornerRadiusBottomStart="10dp"
                        app:boxCornerRadiusTopEnd="10dp"
                        app:boxCornerRadiusTopStart="10dp"
                        app:startIconDrawable="@drawable/ic_search">

                        <com.google.android.material.textfield.TextInputEditText
```

```
                android:id="@+id/etSearchQuery"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="@string/search_service_hint"
                android:imeOptions="actionNext"
                android:inputType="text"
                android:textColor="#000000"
                android:textColorHint="@color/text_hint"
                android:textSize="14sp" />
        </com.google.android.material.textfield.TextInputLayout>

        <!-- Budget / Target Price Input -->
        <com.google.android.material.textfield.TextInputLayout
            style="@style/Widget.Material3.TextInputLayout.OutlinedB
ox"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="10dp"
            app:boxCornerRadiusBottomEnd="10dp"
            app:boxCornerRadiusBottomStart="10dp"
            app:boxCornerRadiusTopEnd="10dp"
            app:boxCornerRadiusTopStart="10dp">

            <com.google.android.material.textfield.TextInputEditText
                android:id="@+id/etTargetPrice"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="@string/target_price_hint"
                android:imeOptions="actionSearch"
                android:inputType="numberDecimal"
                android:textColor="#000000"
                android:textColorHint="@color/text_hint"
                android:textSize="14sp" />
        </com.google.android.material.textfield.TextInputLayout>

        <!-- Search Button -->
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btnDoSearch"
            android:layout_width="match_parent"
            android:layout_height="52dp"
            android:layout_marginTop="16dp"
            android:backgroundTint="@color/orange_primary"
            android:text="Ø¨Ø« Ùˆ¥Ø¸Ù‡Ø§Ø± Ø§Ù„Ù†ØªØ§Ø¦Ø¬ ðŸ”"
            android:textColor="@color/white"
```

```
                        android:textSize="16sp"
                        android:textStyle="bold"
                        app:cornerRadius="12dp" />
                </LinearLayout>
            </com.google.android.material.card.MaterialCardView>

            <!-- Search History Section -->
            <LinearLayout
                android:id="@+id/llSearchHistorySection"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:orientation="vertical"
                android:paddingHorizontal="20dp"
                android:paddingBottom="12dp"
                android:visibility="gone">

                <LinearLayout
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:gravity="center_vertical"
                    android:orientation="horizontal">

                    <TextView
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:text="Ø³Ø¬Ù„ Ø¹Ù…Ù„ÙŠØ§Øª Ø§Ù„Ø¨Ø¯Ø« Ø§Ù„Ø£Ø®ÙŠØ±Ø© ðŸ•"
                        android:textColor="@color/text_primary"
                        android:textSize="14sp"
                        android:textStyle="bold" />

                    <View
                        android:layout_width="0dp"
                        android:layout_height="0dp"
                        android:layout_weight="1" />

                    <TextView
                        android:id="@+id/tvClearHistory"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:clickable="true"
                        android:focusable="true"
                        android:padding="4dp"
                        android:text="Ù…Ø³Ø Ø§Ù„Ø³Ø¬Ù„"
```

```
                        android:textColor="@color/orange_primary"
                        android:textSize="12sp"
                        android:textStyle="bold" />
                </LinearLayout>

                <HorizontalScrollView
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:layout_marginTop="6dp"
                    android:scrollbars="none">

                    <com.google.android.material.chip.ChipGroup
                        android:id="@+id/cgSearchHistory"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        app:singleLine="true" />
                </HorizontalScrollView>
            </LinearLayout>

            <!-- Search Results Section -->
            <LinearLayout
                android:id="@+id/llSearchResultsHeader"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:orientation="vertical"
                android:paddingHorizontal="20dp"
                android:paddingTop="8dp"
                android:paddingBottom="8dp"
                android:visibility="gone">

                <TextView
                    android:id="@+id/tvResultsHeaderTitle"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:text="Ù†ØªØ§Ø¦Ø¬ Ø§Ù„Ø¨ØØ« Ø§Ù„Ù…Ø·Ø§Ø¨Ù‚Ø© Ù„Ù„Ø¨Ø·Ø§Ù‚Ø© Ø§Ù„Ù†Ø´Ø·Ø©:"
                    android:textColor="@color/text_primary"
                    android:textSize="15sp"
                    android:textStyle="bold" />
            </LinearLayout>

            <FrameLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
```

```
                android:minHeight="200dp">

                <androidx.recyclerview.widget.RecyclerView
                    android:id="@+id/rvSearchResults"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:clipToPadding="false"
                    android:paddingBottom="24dp"
                    android:visibility="gone" />

                <TextView
                    android:id="@+id/tvSearchEmptyState"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:layout_gravity="center"
                    android:padding="24dp"
                    android:text="Ø§Ùƒ Ø§Ø³Ù… Ø§Ù„Ø®Ø¯Ù…Ø© Ø£Ùˆ Ø§Ù„Ø³Ø¹Ø± Ø§Ù„Ù…Ø·Ù„ÙˆØ¨ Ù¨Ø§Ù„Ø¬Ø¯ÙŠØ¯ Ø¨Ø« Ù„Ø¥Ø¸Ù‡Ø§Ø± Ø§Ù„Ø®Ø¯Ù…Ø§Øª Ø§Ù„Ù…Ø·Ø§Ø¨Ù‚Ø©"
                    android:textAlignment="center"
                    android:textColor="@color/text_secondary"
                    android:textSize="14sp" />
            </FrameLayout>

        </LinearLayout>
    </ScrollView>
</LinearLayout>
```

activity_add_service.xml .6

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/rootAddServiceLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/bg_app"
    android:clickable="true"
    android:focusable="true"
    android:focusableInTouchMode="true"
```

```
    android:orientation="vertical">

    <!-- Top Action Bar -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/surface_white"
        android:elevation="2dp"
        android:gravity="center_vertical"
        android:orientation="horizontal"
        android:padding="16dp">

        <ImageButton
            android:id="@+id/btnBack"
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:contentDescription="Ø±Ø¬Ùˆ Ø¹"
            android:src="@drawable/ic_arrow_back" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="12dp"
            android:text="@string/add_service_title"
            android:textColor="@color/text_primary"
            android:textSize="18sp"
            android:textStyle="bold" />
    </LinearLayout>

    <ScrollView
        android:id="@+id/scrollViewContainer"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:clickable="true"
        android:fillViewport="true"
        android:focusable="true"
        android:focusableInTouchMode="true">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:clickable="true"
```

```
            android:focusable="true"
            android:focusableInTouchMode="true"
            android:orientation="vertical"
            android:padding="20dp">

            <!-- Service Title -->
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginBottom="6dp"
                android:text="@string/service_name_label"
                android:textColor="@color/text_primary"
                android:textSize="14sp"
                android:textStyle="bold" />

            <com.google.android.material.textfield.TextInputLayout
                style="@style/Widget.Material3.TextInputLayout.OutlinedBox"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginBottom="16dp"
                app:boxCornerRadiusBottomEnd="10dp"
                app:boxCornerRadiusBottomStart="10dp"
                app:boxCornerRadiusTopEnd="10dp"
                app:boxCornerRadiusTopStart="10dp">

                <com.google.android.material.textfield.TextInputEditText
                    android:id="@+id/etTitle"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:hint="@string/service_name_hint"
                    android:imeOptions="actionNext"
                    android:inputType="text"
                    android:textColor="#000000"
                    android:textColorHint="@color/text_hint" />
            </com.google.android.material.textfield.TextInputLayout>

            <!-- Category Dropdown / Spinner -->
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginBottom="6dp"
                android:text="@string/service_category_label"
                android:textColor="@color/text_primary"
                android:textSize="14sp"
```

```
                android:textStyle="bold" />

            <com.google.android.material.textfield.TextInputLayout
                style="@style/Widget.Material3.TextInputLayout.OutlinedBox.Expos
edDropdownMenu"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginBottom="16dp"
                app:boxCornerRadiusBottomEnd="10dp"
                app:boxCornerRadiusBottomStart="10dp"
                app:boxCornerRadiusTopEnd="10dp"
                app:boxCornerRadiusTopStart="10dp">

                <AutoCompleteTextView
                    android:id="@+id/actvCategory"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:hint="Ø§Ø®Øª Ø§Ù„ÙØ¦Ø©"
                    android:inputType="none"
                    android:textColor="#000000"
                    android:textColorHint="@color/text_hint" />
            </com.google.android.material.textfield.TextInputLayout>

            <!-- Price -->
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginBottom="6dp"
                android:text="@string/service_price_label"
                android:textColor="@color/text_primary"
                android:textSize="14sp"
                android:textStyle="bold" />

            <com.google.android.material.textfield.TextInputLayout
                style="@style/Widget.Material3.TextInputLayout.OutlinedBox"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginBottom="16dp"
                app:boxCornerRadiusBottomEnd="10dp"
                app:boxCornerRadiusBottomStart="10dp"
                app:boxCornerRadiusTopEnd="10dp"
                app:boxCornerRadiusTopStart="10dp">

                <com.google.android.material.textfield.TextInputEditText
```

```
            android:id="@+id/etPrice"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="@string/service_price_hint"
            android:imeOptions="actionNext"
            android:inputType="numberDecimal"
            android:textColor="#000000"
            android:textColorHint="@color/text_hint" />
    </com.google.android.material.textfield.TextInputLayout>

    <!-- Description (Multiline Details Field) -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="6dp"
        android:text="@string/service_desc_label"
        android:textColor="@color/text_primary"
        android:textSize="14sp"
        android:textStyle="bold" />

    <com.google.android.material.textfield.TextInputLayout
        style="@style/Widget.Material3.TextInputLayout.OutlinedBox"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"
        app:boxCornerRadiusBottomEnd="10dp"
        app:boxCornerRadiusBottomStart="10dp"
        app:boxCornerRadiusTopEnd="10dp"
        app:boxCornerRadiusTopStart="10dp">

        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/etDescription"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="top"
            android:hint="@string/service_desc_hint"
            android:imeOptions="actionDone"
            android:inputType="textMultiLine"
            android:minLines="4"
            android:textColor="#000000"
            android:textColorHint="@color/text_hint" />
    </com.google.android.material.textfield.TextInputLayout>

    <!-- Provider Name -->
```

```
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginBottom="6dp"
                android:text="@string/provider_name_label"
                android:textColor="@color/text_primary"
                android:textSize="14sp"
                android:textStyle="bold" />

            <com.google.android.material.textfield.TextInputLayout
                style="@style/Widget.Material3.TextInputLayout.OutlinedBox"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginBottom="16dp"
                app:boxCornerRadiusBottomEnd="10dp"
                app:boxCornerRadiusBottomStart="10dp"
                app:boxCornerRadiusTopEnd="10dp"
                app:boxCornerRadiusTopStart="10dp">

                <com.google.android.material.textfield.TextInputEditText
                    android:id="@+id/etProviderName"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:hint="@string/provider_name_hint"
                    android:imeOptions="actionNext"
                    android:inputType="textPersonName"
                    android:textColor="#000000"
                    android:textColorHint="@color/text_hint" />
            </com.google.android.material.textfield.TextInputLayout>

            <!-- Phone / Contact Method -->
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginBottom="6dp"
                android:text="@string/provider_phone_label"
                android:textColor="@color/text_primary"
                android:textSize="13sp"
                android:textStyle="bold" />

            <com.google.android.material.textfield.TextInputLayout
                style="@style/Widget.Material3.TextInputLayout.OutlinedBox"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
```

```
            android:layout_marginBottom="24dp"
            app:boxCornerRadiusBottomEnd="10dp"
            app:boxCornerRadiusBottomStart="10dp"
            app:boxCornerRadiusTopEnd="10dp"
            app:boxCornerRadiusTopStart="10dp">

            <com.google.android.material.textfield.TextInputEditText
                android:id="@+id/etProviderPhone"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="@string/provider_phone_hint"
                android:imeOptions="actionDone"
                android:inputType="text"
                android:textColor="#000000"
                android:textColorHint="@color/text_hint" />
        </com.google.android.material.textfield.TextInputLayout>

        <!-- Service Image Selection Section -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginBottom="6dp"
            android:text="ØµÙˆØ±Ø© Ø§Ù„Ø®Ø¯Ù…Ø© (Ø§Ø®ØªÙŠØ§Ø±ÙŠ)"
            android:textColor="@color/text_primary"
            android:textSize="14sp"
            android:textStyle="bold" />

        <com.google.android.material.card.MaterialCardView
            android:id="@+id/cardSelectServiceImage"
            android:layout_width="match_parent"
            android:layout_height="160dp"
            android:layout_marginBottom="24dp"
            android:clickable="true"
            android:focusable="true"
            app:cardBackgroundColor="@color/surface_white"
            app:cardCornerRadius="12dp"
            app:cardElevation="0dp"
            app:strokeColor="@color/card_stroke"
            app:strokeWidth="1dp">

            <FrameLayout
                android:layout_width="match_parent"
                android:layout_height="match_parent">
```

```
                    <ImageView
                        android:id="@+id/ivSelectedServiceImage"
                        android:layout_width="match_parent"
                        android:layout_height="match_parent"
                        android:scaleType="centerCrop"
                        android:contentDescription="ØµÙˆØ±Ø© Ø§Ù„Ø®Ø¯Ù…Ø© Ø§Ù„Ù…Ø®ØªØ§Ø±Ø©" />

                    <LinearLayout
                        android:id="@+id/llImagePlaceholder"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:layout_gravity="center"
                        android:gravity="center"
                        android:orientation="vertical">

                        <ImageView
                            android:layout_width="40dp"
                            android:layout_height="40dp"
                            android:src="@drawable/ic_add"
                            app:tint="@color/orange_primary" />

                        <TextView
                            android:layout_width="wrap_content"
                            android:layout_height="wrap_content"
                            android:layout_marginTop="8dp"
                            android:text="Ø§Ø¶ØºØ· Ù‡Ù†Ø§ Ù„Ø¥Ø¶Ø§ÙØ© ØµÙˆØ± Ù„Ù„Ø®Ø¯Ù…Ø© Ùˆ Ù…Ø§ Ø§Ù„Ù…Ø¹Ø±Ø¶ ðŸ–¼ï¸"
                            android:textColor="@color/text_secondary"
                            android:textSize="13sp" />
                    </LinearLayout>
                </FrameLayout>
            </com.google.android.material.card.MaterialCardView>

            <!-- Publish Button -->
            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnPublish"
                android:layout_width="match_parent"
                android:layout_height="56dp"
                android:backgroundTint="@color/orange_primary"
                android:text="@string/btn_publish_service"
                android:textColor="@color/white"
                android:textSize="16sp"
                android:textStyle="bold"
```

```
                app:cornerRadius="12dp" />
        </LinearLayout>
    </ScrollView>
</LinearLayout>
```

activity_service_detail.xml .7

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/bg_app"
    android:orientation="vertical">

    <!-- Top Action Bar -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/surface_white"
        android:elevation="2dp"
        android:gravity="center_vertical"
        android:orientation="horizontal"
        android:padding="16dp">

        <ImageButton
            android:id="@+id/btnBack"
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:contentDescription="Ø±Ø¬Ù‘Ø¹"
            android:src="@drawable/ic_arrow_back" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="12dp"
            android:text="@string/service_details_title"
            android:textColor="@color/text_primary"
            android:textSize="18sp"
```

```
            android:textStyle="bold" />
    </LinearLayout>

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:fillViewport="true">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:padding="20dp">

            <!-- Featured Work Sample Photo Card -->
            <com.google.android.material.card.MaterialCardView
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginBottom="16dp"
                app:cardBackgroundColor="@color/surface_white"
                app:cardCornerRadius="16dp"
                app:cardElevation="3dp"
                app:strokeWidth="0dp">

                <LinearLayout
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:orientation="vertical">

                    <ImageView
                        android:id="@+id/ivDetailWorkPhoto"
                        android:layout_width="match_parent"
                        android:layout_height="180dp"
                        android:contentDescription="ØµÙˆØ±Ø© Ø§Ù„Ø¹Ù…Ù„ Ø§Ù„Ø´Ø§
Ù…Ù„Ø©"
                        android:scaleType="centerCrop"
                        android:src="@drawable/img_work_code" />

                    <TextView
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:padding="12dp"
                        android:text="ðŸ–¼ï¸ Ù†Ù…Ùˆ Ø°Ù‡Ø¨ÙŠ Ù…Ø¹ Ø£Ø¹Ù…Ø§Ù
```

```
„ Ùˆ Ù…Ø´Ø§Ø±ÙŠØ¹ Ù‡Ø°Ø§ Ø§Ù„Ù…Ø²Ø§Ø¬ Ø§Ù„Ø¹Ø§Ù…"
                        android:textColor="@color/navy_secondary"
                        android:textSize="12sp"
                        android:textStyle="bold" />
                </LinearLayout>
            </com.google.android.material.card.MaterialCardView>

            <!-- Main Service Card -->
            <com.google.android.material.card.MaterialCardView
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginBottom="16dp"
                app:cardBackgroundColor="@color/surface_white"
                app:cardCornerRadius="16dp"
                app:cardElevation="2dp"
                app:strokeColor="@color/card_stroke"
                app:strokeWidth="1dp">

                <LinearLayout
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:orientation="vertical"
                    android:padding="20dp">

                    <LinearLayout
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:gravity="center_vertical"
                        android:orientation="horizontal">

                        <TextView
                            android:id="@+id/tvDetailCategory"
                            android:layout_width="wrap_content"
                            android:layout_height="wrap_content"
                            android:background="@drawable/bg_category_badge"
                            android:paddingHorizontal="12dp"
                            android:paddingVertical="6dp"
                            android:text="Ø¨Ø±Ù…Ø¬Ø©"
                            android:textColor="@color/navy_secondary"
                            android:textSize="13sp"
                            android:textStyle="bold" />

                        <View
                            android:layout_width="0dp"
```

```
                android:layout_height="0dp"
                android:layout_weight="1" />

            <ImageView
                android:layout_width="18dp"
                android:layout_height="18dp"
                android:contentDescription="Rating"
                android:src="@drawable/ic_star" />

            <TextView
                android:id="@+id/tvDetailRating"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginStart="4dp"
                android:text="4.9"
                android:textColor="@color/text_primary"
                android:textSize="14sp"
                android:textStyle="bold" />
        </LinearLayout>

        <TextView
            android:id="@+id/tvDetailTitle"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="12dp"
            android:text="Ø¹Ù†Ùˆ Ø§Ù„Ø®Ø¯Ù…Ø© Ø§Ù„Ø¥ÙˆÙ„Ù‰"
            android:textColor="@color/text_primary"
            android:textSize="20sp"
            android:textStyle="bold" />

        <!-- Price Card -->
        <LinearLayout
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="16dp"
            android:background="@drawable/bg_price_tag"
            android:gravity="center"
            android:orientation="horizontal"
            android:paddingHorizontal="16dp"
            android:paddingVertical="8dp">

            <TextView
                android:id="@+id/tvDetailPrice"
                android:layout_width="wrap_content"
```

```
                        android:layout_height="wrap_content"
                        android:text="50"
                        android:textColor="@color/orange_primary"
                        android:textSize="24sp"
                        android:textStyle="bold" />

                    <TextView
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:layout_marginStart="6dp"
                        android:text="@string/currency"
                        android:textColor="@color/orange_primary"
                        android:textSize="16sp"
                        android:textStyle="bold" />
                </LinearLayout>
            </LinearLayout>
        </com.google.android.material.card.MaterialCardView>

        <!-- Description Card -->
        <com.google.android.material.card.MaterialCardView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginBottom="16dp"
            app:cardBackgroundColor="@color/surface_white"
            app:cardCornerRadius="16dp"
            app:cardElevation="2dp"
            app:strokeColor="@color/card_stroke"
            app:strokeWidth="1dp">

            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:orientation="vertical"
                android:padding="20dp">

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Ùˆ Ø§Ù„Ø®Ø¯Ù…Ø©"
                    android:textColor="@color/text_primary"
                    android:textSize="16sp"
                    android:textStyle="bold" />

                <TextView
```

```
                        android:id="@+id/tvDetailDescription"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:layout_marginTop="8dp"
                        android:lineSpacingExtra="4dp"
                        android:text="Ù‡Ù†Ø§ ÙŠÙ…ÙƒÙ†Ùƒ Ø§Ù„Ø¹Ø«ÙˆØ±
„ Ù„Ù„Ø®Ø¯Ù…Ø© Ù‚Ù…Ø§ ÙŠÙ…ÙƒÙ†Ùƒ Ø¨Ø§Ù„Ø§ØªØµ..."
                        android:textColor="@color/text_secondary"
                        android:textSize="14sp" />
                </LinearLayout>
            </com.google.android.material.card.MaterialCardView>

            <!-- Provider Info & Contact Options Card -->
            <com.google.android.material.card.MaterialCardView
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginBottom="20dp"
                app:cardBackgroundColor="@color/surface_white"
                app:cardCornerRadius="16dp"
                app:cardElevation="2dp"
                app:strokeColor="@color/card_stroke"
                app:strokeWidth="1dp">

                <LinearLayout
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:orientation="vertical"
                    android:padding="20dp">

                    <TextView
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:text="@string/provider_info"
                        android:textColor="@color/text_primary"
                        android:textSize="16sp"
                        android:textStyle="bold" />

                    <TextView
                        android:id="@+id/tvDetailProviderName"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:layout_marginTop="8dp"
                        android:text="Ù…Ù‚Ø¯Ù… Ø§Ù„Ø®Ø¯Ù…Ø©: Ø£Ø³Ø§Ù…Ø© Ø£Ø Ù…Ø¯"
                        android:textColor="@color/text_primary"
```

```
                        android:textSize="14sp"
                        android:textStyle="bold" />

                    <TextView
                        android:id="@+id/tvDetailProviderPhone"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:layout_marginTop="4dp"
                        android:text="Ø±Ù‚Ù… / Ù‘Ø³ÙŠÙ„Ø© Ø§Ù„Ø§Ø‘Ø§Ø§Ù„: 079123
4567"
                        android:textColor="@color/text_secondary"
                        android:textSize="14sp" />

                    <View
                        android:layout_width="match_parent"
                        android:layout_height="1dp"
                        android:layout_marginVertical="12dp"
                        android:background="@color/card_stroke" />

                    <TextView
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:layout_marginBottom="8dp"
                        android:text="Ø§Ø®Ø±Ø· Ø·Ø±ÙŠÙ‚Ø© Ø§Ù„Ø§Ø‘Ø§Ø§Ù„ Ø§Ù„Ù…Ù…Ù†
†Ø§Ø³Ø¨Ø© Ù„Ùf:"
                        android:textColor="@color/text_primary"
                        android:textSize="13sp"
                        android:textStyle="bold" />

                    <!-- WhatsApp Button -->
                    <com.google.android.material.button.MaterialButton
                        android:id="@+id/btnContactWhatsApp"
                        style="@style/Widget.Material3.Button.OutlinedButton"
                        android:layout_width="match_parent"
                        android:layout_height="50dp"
                        android:layout_marginBottom="8dp"
                        android:text="@string/btn_contact_whatsapp"
                        android:textColor="#16A34A"
                        app:cornerRadius="10dp"
                        app:icon="@drawable/ic_whatsapp"
                        app:iconGravity="textStart"
                        app:iconPadding="8dp"
                        app:iconTint="#16A34A"
                        app:strokeColor="#16A34A" />
```

```
            <!-- Phone Call Button -->
            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnContactPhone"
                style="@style/Widget.Material3.Button.OutlinedButton"
                android:layout_width="match_parent"
                android:layout_height="50dp"
                android:layout_marginBottom="8dp"
                android:text="@string/btn_contact_phone"
                android:textColor="@color/navy_secondary"
                app:cornerRadius="10dp"
                app:icon="@drawable/ic_phone"
                app:iconGravity="textStart"
                app:iconPadding="8dp"
                app:iconTint="@color/navy_secondary"
                app:strokeColor="@color/navy_secondary" />

            <!-- Direct Message Chat Dialog Button -->
            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnSendDirectMessage"
                style="@style/Widget.Material3.Button.OutlinedButton"
                android:layout_width="match_parent"
                android:layout_height="50dp"
                android:text="@string/btn_send_direct_message"
                android:textColor="@color/orange_primary"
                app:cornerRadius="10dp"
                app:icon="@drawable/ic_chat_message"
                app:iconGravity="textStart"
                app:iconPadding="8dp"
                app:iconTint="@color/orange_primary"
                app:strokeColor="@color/orange_primary" />

        </LinearLayout>
    </com.google.android.material.card.MaterialCardView>

    <!-- Order Button -->
    <com.google.android.material.button.MaterialButton
        android:id="@+id/btnOrderNow"
        android:layout_width="match_parent"
        android:layout_height="56dp"
        android:backgroundTint="@color/orange_primary"
        android:text="@string/btn_order_now"
        android:textColor="@color/white"
        android:textSize="16sp"
```

```
                android:textStyle="bold"
                app:cornerRadius="12dp" />

        </LinearLayout>
    </ScrollView>
</LinearLayout>
```

activity_chat.xml .8

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/bg_app"
    android:orientation="vertical">

    <!-- Top Action Bar -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/surface_white"
        android:elevation="2dp"
        android:gravity="center_vertical"
        android:orientation="horizontal"
        android:padding="12dp">

        <ImageButton
            android:id="@+id/btnBack"
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:contentDescription="Ø±Ø¬ÙˆØ¹"
            android:src="@drawable/ic_arrow_back" />

        <LinearLayout
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_marginStart="8dp"
            android:layout_weight="1"
```

```
            android:orientation="vertical">

            <TextView
                android:id="@+id/tvChatReceiverName"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Ø§Ø³Ù… Ø§Ù„Ù…Ø³ØªÙ‚Ø¨Ù„"
                android:textColor="@color/text_primary"
                android:textSize="16sp"
                android:textStyle="bold" />

            <TextView
                android:id="@+id/tvChatServiceTitle"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:ellipsize="end"
                android:maxLines="1"
                android:text="Ø¨Ø´Ø£Ù† Ø§Ù„Ø®Ø¯Ù…Ø©..."
                android:textColor="@color/text_secondary"
                android:textSize="12sp" />
        </LinearLayout>
    </LinearLayout>

    <!-- Chat Messages List -->
    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/rvChatMessages"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:clipToPadding="false"
        android:paddingVertical="8dp" />

    <!-- Bottom Message Input Bar -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/surface_white"
        android:elevation="4dp"
        android:gravity="center_vertical"
        android:orientation="horizontal"
        android:padding="8dp">

        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/etChatMessageInput"
```

```
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:background="@null"
            android:hint="Ø§Ùƒ Ø±Ø³Ø§Ù„Ø© Ù‡Ù†Ø§..."
            android:inputType="textMultiLine"
            android:maxLines="4"
            android:paddingHorizontal="16dp"
            android:paddingVertical="10dp"
            android:textColor="@color/text_primary"
            android:textColorHint="@color/text_hint"
            android:textSize="14sp" />

        <com.google.android.material.button.MaterialButton
            android:id="@+id/btnSendChatMessage"
            android:layout_width="wrap_content"
            android:layout_height="48dp"
            android:layout_marginStart="8dp"
            android:backgroundTint="@color/orange_primary"
            android:text="Ø¥Ø±Ø³Ø§Ù„"
            android:textColor="@color/white"
            android:textSize="14sp"
            android:textStyle="bold"
            app:cornerRadius="12dp" />
    </LinearLayout>
</LinearLayout>
```

9. item_service.xml

```
<?xml version="1.0" encoding="utf-8"?>
<com.google.android.material.card.MaterialCardView xmlns:android="http://schema
s.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginHorizontal="16dp"
    android:layout_marginVertical="8dp"
    android:clickable="true"
    android:focusable="true"
    app:cardBackgroundColor="@color/surface_white"
```

```
    app:cardCornerRadius="16dp"
    app:cardElevation="2dp"
    app:strokeColor="@color/card_stroke"
    app:strokeWidth="1dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <!-- Work Image Sample Preview -->
        <ImageView
            android:id="@+id/ivServiceThumbnail"
            android:layout_width="match_parent"
            android:layout_height="140dp"
            android:contentDescription="ØµÙˆØ±Ø© Ø§Ù„Ø¹Ù…Ù„"
            android:scaleType="centerCrop"
            android:src="@drawable/img_work_code" />

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:padding="16dp">

            <!-- Top Header: Category Badge & Rating -->
            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:gravity="center_vertical"
                android:orientation="horizontal">

                <TextView
                    android:id="@+id/tvCategoryBadge"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:background="@drawable/bg_category_badge"
                    android:paddingHorizontal="10dp"
                    android:paddingVertical="4dp"
                    android:text="Ø¨Ø±Ù…Ø¬Ø©"
                    android:textColor="@color/navy_secondary"
                    android:textSize="12sp"
                    android:textStyle="bold" />
```

```
                <View
                    android:layout_width="0dp"
                    android:layout_height="0dp"
                    android:layout_weight="1" />

                <ImageView
                    android:layout_width="16dp"
                    android:layout_height="16dp"
                    android:contentDescription="Rating"
                    android:src="@drawable/ic_star" />

                <TextView
                    android:id="@+id/tvRating"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:layout_marginStart="4dp"
                    android:text="4.9"
                    android:textColor="@color/text_primary"
                    android:textSize="13sp"
                    android:textStyle="bold" />

                <TextView
                    android:id="@+id/tvReviewsCount"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:layout_marginStart="2dp"
                    android:text="(38)"
                    android:textColor="@color/text_secondary"
                    android:textSize="12sp" />
            </LinearLayout>

            <!-- Service Title -->
            <TextView
                android:id="@+id/tvServiceTitle"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginTop="10dp"
                android:ellipsize="end"
                android:maxLines="2"
                android:text="ØªØµÙ…ÙŠÙ… Ø´Ø¹Ø§Ø± ÙˆÙ‡ÙˆÙŠØ© Ø¨ØµØ±ÙŠØ© Ø§Ø§ØªØ±Ø§
§ÙÙŠØ©"
                android:textColor="@color/text_primary"
                android:textSize="16sp"
                android:textStyle="bold" />
```

```
<!-- Provider Name -->
<TextView
    android:id="@+id/tvProviderName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="4dp"
    android:text="Ø¨Ù˜Ø§Ø³Ø·Ø©: Ø£ØÙ…Ø¯ Ø§Ù„Ù„Ù…Ù‡Ø¯Ø³"
    android:textColor="@color/text_secondary"
    android:textSize="13sp" />

<!-- Price Similarity Badge -->
<TextView
    android:id="@+id/tvPriceSimilarityBadge"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="8dp"
    android:background="@color/green_light"
    android:paddingHorizontal="8dp"
    android:paddingVertical="3dp"
    android:text="Ø³Ø¹Ø± Ù…Ù…Ø§Ø«Ù„ Ù„Ù„Ù…Ù„Ø§Ø¦Ù…! ðŸŽ¯"
    android:textColor="@color/green_success"
    android:textSize="11sp"
    android:textStyle="bold"
    android:visibility="gone" />

<View
    android:layout_width="match_parent"
    android:layout_height="1dp"
    android:layout_marginVertical="12dp"
    android:background="@color/card_stroke" />

<!-- Bottom Footer: Price & Details Action -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center_vertical"
    android:orientation="horizontal">

    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:background="@drawable/bg_price_tag"
        android:gravity="center"
```

```
                    android:orientation="horizontal"
                    android:paddingHorizontal="12dp"
                    android:paddingVertical="6dp">

                    <TextView
                        android:id="@+id/tvPriceValue"
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:text="25"
                        android:textColor="@color/orange_primary"
                        android:textSize="18sp"
                        android:textStyle="bold" />

                    <TextView
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:layout_marginStart="4dp"
                        android:text="@string/currency"
                        android:textColor="@color/orange_primary"
                        android:textSize="14sp"
                        android:textStyle="bold" />
                </LinearLayout>

                <View
                    android:layout_width="0dp"
                    android:layout_height="0dp"
                    android:layout_weight="1" />

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Ø¹Ø±Ø¶ Ø§Ù„Ø§ØµÙŠÙ„ Ùˆ Ø§Ù„Ø£Ø¹Ù…Ø§Ù„ â†"
                    android:textColor="@color/navy_secondary"
                    android:textSize="13sp"
                    android:textStyle="bold" />
            </LinearLayout>

        </LinearLayout>
    </LinearLayout>
</com.google.android.material.card.MaterialCardView>
```

item_order.xml .10

```
<?xml version="1.0" encoding="utf-8"?>
<com.google.android.material.card.MaterialCardView xmlns:android="http://schema
s.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginHorizontal="16dp"
    android:layout_marginVertical="8dp"
    app:cardBackgroundColor="@color/surface_white"
    app:cardCornerRadius="16dp"
    app:cardElevation="2dp"
    app:strokeColor="@color/card_stroke"
    app:strokeWidth="1dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <!-- Top Order Status & Date -->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center_vertical"
            android:orientation="horizontal">

            <TextView
                android:id="@+id/tvOrderStatus"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:background="@drawable/bg_category_badge"
                android:paddingHorizontal="10dp"
                android:paddingVertical="4dp"
                android:text="ÙŠÙ†Ø²Ø¸Ø±Ø§ Ù…Ù‘Ø§Ù‹Ù,Ø© Ù…Ù‚Ù… Ø§Ù„Ø„Ø®Ø¯Ù…Ø©
â³"
                android:textColor="@color/brand_primary"
                android:textSize="12sp"
                android:textStyle="bold" />
```

```
        <View
            android:layout_width="0dp"
            android:layout_height="0dp"
            android:layout_weight="1" />

        <TextView
            android:id="@+id/tvOrderDate"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Ø³Ø§Ø±ÙŠØ® Ø§Ù„Ø·Ù„Ø¨: 2026-10-09"
            android:textColor="@color/text_secondary"
            android:textSize="12sp" />
    </LinearLayout>

    <!-- Service Title -->
    <TextView
        android:id="@+id/tvOrderServiceTitle"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:ellipsize="end"
        android:maxLines="2"
        android:text="Ø¹Ù†Ù‘Ø§Ù† Ø§Ù„Ø®Ø¯Ù…Ø© Ø§Ù„Ø·Ù„Ø¨"
        android:textColor="@color/text_primary"
        android:textSize="16sp"
        android:textStyle="bold" />

    <!-- Provider Info / Client Info & Price -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:gravity="center_vertical"
        android:orientation="horizontal">

        <LinearLayout
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:orientation="vertical">

            <TextView
                android:id="@+id/tvOrderProviderName"
```

```
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Ø§Ù„Ù…Ø²Ù‚Ø¯: Ø£Ø…Ø¯ Ø§Ù„Ù…Ù‡Ù†Ø¯Ø³"
                android:textColor="@color/text_secondary"
                android:textSize="13sp" />

            <TextView
                android:id="@+id/tvOrderClientName"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginTop="2dp"
                android:text="Ø§Ù„Ø¹Ù…ÙŠÙ„: Ø§Ù„Ø·Ù„Ø¨"
                android:textColor="@color/text_primary"
                android:textSize="13sp"
                android:textStyle="bold"
                android:visibility="gone" />
        </LinearLayout>

        <TextView
            android:id="@+id/tvOrderPrice"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="$25"
            android:textColor="@color/orange_primary"
            android:textSize="18sp"
            android:textStyle="bold" />
    </LinearLayout>

    <View
        android:layout_width="match_parent"
        android:layout_height="1dp"
        android:layout_marginVertical="12dp"
        android:background="@color/card_stroke" />

    <!-- Provider Accept / Reject Action Buttons -->
    <LinearLayout
        android:id="@+id/llProviderActions"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="8dp"
        android:orientation="horizontal"
        android:visibility="gone">

        <com.google.android.material.button.MaterialButton
```

```
                android:id="@+id/btnAcceptOrder"
                android:layout_width="0dp"
                android:layout_height="44dp"
                android:layout_weight="1"
                android:layout_marginEnd="6dp"
                android:backgroundTint="@color/orange_primary"
                android:text="Ù‚Ø¨Ù‘Ù„ Ø§Ù„Ø·Ù„Ø¨ âœ…"
                android:textColor="@color/white"
                android:textSize="13sp"
                android:textStyle="bold"
                app:cornerRadius="10dp" />

            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnRejectOrder"
                style="@style/Widget.Material3.Button.OutlinedButton"
                android:layout_width="0dp"
                android:layout_height="44dp"
                android:layout_weight="1"
                android:layout_marginStart="6dp"
                android:text="Ø±ÙØ¶ Ø§Ù„Ø·Ù„Ø¨ âŒ"
                android:textColor="#DC2626"
                android:textSize="13sp"
                app:cornerRadius="10dp"
                app:strokeColor="#DC2626" />
        </LinearLayout>

        <!-- Provider Complete Action Button -->
        <com.google.android.material.button.MaterialButton
            android:id="@+id/btnCompleteOrder"
            android:layout_width="match_parent"
            android:layout_height="44dp"
            android:layout_marginBottom="8dp"
            android:backgroundTint="#16A34A"
            android:text="Ø¥Ù†Ù‡Ø§Ø¡ Ø§Ù„Ø®Ø¯Ù…Ø© Ù‘ØªÙ…Ù‘Ø© ðŸ"
            android:textColor="@color/white"
            android:textSize="13sp"
            android:textStyle="bold"
            android:visibility="gone"
            app:cornerRadius="10dp" />

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">
```

```
            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnInAppChat"
                android:layout_width="0dp"
                android:layout_height="44dp"
                android:layout_weight="1"
                android:layout_marginEnd="4dp"
                android:backgroundTint="@color/navy_secondary"
                android:text="Ù…Ø±Ø§Ø³Ù„Ø© Ø¯Ø§Ø®Ù„Ø§Ø·Ø·Ø¨Ù, ðŸ’¬"
                android:textColor="@color/white"
                android:textSize="12sp"
                android:textStyle="bold"
                app:cornerRadius="10dp" />

            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnOrderContactProvider"
                style="@style/Widget.Material3.Button.OutlinedButton"
                android:layout_width="0dp"
                android:layout_height="44dp"
                android:layout_weight="1"
                android:layout_marginStart="4dp"
                android:text="Ù‚Ø§Ø³Ø§Ø¨ / Ø§Ø²Ø§Ø§Ù„Ù„ ðŸ“ž"
                android:textColor="@color/text_primary"
                android:textSize="12sp"
                app:cornerRadius="10dp"
                app:strokeColor="@color/card_stroke" />
        </LinearLayout>

    </LinearLayout>
</com.google.android.material.card.MaterialCardView>
```

item_chat_message.xml .11

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/llMessageContainer"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:paddingHorizontal="16dp"
```

```
    android:paddingVertical="6dp"
    android:orientation="vertical">

    <com.google.android.material.card.MaterialCardView
        android:id="@+id/cardMessageBubble"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        app:cardCornerRadius="16dp"
        app:cardElevation="1dp"
        app:strokeWidth="0dp">

        <LinearLayout
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:paddingHorizontal="14dp"
            android:paddingVertical="10dp">

            <TextView
                android:id="@+id/tvMessageSender"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:textColor="@color/orange_primary"
                android:textSize="11sp"
                android:textStyle="bold" />

            <TextView
                android:id="@+id/tvMessageText"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginTop="2dp"
                android:textColor="@color/text_primary"
                android:textSize="14sp" />

            <TextView
                android:id="@+id/tvMessageTime"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_gravity="end"
                android:layout_marginTop="4dp"
                android:textColor="@color/text_secondary"
                android:textSize="10sp" />
        </LinearLayout>
```

```
    </com.google.android.material.card.MaterialCardView>
</LinearLayout>
```